

Monodromy of Hypergeometric Functions and Non-Lattice Integral Monodromy

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Introduction

The hypergeometric series

$$F(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a, n)(b, n)}{(c, n) n!} x^n, \quad |x| < 1 \quad (0.1)$$

where $(a, n) = \prod_{i=0}^{n-1} (a + i)$, defined for c not an integer ≤ 0 , was first introduced by Euler in 1778 as a solution of the hypergeometric differential equation

$$x(1-x)y'' + (c - (a+b+1)x)y' - aby = 0. \quad (0.2)$$

$F(a, b; c; x)$ represents the unique solution of (??) which is holomorphic at $x = 0$ and takes the value 1 at $x = 0$. If neither b nor $c - b$ is an integer < 0 , Euler knew the integral representation

$$F(a, b; c; x) = \frac{\Gamma(b)\Gamma(c-b)}{\Gamma(c)} \int_0^1 z^{b-1}(1-z)^{c-b-1}(1-zx)^{-a} dz. \quad (0.3)$$

Replacing z by u^{-1} , (??) also has the form

$$\int_1^{\infty} u^{-a}(u-1)^{c-b-1}(u-x)^{-b} du. \quad (3')$$

If we integrate instead from g to h with g and h in $\{0, 1, \infty, x\}$ we get other solutions of (??), a fact discovered independently by Hermite [?], Pochhammer [?], and Schäfli [?]. Even when the integral diverges, it yields solutions of (??) when taken as its Hadamard “finite part”, provided that the integrand does not have a pole of integral order at g or h .

The hypergeometric equation (??) is the unique second order linear differential equation with regular singularities, singular only at $0, 1, \infty$ with exponents $(0, 1-c)$, $(0, c-a-b)$, (a, b) respectively. “Exponents (α, α') at a singularity p ” means that suitable linearly independent linear combinations of the branches of a solution have the form $(x-p)^\alpha f_1(x)$, $(x-p)^{\alpha'} f_2(x)$ with f holomorphic around p ; if $\alpha - \alpha' \in \mathbb{Z}$ an additional logarithmic term is allowed. In 1857, Riemann proved that the solutions of (??) are the only multivalued functions with exactly two linearly independent branches, branching only at $0, 1, \infty$ with exponents as above (cf. [?]). Riemann’s proof assumed that none of the exponent

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differences at a singularity was an integer and proceeded by first computing global monodromy. The above characterization of the hypergeometric equation is the basis of Fuchs' proof of Riemann's theorem (cf. [?]).

In his seminal paper [?], which seeks to determine the values a, b, c for which the hypergeometric function is an algebraic function of x , Schwarz considered the map $x \mapsto w_2(x)/w_1(x)$ where w_1 and w_2 form a base of W , the two dimensional linear space of all solutions of (??). Let $\tilde{Q}$ denote the universal covering space of $Q := \mathbb{P}^1(\mathbb{C}) - \{0, 1, \infty\}$. The map

$$w : x \mapsto w_2(x)/w_1(x) \quad (0.4)$$

is a multivalued map from Q to the projective space $\mathbb{P}(W^*)$ of lines in the dual space W^* of W , i.e. w may be construed as a single-valued map

$$\tilde{w} : \tilde{Q} \rightarrow \mathbb{P}(W^*). \quad (0.5)$$

The fundamental group $\pi_1(Q)$ acts on $\mathbb{P}(W^*)$ ("monodromy action") and the map is $\pi_1(Q)$ -equivariant. Let Γ denote the image of $\pi_1(Q)$ via the monodromy action. For Schwarz's original problem, the question reduces to "when is Γ finite?" Schwarz also solved the problem : when can (??) be inverted to provide a univalued map from an open domain to Q ? When this happens Γ has a fundamental domain for its action on either (i) $\mathbb{P}(W^*)$, (ii) $\mathbb{P}(W^*)$ minus a point, or (iii) a disc in $\mathbb{P}(W^*)$, and is consequently discrete in $\text{PGL}_2(\mathbb{C})$.

It is case (iii) of this latter question which Picard generalized in [?] to a two variable analogue of the hypergeometric function. In the more general d variable case, this function is best defined by its integral representation

$$F(x_2, \dots, x_{d+1}) = \int_1^\infty u^{-\mu_0} (u-1)^{-\mu_1} \prod_{i=2}^{d+1} (u-x_i)^{-\mu_i} du. \quad (0.6)$$

Let b_∞ be the order of the pole of the integrand at ∞ . When the sum is extended over all the μ 's, one has $\sum b_\mu = 2$. In this introduction, we assume that none of the μ_i is an integer.

The function G obtained from F by holding fixed all variables but one was investigated by Pochhammer [?]. He formed the $(d+1)$ -order linear differential equation satisfied by G and characterized its solutions as the multivalued functions with exactly $d+1$ linearly independent branches and ramification of a prescribed type. At each finite ramification point, d branches of G are holomorphic, and at ∞ the same holds true after multiplying G by a suitable power of the variable.

The function F defined by (??) has a power series expansion

$$\sum_{\substack{n_i \geq 0 \\ 2 \leq i \leq d+1}} \frac{\prod_{i=2}^{d+1} (\mu_i, n_i) \cdot (1 - \sum \mu_i, \sum n_i)}{(1 - \mu_0, \sum n_i) \cdot (\sum n_i)!} \prod_{i=2}^{d+1} x_i^{n_i}. \quad (0.7)$$

For $d = 2$, this series has been investigated by Appell [?] who showed that it satisfies a system of linear partial differential equations expressing each second derivative of F in terms of its first derivatives. In [?] Picard characterized the solutions of this system as the multivalued functions of two variables x, y with exactly three linearly independent branches and with ramification of a prescribed type along the seven lines x or $y = 0, 1, \infty$, $x = y$. The function F is the only solution holomorphic at $(0, 0)$.

For d arbitrary, the series (??) has been investigated by Lauricella [?], and Terada [?] obtained results parallel to those of Picard [?].

The existence of differential or partial differential equations of Fuchsian type as above is to be expected if one considers (??) as a period integral. The equations are satisfied not only by F but also by any integral of the same integrand taken from g to h where g and h are in $\{\infty, 0, 1, x_2, \dots, x_{d+1}\}$. There are $d + 1$ linearly independent such integrals, and, following Schwarz, it is natural to take them as the projective coordinates of a point in projective d -space $\mathbb{P}^d$. This yields a map

$$\tilde{Q} \rightarrow \mathbb{P}^d \quad (4'')$$

where $\tilde{Q}$ is the universal covering of the space $Q \subset (\mathbb{P}^1)^d$ defined as

$$Q = \{(x_i) \mid x_i \neq 0, 1, \infty \text{ and } x_i \neq x_j \text{ for } i \neq j\}.$$

The action of $\pi_1(Q)$ on $\mathbb{P}^d$ is called the monodromy action. The map (4'') is $\pi_1(Q)$ -equivariant. Let Γ denote the image of $\pi_1(Q)$ in $\mathrm{PGL}(d + 1, \mathbb{C})$. In [?] and [?] Picard gives a criterion for the multivalued map $w : \tilde{Q} \rightarrow \mathbb{P}^d$ ($d = 2$) defined by (4'') to invert as a univalued map from a ball in $\mathbb{P}^d$ to a partial compactification of Q . When this happens, Γ is discrete in a $\mathrm{PU}(1, d)$ subgroup of $\mathrm{PGL}(d + 1, \mathbb{C})$.

As pointed out in [?] the proof of discreteness of Γ that Picard sketches in [?] leads to an obstacle and is inadequate. Our first objective in this paper is to give a correct proof for $d = 2$ and also for general d . In order to carry this out in modern concepts, we need only deal with rational μ_i and multivalued integrands on $\mathbb{P}^1$ which are single valued on a finite ramified covering of $\mathbb{P}^1$. However, in order to provide a framework for dealing with arbitrary parameters μ_i , we have introduced local systems on $\mathbb{P}^1 - \{\infty, 0, 1, x_2, \dots, x_{d+1}\}$. This necessitates an *ab ovo* development of holomorphic cohomology in order to validate the corresponding Hodge decomposition. In the end, this viewpoint does have the advantage of directness.

The two main theorems of this paper are Theorems (11.4) and (12.11). The first says in effect : If $0 < \mu_i < 1$ for all i ($0 \leq i \leq d + 1$ or $i = \infty$) and $(1 - \mu_i - \mu_j)^{-1}$ is an integer for all $i \neq j$ such that $\mu_i + \mu_j < 1$, then the monodromy group of the hypergeometric function (??) is a lattice in the projective unitary group $\mathrm{PU}(1, d)$, i.e. is discrete and of finite covolume.

Theorem (12.11) combined with (12.9) says in effect : For each non-arithmetic lattice arising in Theorem (11.4), there is an algebraic family defined over $\mathbb{Q}$ of algebraic curves X whose monodromy group is a subgroup of $\mathrm{Aut} H^1(X_0, \mathbb{Z})$ which is not of finite covolume in its Zariski-closure.

The example 15 of §(14.3) corresponds for instance to the family of curves, depending on the parameters x, y , with equation

$$U_1^{19} = U_0^6(U - 1)^5(U - x)^4(U - y)^4$$

(a cyclic covering of order 19 of $\mathbb{P}^1$).

The integrality condition that $(1 - \mu_i - \mu_j)^{-1}$ be an integer ensures that the key map of this paper is

'etale in codimension one. A model situation in which integrality is used in such a way is the following. Let D be the unit disc, $D^* := D - \{0\}$, and $\zeta = r/s$ a rational number > 0 , written as a reduced fraction. Let $\tilde{D}^*$ be the finite covering of D^* on which the multivalued function $z \mapsto z^\zeta$ is defined. We complete it to the ramified covering $\tilde{D} = \tilde{D}^* \cup \{0\}$ of D .

The multivalued function $z^{1/s}$ on D^* “is” a uniformizing parameter for $\tilde{D}$ at 0. The map $z \mapsto z^\zeta$ from $\tilde{D}$ to D is
 ’etale at 0 (i.e.

’etale in codimension 1) if and only if ζ^{-1} is an integer. Indeed $z^\zeta = (z^{1/s})^r$. For a description of how this enters the proof, we refer to the comments after (3.11).

The lists in Section 14 provide examples of non-arithmetic lattices in $\mathrm{PU}(1, d)$ for $d = 2, 3$ —both cocompact and non-cocompact for $d = 2$. For $d > 5$, Theorem (11.4) yields no lattice at all, by (14.2).

The case $d = 1$ is treated in (12.5.5) and (12.6.3). The lattices arising from Theorem (11.4) in this case are the triangle groups (p, q, r) generated by rotations through double the angles with centers the vertex of a geodesic triangle in the Poincaré disc, with angles $\pi/p, \pi/q, \pi/r$ when p, q, r are positive integers (or ∞) satisfying

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \leq 1.$$

These lattices are described in Fricke-Klein [?] and are infinite in number.

However, the number of arithmetic lattices arising is finite. An explicit list can be deduced from results of Takeuchi on arithmetic triangle groups [?].

In the final Section 15, we make some comments about the list of 102 solutions of R. Levavasseur in [?] to Picard’s integrality conditions for discreteness in the case $d = 2$.

In the paper “Generalized Picard lattices arising from half-integral conditions”, printed after this one, the second named author is able to relax the integrality condition on $(1 - \mu_i - \mu_j)^{-1}$ and obtains lattices for $d < 9$ [?].

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Notations

P	a complex projective line
N	an integer ≥ 0
S	a set of N points of P
$\alpha = (\alpha_s)_{s \in S}$	a family of complex numbers indexed by S
L	rank one local system
$H^*(P - S, L)$	cohomology with coefficients in L
$H_c^*(P - S, L)$	cohomology with compact support
M	space of injective maps $m : S \rightarrow P$
Q	quotient $\mathrm{PGL}(2)$
$\backslash M$	
$B(\alpha)_M$	flat projective space bundle
$\tilde{w}$	holomorphic section
Γ	monodromy group
Q_{st}	stable compactification
Q_{sst}	semi-stable compactification
$B^+(\alpha)$	complex ball bundle
INT	integrality condition

1 Cohomology of a rank one local system on a punctured projective line

1.1

Let us start with the following data :

- P : a complex projective line ;
- N : an integer ≥ 0 ;
- S : a set of N points of P ;
- $\alpha = (\alpha_s)_{s \in S}$: a family of complex numbers indexed by S , satisfying the condition $\prod_{s \in S} \alpha_s = 1$.

We will be mainly interested in the case when $N > 3$ and none of the α_s is 1.

If a base point $o \in P - S$ is given, the functor “fibre at o ” is an equivalence of the category of complex local systems (= flat vector bundles) on $P - S$ with that of complex vector spaces provided with an action of $\pi_1(P - S, o)$. In particular, rank one local systems correspond to homomorphisms $\mathbb{H}_1(P - S) = \pi_1(P - S, o)^{\mathrm{ab}} \rightarrow \mathbb{C}^*$. The group $\mathbb{H}_1$ is generated by small positive loops γ_s around each $s \in S$, with the relation $\prod \gamma_s = 1$ as the only relation. Up to isomorphism, there is hence a unique one dimensional complex local system L on $P - S$ such that the monodromy of L around each $s \in S$ is multiplication by α_s , or, as we will say, of monodromy α . By definition, if v_0 is in the fibre of L at a point x near $s \in S$, and if we let x turn counter clockwise once around s , and push v_0 horizontally along the path of x , then when coming back, v_0 becomes $\alpha_s \cdot v_0$. In other words, in term of a local coordinate z centered at s , and of a multivalued section $e(z)$ of L defined in a neighborhood of s , one has

$$e(\exp(2\pi i u) \cdot z)_{u=1} = \alpha_s \cdot e(\exp(2\pi i u) \cdot z)_{u=0}.$$

The complex local system L has automorphisms : $\text{Aut}(L) = \mathbb{C}^*$, with $\nu \in \mathbb{C}^*$ corresponding to multiplication by the scalar ν . Because of them, even though the isomorphism class of L is uniquely determined by the α_s , L is not determined up to unique isomorphism.

Let us fix one L . In (2.2)–(2.20), we will review some of the descriptions of the cohomology of $P - S$ with coefficients in L .

1.2 Combinatorial description

Let us fix a triangulation of $P - S$. One can then identify $H^*(P - S, L)$ with the cohomology of the complex of L -valued cochains : cochains c for which the value of c on an oriented simplex σ is a horizontal section of L on σ (thus $c(\sigma) \in H^0(\sigma, L)$). To make sense of the formula $(dc)(\tau) = c(\partial\tau)$, one uses that a horizontal section of L on a face of τ extends uniquely as a horizontal section on τ . This complex is the dual of the complex of chains of with coefficients in the dual local system $L^\vee$: chains C which are finite sums $\sum c_\sigma \cdot \sigma$ with $c_\sigma \in H^0(\sigma, L^\vee)$ and σ an oriented simplex of . The sign rule $\langle dc, C \rangle = -\langle c, \partial C \rangle$ is however unusual for duality of complexes. We write $H_*(P - S, L^\vee)$ for the corresponding homology.

The cohomology with compact support $H_c^*(P - S, L)$ is the cohomology of the complex of compactly supported L -valued cochains of . The dual complex is the complex of locally finite chains with coefficient in $L^\vee$: chains C which are possibly infinite “sums” $\sum c_\sigma \cdot \sigma$. We write $H_*^{\text{lf}}(P - S, L^\vee)$ for the corresponding homology.

Let X be obtained from P by deleting a small open disc around each $s \in S$. “Small” means “small enough”. What matters here is that the closure of the discs be disjoint. A homotopy argument shows that ($\dot{X}$ denoting $X - \partial X$)

$$H^*(P - S, L) \xrightarrow{\sim} H^*(X, L) \xrightarrow{\sim} H^*(\dot{X}, L)$$

and that $H^*(X \bmod \partial X, L) \xrightarrow{\sim} H_c^*(\dot{X}, L) \xrightarrow{\sim} H_c^*(P - S, L)$.

To compute cohomology, one can hence use a triangulation of X , instead of one of $P - S$. As X is compact, the triangulation is finite, and this combinatorial description makes clear that the Euler-Poincaré characteristic

$$\chi(P - S, L) := \sum (-1)^i \dim H^i(P - S, L)$$

(resp. $\chi_c(P - S, L) := \sum (-1)^i \dim H_c^i(P - S, L)$) is independent of L . Each H^i (resp. H_c^i) is indeed expressed as the i -th cohomology group of a finite dimensional chain complex K , and $\dim K^i$ is independent of L . In the case of cohomology with compact support, the additivity of χ_c (deduced from the long exact sequence $\cdots \rightarrow H_c^i(P - S, \mathbb{C}) \rightarrow H^i(P, \mathbb{C}) \rightarrow H^i(S, \mathbb{C}) \rightarrow \cdots$) gives

$$\chi_c(P - S, L) = \chi_c(P - S, \mathbb{C}) = \chi_c(P) - \chi_c(S) = 2 - N. \quad (1.1)$$

For an algebraic variety Y , one always has $\chi_c(Y) = \chi(Y)$. In the case at hand ($Y = P - S$), this can be deduced from Poincaré duality. One gets

$$\chi(P - S, L) = 2 - N. \quad (1.2)$$

1.3 De Rham (C^∞) description

$H^*(P - S, L)$ is the cohomology of the de Rham complex of L -valued C^∞ differential forms on $P - S$, and $H_c^*(P - S, L)$ that of the subcomplex of compactly supported forms.

If the triangulation of (2.2) is smooth, integration : $\omega \mapsto c$, $c(\sigma) = \int_{\sigma} \omega$ is defined. It is a map of complexes from the de Rham complex to the cochain complex (both with or without support condition), inducing an isomorphism on cohomology.

The Poincar

'e duality pairing obtained by integration on $P - S$: $\alpha, \beta \mapsto \int_{P-S} \alpha \wedge \beta$ induces a perfect pairing

$$H^i(P - S, L) \otimes H_c^{2-i}(P - S, L^\vee) \rightarrow \mathbb{C},$$

the composite of cup product with value in $H_c^2(P - S, \mathbb{C})$, and of the trace, or integration, map : $H_c^2(P - S, \mathbb{C}) \rightarrow \mathbb{C}$.

If at least one of the α_s is not 1, there can be no global horizontal section : $H^0(P - S, L) = 0$, and a fortiori none with compact support : $H_c^0(P - S, L) = 0$ (for this, $i = 0$ suffices). The same applies to the dual $L^\vee$ of L , with monodromy α^{-1} , and Poincar 'e duality gives vanishing for H^2 and H_c^2 . The Euler-Poincar 'e characteristic being known ((??) and (??)), we get

Proposition 1.1. *If $\alpha_s \neq 1$ for at least one s , then $H^i(P - S, L)$ and $H_c^i(P - S, L)$ vanish for $i \neq 1$, and*

$$\dim H^1(P - S, L) = \dim H_c^1(P - S, L) = N - 2.$$

1.4 Currents

Instead of using the C^∞ de Rham complex, consisting of forms written in local coordinates as $\sum_i f_I dx_{i_1} \wedge \cdots \wedge dx_{i_k} \cdot e$, with e an horizontal section of L and f a C^∞ function, one can as well use the complex of currents, where f is allowed to be a generalized function (distribution). This complex can be used to express Poincar 'e duality as a cap-product isomorphism between homology and cohomology : for C a $L^\vee$ -valued chain, there is a unique current $\delta(C)$ such that $\int_C \omega = \int \delta(C) \wedge \omega$; one has $d\delta(C) = (-1)^k \delta(\partial C)$, and the map $C \mapsto \delta(C)$ provides isomorphisms

$$H_k(P - S, L^\vee) \xrightarrow{\sim} H_c^{2-k}(P - S, L) \quad \text{and} \quad H_k^{\text{lf}}(P - S, L^\vee) \xrightarrow{\sim} H^{2-k}(P - S, L).$$

It is often more convenient to use currents than the chains of a fixed triangulation.

If β is a rectifiable proper map from an open, semi-open or closed interval I to $P - S$, and for $e \in H^0(I, \beta^* L^\vee)$, we let $(e \cdot \beta)$ be the $L^\vee$ -valued current for which

$$\int (e \cdot \beta) \wedge \omega = \int_I \langle e, \beta^* \omega \rangle.$$

If $\beta : [0, 1] \rightarrow P$ maps 0 and 1 to S , and $]0, 1[$ to $P - S$, then for $e \in H^0(]0, 1[, \beta^* L^\vee)$, $e \cdot \beta$ is a cycle and as such defines an homology class in $H_1^{\text{lf}}(P - S, L^\vee)$. We will use such cycles to construct convenient bases of $H_1^{\text{lf}}(P - S, L^\vee)$.

1.5

Assume the following is given :

- a) a partition of S into two subsets S_1 and S_2 ;
- b) trees T_1 and T_2 (a tree is a contractible CW complex of dimension ≤ 1), and an embedding $\beta : T_1 \amalg T_2 \hookrightarrow P$, mapping the set of vertices of T_1 (resp. T_2) onto S_1 (resp. S_2);

c) for each (open) edge a of T_1 or T_2 , an orientation of a , and $t(a) \in H^0(a, \beta^* L^\vee)$.
For each edge a , $t(a) \cdot \beta|_a$ is then a locally finite cycle on $P - S$, with coefficient in L .

Proposition 1.2. *If $\prod_{s \in S_i} \alpha_s \neq 1$, the elements $t(a) \cdot \beta|_a$, for a an edge of T_1 or T_2 , form a basis of $H_1^{\text{lf}}(P - S, L^\vee)$.*

Démonstration. Let T'_i be the disjoint union of the (open) edges of T_i . We have $(P - S) - \beta(T'_1 \amalg T'_2) = P - \beta(T_1 \amalg T_2)$, hence a long exact sequence

$$\cdots \rightarrow H_k^{\text{lf}}(T'_1 \amalg T'_2, \beta^* L^\vee) \rightarrow H_k^{\text{lf}}(P - S, L) \rightarrow H_k(P - \beta(T_1 \amalg T_2), L^\vee) \rightarrow \cdots$$

The group $H_1^{\text{lf}}(T'_1 \amalg T'_2, \beta^* L^\vee)$ has as basis the $t(a) \cdot a$. It remains to be proved that $H_1^{\text{lf}}(P - \beta(T_1 \amalg T_2), L) = 0$.

The space $P - \beta(T_1) - \beta(T_2)$ is the complement of two disjoint trees in S^2 . It is essentially an annulus. The monodromy of “turning around T_1 ” is the product of the α_s ($s \in S_1$). In $P - S$, a loop around T_1 can indeed be homotoped to a product of loops around each $s \in S_1$.

The group $H_1^{\text{lf}}(P - \beta(T_1 \amalg T_2), L^\vee)$ is dual to $H_c^1(P - \beta(T_1 \amalg T_2), L)$, which is the Poincaré dual of $H^{2-1}(P - \beta(T_1 \amalg T_2), L^\vee)$. One can homotope $P - \beta(T_1 \amalg T_2)$ to S^1 , and the group becomes $H^{2-1}(S^1, L^\vee)$, for $L^\vee$ a non trivial local system on S^1 . It remains to use the

Lemme 1.3. *If L is a non trivial rank one local system on S^1 , then $H^*(S^1, L) = 0$.*

The H^0 is 0 by non triviality, and $\chi(S^1, L) = \chi(S^1) = 0$. □

Remarque 1.4. As $T_1 \amalg T_2$ has $N-2$ edges, (2.5.2) reproves that $\dim H_c^1(P - S, L) = N-2$ when the α_s are not all 1.

1.6 Sheaf cohomology

We will always identify a local system L with its sheaf of locally constant sections. Let j denote the inclusion of $P - S$ in P . Then, $H_c^*(P - S, L)$ is the cohomology of P with coefficients in $j_! L$, the extension of L by 0. It is the hypercohomology on P of any complex of sheaves K with ${}^0(K) = j_! L$ and ${}^i(K) = 0$ for $i \neq 0$. On the other hand, if L^* is any resolution of L whose components are acyclic for j_* ($R^q j_* L^k = 0$ for $q > 0$), then $H^*(P - S, L)$ is the hypercohomology on P of $j_* L^*$. To prove it, one reduces to the case where L^* (and hence $j_* L^*$) is soft, one has $\Gamma(P - S, L^*) = \Gamma(P, j_* L^*)$, and

$$H^*(P - S, L) := H^* \Gamma(P - S, L^*) = H^* \Gamma(P, j_* L^*) =: H^*(P, j_* L^*).$$

Proposition 1.5. *If all α_s are different from 1, then*

$$H_c^*(P - S, L) \xrightarrow{\sim} H^*(P - S, L).$$

Démonstration. Let L^* be a soft resolution of L , for instance the L -valued C^∞ de Rham complex of sheaves. It suffices to check that $j_! L^*$ is a resolution of $j_! L$, and this results from the local fact that for D any small disc around $s \in S$, one has $H^* \Gamma(D - s, L^*) = H^*(D - s, L) = 0$. To check this vanishing, one can replace $D - s$ by a homotopic S^1 and apply (2.5.2). □

From the point of view of homology, this implies that a locally finite 1-cycle with coefficient in $L^\vee$ is homologous to a compactly supported one. Here is a drawing of how it can be done :

(figure : a locally finite cycle being pushed to a compactly supported one)

This is reminiscent of the arguments used in Whittaker and Watson (12.22) (Hankel's expression of $\Gamma(z)$ as a contour integral) or (12.43) [?].

1.7 The de Rham (holomorphic) description

The holomorphic L -valued de Rham complex $\Omega^*(L) : \mathcal{O}(L) \rightarrow \Omega^1(L)$ is a resolution of L on $P - S$. One can hence interpret $H^*(P - S, L)$ as the hypercohomology on $P - S$ of $\Omega^*(L)$. As $S \neq \emptyset$, $P - S$ is Stein, hence $H^q(P - S, \Omega^p(L)) = 0$ for $q > 0$, and this gives

$$H^*(P - S, L) = H^*\Gamma(P - S, \Omega^*(L)). \quad (1.3)$$

Similarly, $R^q j_* \Omega^p(L) = 0$ for $q > 0$ hence

$$H^*(P - S, L) = H^*(P, j_* \Omega^*(L))$$

(hypercohomology of P , with coefficients in the complex $j_* \Omega^*(L)$; see (2.6)).

After the preliminaries (2.8), (2.9), we will show how to replace $j_* \Omega^*(L)$ by smaller complexes of sheaves.

1.8

We will often describe a section of $\mathcal{O}(L)$ on a connected open set U as the product of a multivalued function with a multivalued section of L . This has the following meaning : U should be provided with a base point 0, and a multivalued section of a sheaf ($\mathcal{O}$, or L , ...) is a section of the pull back of on the universal covering $(\tilde{U}, 0)$ of $(U, 0)$. Products are taken on $\tilde{U}$. A multivalued section, which is the pull-back of a section on U , is then identified with that section. The role of the base point can be played by a contractible subset of U . A section of L at 0 extends uniquely to a multivalued section and will be denoted by the same symbol. A multivalued section of $\mathcal{O}$ is determined by its germ at 0, and will be denoted by the same symbol. As multivalued sections of $\mathcal{O}$, we will mainly use products $\prod g^{\mu_i}$. When g is not real and negative, the principal determination of g^μ is $\exp(2\pi i \mu \log g)$, with $|\operatorname{Im} \log g| < \pi$. As a rule, g^μ will denote the multivalued function whose germ at the base contractible set is the principal determination when g^μ is not real and negative on it. When the determination used is irrelevant, we will not take the trouble of specifying which one we mean.

1.9

Fix $s \in S$ and let z be a local coordinate, which we take to be an isomorphism, carrying s to 0, of a small neighborhood D of s with a disc in $\mathbb{C}$ centered at 0. If $\mu \in \mathbb{C}$ is such that $\alpha_s = \exp(2\pi i \mu)$, the monodromy of $z^{-\mu}$ around s is the inverse of that of a horizontal section of L . Any section u of $\mathcal{O}(L)$ (resp. $\Omega^1(L)$) on D^* can hence be written $u = z^{-\mu} \cdot e \cdot f$ (resp. $u = z^{-\mu} \cdot e \cdot f dz$) with e a non zero multivalued section of L and f a

holomorphic function on D^* . We define u to be meromorphic at s iff f is, and we define its valuation by

$$v_s(u) = v_0(f).$$

These definitions are independent of the choices of local coordinate and of μ .

1.10 The de Rham (meromorphic at S) description

Let us write $j_*^m \Omega^*(L)$ for the subcomplex of $j_* \Omega^*(L)$ of meromorphic forms. A local computation around each $s \in S$ shows that the inclusion of $j_*^m \Omega^*(L)$ in $j_* \Omega^*(L)$ induces an isomorphism on the cohomology sheaves. This implies that

$$H^*(P, j_*^m \Omega^*(L)) \xrightarrow{\sim} H^*(P, j_* \Omega^*(L)) = H^*(P - S, L). \quad (1.4)$$

It follows from (2.11) below that $j_*^m \Omega^p(L)$ is an inductive limit of line bundles whose degrees tends to ∞ . From this one concludes first that $H^q(P, j_*^m \Omega^p(L)) = 0$ for $q > 0$, and then that $H^*(P, j_*^m \Omega^*(L))$ is simply $H^* \Gamma(P, j_*^m \Omega^*(L))$, the cohomology of the complex of L -valued forms holomorphic on $P - S$ and meromorphic at each $s \in S$:

$$H^*(P - S, L) = H^* \Gamma(P, j_*^m \Omega^*(L)). \quad (1.5)$$

1.11

Let $(\mu_s)_{s \in S}$ be a family of complex numbers, such that $\exp(2\pi i \mu_s) = \alpha_s$. We define the line bundle $\mathcal{O}(\sum \mu_s \cdot s)(L)$ as the subsheaf of $j_*^m \mathcal{O}(L)$ whose local holomorphic sections are the local sections u of $j_*^m \mathcal{O}(L)$ such that for $s \in S$, the integer $v_s(u) + \mu_s$ is > 0 (in short : $v_s(u) \geq -\mu_s$). With the local coordinate notation (2.9), $z^{-\mu_s} \cdot e$ determines an invertible section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ near $s \in S$. The degree of a meromorphic section u of a line bundle at a point x is the order of the zero (or minus the order of the pole) of u at x ; for z a local coordinate centered at x , it is the supremum of the integers n such that $z^{-n}u$ is a holomorphic section of at x . The degree of is the sum of the degrees at all points of any non zero meromorphic section of . If u is a meromorphic section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ and $x \notin S$, one has $\deg_x(u) = v_x(u)$. If one defines $\mu_x = 0$ for $x \in P - S$, the same holds at any $x \in P$.

Proposition 1.6. *The line bundle $\mathcal{O}(\sum \mu_s \cdot s)(L)$ is of degree $\sum \mu_s$.*

Démonstration. We may assume $P = \mathbb{P}^1$ and $\infty \notin S$. Let e be a multivalued horizontal section of L . The product

$$u = \prod_{s \in S} (z - s)^{-\mu_s} \cdot e$$

is an invertible section of $\mathcal{O}(\sum \mu_s \cdot s)(L)$ on the affine line. At ∞ , u is meromorphic, of valuation $\sum \mu_s$. The degree of $\mathcal{O}(\sum \mu_s \cdot s)(L)$, equal to the sum at all $x \in P$ of the degree of any non zero meromorphic section, is hence $\sum \mu_s$. $\square$

We define $\Omega^1(\sum \mu_s \cdot s)(L) := \mathcal{O}(\sum \mu_s \cdot s)(L) \otimes \Omega^1$. It results from (2.11.1) that

$$\deg \Omega^1(\sum \mu_s \cdot s)(L) = -2 + \sum \mu_s. \quad (1.6)$$

In particular, if $\sum \mu_s = 2$, the line bundle $\Omega^1(\sum \mu_s \cdot s)(L)$ is of degree 0, hence isomorphic to $\mathcal{O}$, the trivial line bundle.

Corollaire 1.7. *If $\sum \mu_s = 2$, there is up to a constant factor one and only one non zero form $\omega \in \Gamma(P, j_*^m \Omega^1(L))$ whose valuation at $s \in S$ is $\geq -\mu_s$. One has $v_s(\omega) = -\mu_s$, and ω is invertible on $P - S$.*

For $P = \mathbb{P}^1$, $\infty \notin S$, and e a multivalued horizontal section of L , one has (up to a factor) $\omega = \prod_{s \in S} (z - s)^{-\mu_s} \cdot e \, dz$. If $\infty \in S$, one has $\omega = \prod_{s \in S - \{\infty\}} (z - s)^{-\mu_s} \cdot e \, dz$.

Proposition 1.8. *Assume that $\sum \mu_s = 2$, $N \geq 3$, and that none of the α_s is 1 (i.e. that none of the μ_s is an integer). Then the cohomology class of the form ω of (2.12) is not zero.*

Démonstration. One has $\sum v_s(\omega) = \sum -\mu_s = -2 > 1 - N$. The proposition hence results from the

Proposition 1.9. *Assume that none of the α_s is 1. If a non zero form ω is such that $\sum v_s(\omega) \geq 1 - N$, its cohomology class is not zero.*

Démonstration. By (2.10.2), it suffices to verify that the equation $\omega = du$ has no solution $u \in \Gamma(P, j_*^m \mathcal{O}(L))$.

For any local section u of $j_*^m \mathcal{O}(L)$ near s , one has $v_s(du) > v_s(u) - 1$, with equality if $v_s(u)$ is not 0, or if $v_s(du)$ is not a positive (meaning > 0) integer. The integer case being excluded, a solution u would be a section of $\mathcal{O}(-\sum (v_s(\omega) + 1)s)(L)$, a line bundle of degree $-\sum (v_s(\omega) + 1) \leq -1$, a contradiction. $\square$

$\square$

1.12

Here is how (2.13), (2.14) has to be modified for integral μ 's. As the result will not be needed, except for some historical comments in §15, we will be sketchy.

(2.15.1) When some α_s are 1, (2.6.1) is not true, one has to distinguish between H_c^1 and H^1 , and even to introduce some intermediate groups : for $S' \subset S$, the family $\Phi(S')$ of subsets of $P - S$ closed in $P - S'$ is a family of supports, and $H_{\Phi(S')}^*(P - S, L)$ is defined. For j' the inclusion of $P - S$ into $P - S'$, it is $H^*(P - S', j'_! L)$. For $T \subset S$, define $T(1) := \{s \in T \mid \alpha_s = 1\}$. The proof of (2.6.1) shows that

$$H_{\Phi(S')}^*(P - S, L) \xrightarrow{\sim} H_{|S'|}^*(P - S, L).$$

These groups are also the hypercohomology groups, on P , of the subcomplex of $j_*^m \Omega^*(L)$ consisting of the u in $j_*^m \mathcal{O}(L)$ and of the ω in $j_*^m \Omega^1(L)$ such that $v_s(\omega) > 0$ and $v_s(u) > 0$ for $s \in (S - S')(1)$. If the μ_s are such that $\alpha_s = \exp(2\pi i \mu_s)$, and that $S'(1)$ is the set of s for which μ_s is integral > 0 , the groups $H_{\Phi(S')}^*(P - S, L)$ are also the hypercohomology groups of the sub-complex $\mathcal{O}(\sum (\mu_s - 1)s)(L) \rightarrow \Omega^1(\sum \mu_s \cdot s)(L)$ of $j_*^m \Omega^*(L)$.

(2.15.2) If $\omega \in \Gamma(P, j_*^m \Omega^1(L))$, and if all s with $\alpha_s = 1$ and $v_s(\omega) < 0$ are in S' , then ω defines a cohomology class in $H_{\Phi(S')}^*(P - S, L)$. If in addition S' contains none of the s for which simultaneously $\alpha_s = 1$ and $v_s(\omega) \geq 0$, and if further $\sum v_s(\omega) > 1 - N$ (which amounts to ω having at most $N - 3$ zeros, counted with multiplicities, on $P - S$), the proof of (2.14) still shows that the class of ω in $H_{\Phi(S')}^*(P - S, L)$ is not zero. This applies to the form ω of (2.12), with $S' = S$, provided $N > 3$ and $\mu_s > 0$ whenever $\mu_s \in \mathbb{Z}$.

Remarque 1.10. Let us assume that none of the α_s is 1. If ω is an holomorphic section of $\Omega^1(L)$ on $P - S$, by (2.6.1), the class $[\omega]$ of ω in $H^1(P - S, L)$ is the image of a unique class, again denoted $[\omega]$, in $H_c^1(P - S, L)$. For any locally finite cycle or current C with coefficient in $L^\vee$, $\langle [C], [\omega] \rangle$ is defined : $H_1^{\text{lf}}(P - S, L^\vee)$ is indeed paired with $H_c^1(P - S, L)$. We want to prove that

$$\langle [C], [\omega] \rangle = \int_C \omega.$$

This is not completely obvious, especially since the integral may be divergent. The problem is rather to understand which value the cohomological formalism attaches to the integral.

1st answer : One replaces C by a finite homologous cycle C' , as pictured in (2.6), and $\langle [C], [\omega] \rangle = \int_{C'} \omega$.

2nd answer : If D_s is a small open disk around $s \in S$, one has $H^1(D_s - s, L) = 0$ (cf. (2.6)). On $D_s^* := D_s - s$, ω is hence of the form du_s , for a unique section u_s of $\mathcal{O}(L)$ on D_s^* . Let φ_s be a compactly supported C^∞ function on D_s , equal to 1 in a neighborhood of s . The C^∞ 1-form $\omega - \sum d(\varphi_s u_s)$ is cohomologous to ω and is compactly supported. One has hence

$$\langle [C], [\omega] \rangle = \int_C \omega - \sum \int_C d(\varphi_s u_s).$$

It is sometimes more convenient to take for φ_s the (discontinuous) characteristic function of D_s . The current $\omega - \sum d(\varphi_s u_s)$ is again compactly supported and cohomologous to ω .

These formulae can also be used to define the integral of ω on a locally finite chain or current, which is a cycle only near each $s \in S$. Given for instance a path $\beta : [0, 1] \rightarrow P$, with $\beta([0, 1])$ in $P - S$ and $\beta(1) = s \in S$, and given $e \in H^0([0, 1[, \beta^* L^\vee)$, the formula for the “finite part” $\int_{[0, 1]} e \cdot \beta^* \omega - d(\varphi_s u_s)$ can be rewritten

$$\text{P.V.} \int_{[0, 1]} \langle e, \beta^* \omega \rangle - \langle e(1 - \varepsilon), \beta^* u_s(1 - \varepsilon) \rangle \quad (\varepsilon \rightarrow 0).$$

For a meromorphic ω , this agrees with Hadamard’s “finite part” of a divergent integral.

Exemple 1.11. We take $P = \mathbb{P}^1$, $S = \{0, 1, \infty\}$ and we assume that none of the μ_s is an integer, and that $\sum \mu_s = 2$. For $\lambda_s = 1 - \mu_s$, this means $\sum \lambda_s = 1$. We normalize L by fixing a section e of it on $]0, 1[$, and we let $e^\vee$ be the dual base of $L^\vee$ on $]0, 1[$. Take

$$\omega = z^{\lambda_0 - 1} (1 - z)^{\lambda_1 - 1} e dz;$$

the “principal” determination of $z^{\lambda_0 - 1}$ and of $(1 - z)^{\lambda_1 - 1}$ on $]0, 1[$ is used (cf. (2.8)). By (2.5.1), the homology $H_1^{\text{lf}}(P - S, L^\vee)$ is one-dimensional, generated by the current $e^\vee \cdot]0, 1[$. The content of (2.13) here is that

$$\text{P.V.} \int_0^1 z^{\lambda_0 - 1} (1 - z)^{\lambda_1 - 1} dz \neq 0. \quad (1.7)$$

The generalization (2.15) of (2.13) can be used to extend this to the case when λ_0 and λ_1 , if integers, are > 0 , and $\lambda_\infty = 1 - \lambda_0 - \lambda_1$, if an integer, is < 0 .

Of course, (2.17.1) is easy to deduce from the formula for the B-function in terms of Γ -functions.

1.13

We now assume that each α_s is of absolute value 1, and not equal to 1. There is then a horizontal positive Hermitian structure $(,)$ on L . Let us choose one. One can view $(,)$ as a perfect pairing between L and the complex conjugate local system $\bar{L}$. As such, it induces a perfect pairing

$$\psi_0 : H_c^1(P - S, L) \otimes \overline{H_c^1(P - S, L)} \rightarrow H_c^2(P - S, \mathbb{C}) = \mathbb{C}$$

((2.3), (2.6.1)). The vector space $\overline{H_c^1(P - S, L)}$ is the complex conjugate of $H_c^1(P - S, L)$, and $\psi_0(u, \bar{v})$ defines a non degenerate skew-Hermitian form ψ on $H_c^1(P - S, L)$. Anticommutativity of the cup product shows that

$$\psi(v, u) = -\overline{\psi(u, v)}, \quad \text{i.e.} \quad \psi(u, v) = -\overline{\psi(v, u)}. \quad (1.8)$$

Writing $(u, v) := \frac{1}{2\pi i} \psi(u, v) = \frac{1}{2\pi i} \psi_0(u, \bar{v})$, we have a Hermitian form

$$(u, v) = \overline{(v, u)}. \quad (1.9)$$

A section ω of $j_*^m \Omega^1(L)$ is said to be of the *first kind* if $v_s(\omega) > -1$ for each $s \in S$. Those are the forms for which the integral $\int_{P-S} \omega \wedge \bar{\omega}$ is convergent. The integrand is defined as follows : if, locally, $\omega_i = e_i \cdot f_i$ ($i = 1, 2$, f_i a 1-form, e_i a section of L), then $\omega_1 \wedge \bar{\omega}_2$ is the $(1, 1)$ -form $(e_1, e_2) f_1 \wedge \bar{f}_2$.

We define $H^{1,0}(P - S, L)$ as the vector space of forms of the first kind in $\Gamma(P, j_*^m \Omega^1(L))$, and $H^{0,1}(P - S, L)$ as the complex conjugate of $H^{1,0}(P - S, L)$. It can be viewed as the space of antiholomorphic L -valued forms, whose complex conjugate is of the first kind. Such forms also define cohomology classes.

Proposition 1.12. *If ω_1 and ω_2 are each either in $H^{1,0}(P - S, L)$ or in $\overline{H^{0,1}(P - S, L)}$, one has, for the corresponding cohomology classes $[\omega_1]$ and $[\omega_2]$,*

$$\psi([\omega_1], [\omega_2]) = \int_{P-S} \omega_1 \wedge \omega_2.$$

Démonstration. Let us assume, for definiteness, that ω_1 is in $H^{1,0}(P - S, L)$. Choose D_s , $(u_s)_{s \in S}$, as in (2.16), and let φ_s be the characteristic function of D_s . It follows from (2.16) that the class of ω_1 in $H_c^1(P - S, L)$ is also the class of the compactly supported current $\omega_1 - \sum d(\varphi_s u_s)$. Hence (2.3)

$$\psi([\omega_1], [\omega_2]) = \psi_0([\omega_1], [\bar{\omega}_2]) = \int_{P-S} (\omega_1 - \sum d(\varphi_s u_s)) \wedge \omega_2.$$

When we let the D_s shrink, the first term converges to $\int_{P-S} \omega_1 \wedge \omega_2$. We want to show that all others tend to 0. When the radius r of D_s tends to zero, the size of u_s on ∂D_s is $O(r^{v_s(u_s)+\varepsilon})$. One has $v_s(u_s) = v_s(\omega_1) + 1$. The forms ω_1 and ω_2 being of the first kind, the size is $O(r^\beta)$, with $\beta > -1$. On the other hand, the length of ∂D_s is $O(r)$; the boundary term is hence $O(r^{\beta+1})$ and tends to zero. $\square$

Proposition 1.13. *Under the assumption (2.18) that $|\alpha_s| = 1$, $\alpha_s \neq 1$ for $s \in S$, the natural map*

$$H^{1,0}(P - S, L) \oplus \overline{H^{0,1}(P - S, L)} \rightarrow H^1(P - S, L) = H_c^1(P - S, L)$$

is an isomorphism. The Hermitian form $(,)$ is positive definite on $H^{1,0}$, negative definite on $H^{0,1}$ and the decomposition is orthogonal.

Démonstration. If ω is in $H^{1,0}(P - S, L)$ (resp. $\overline{H^{0,1}(P - S, L)}$), the integrand $\frac{1}{2\pi i} \omega \wedge \bar{\omega}$ is > 0 (resp. < 0), and vanishes only for $\omega = 0$. If ω_1 and ω_2 are one in $H^{1,0}$ and the other in $H^{0,1}$, the integrand $\frac{1}{2\pi i} \omega_1 \wedge \bar{\omega}_2$ vanishes. This implies that $H^{1,0} \oplus H^{0,1}$ injects into $H^1(P - S, L)$. It remains only to prove the surjectivity of $H^{1,0} \oplus H^{0,1} \rightarrow H^1$.

We will check surjectivity by counting dimensions. Let μ_s be the number between 0 and 1 such that $\alpha_s = \exp(2\pi i \mu_s)$. From the definitions it follows that $H^{1,0}(P - S, L) = \Gamma(P, \Omega^1(\sum \mu_s \cdot s)(L))$. By (2.8), $\deg \Omega^1(\sum \mu_s \cdot s)(L) = -2 + \sum \mu_s$. It is an integer > -2 and hence

$$\dim H^{1,0}(P - S, L) = \deg \Omega^1(\sum \mu_s \cdot s)(L) + 1 = -1 + \sum \mu_s. \quad (1.10)$$

Let us apply this to $\bar{L}$. One has to replace α_s by $\bar{\alpha}_s = \alpha_s^{-1}$ and μ_s by $1 - \mu_s$. One gets

$$\dim H^{0,1}(P - S, L) = -1 + \sum (1 - \mu_s). \quad (1.11)$$

The sum of the dimensions is $-2 + N = N - 2$, as required (2.3.1). $\square$

Corollaire 1.14. *Notation being as above ($\alpha_s = \exp(2\pi i \mu_s)$ with $0 < \mu_s < 1$), the signature of the Hermitian form $(\cdot, \cdot)$ on $H^1(P - S, L)$ is $(\sum \mu_s - 1, \sum (1 - \mu_s) - 1)$.*

Remarque 1.15. (2.20) is a special case of Hodge theory for the cohomology of a curve with values in a polarized variation of Hodge structures : a local system, provided with an horizontal positive Hermitian structure, can be viewed as a polarized variation of type $(0, 0)$. The artificial counting argument above enabled us to shortcut the general theory, for which the reader may consult S. Zucker [?].

1.14

In our applications, the α_s will be roots of unity. When this is the case, (2.20) can also be deduced from the Hodge theory of suitable coverings of P . Let X be an irreducible abelian covering of P , with covering group G , ramified only at S . If, for definiteness, we take $P = \mathbb{P}^1$ and $\infty \notin S$, this means that the function field $\mathbb{C}(X)$ of X is a subextension of the extension $\mathbb{C}(\mathbb{P}^1)((z - s)_{s \in S - \{\infty\}})^{1/d}$ of $\mathbb{C}(z) = \mathbb{C}(\mathbb{P}^1)$, for suitable d .

Let π denote the projection of X onto P . The Galois group $G = \text{Aut}(X/P)$ acts (by transport of structure) on $\pi_* \mathbb{C}$ (by abuse of notation, we also denote by $\mathbb{C}$ the constant sheaf with fiber $\mathbb{C}$), and at a point $z \in P$ the representation of G on $(\pi_* \mathbb{C})_z = H^0(\pi^{-1}(z), \mathbb{C})$ is a regular representation of G . For each character χ of G , let L_χ be the subsheaf of $\pi_* \mathbb{C}$ on which G acts by χ . One has

$$\pi_* \mathbb{C} = \bigoplus_{\chi} L_{\chi}$$

and, outside of S , L_χ is a rank one local system.

Let $g_s \in G$ be the natural generator of the inertial (= decomposition) group at s : if $x(t)$ ($0 < t < 1$) is a path in X , such that $\pi x(t)$ in P stays near s in $P - S$, and turn once around s , then $x(1) = g_s x(0)$. Let d_s be the order of g_s . If t is a local coordinate centered at s , near s , X is a sum of copies of the Riemann surface of t^{1/d_s} . If $\chi(g_s) = 1$, L_χ is a local system at s . If $\chi(g_s) \neq 1$, the fibre of L_χ at s is zero, and the monodromy of L_χ around s is the d_s -th root of unity $\chi(g_s)$.

If we take X to be the largest covering such that G is killed by d , G is the abelian group generated by the g_s , with the only relations $g_s^{d_s} = 1$ and $\prod g_s = 1$: $G = (\mathbb{Z}/d)^S / (\mathbb{Z}/d)$.

If $p \in \mathbb{P}^1$ and $\infty \in S$, X corresponds to the extension $\mathbb{C}(\mathbb{P}^1)((z-s)_{s \in S-\{\infty\}})^{1/d}$ of $\mathbb{C}(\mathbb{P}^1)$. This shows that any system of roots of unity (α_s) , with $\prod \alpha_s = 1$, is of the form $(\chi(g_s))_{s \in S}$ for suitable X and χ .

Let $H^*(X, \mathbb{C})_\chi$ be the subspace of $H^*(X, \mathbb{C})$ on which G acts (by transport of structure) as χ . The isomorphism $H^*(X; \mathbb{C}) = H^*(P, \pi_* \mathbb{C})$ is compatible with the action of G , hence induces isomorphisms

$$H^*(X, \mathbb{C})_\chi = H^*(P, L_\chi). \quad (1.12)$$

On $P-S$, the sheaf $\mathcal{O}(L_\chi)$ is the direct factor $(\pi_* \mathcal{O})_\chi$ of $\pi_* \mathcal{O}$ consisting of the holomorphic functions on X such that $f(g^{-1}x) = \chi(g)f(x)$. Similarly, $\Omega^1(L_\chi) = (\pi_* \Omega^1)_\chi$. Let j' be the inclusion of $X - \pi^{-1}(S)$ in X . On P , one has again

$$j_*^m(L_\chi) = (\pi_* j'^m \mathcal{O})_\chi \quad \text{and} \quad (1.13)$$

$$j_*^m \Omega^1(L_\chi) = (\pi_* j'^m \Omega^1)_\chi. \quad (1.14)$$

Let us assume $\chi(g_s) \neq 1$ for $s \in S$, so that $H^*(P, L_\chi) = H_c^*(P-S, L_\chi)$. Differential forms of the first kind on $P-S$ correspond then, by (2.23.3), to the differential forms of the first kind on X , on which G acts by χ :

$$H^{1,0}(P-S, L_\chi) = \Gamma(X, \Omega_X^1)_\chi = H^{1,0}(X)_\chi$$

and (2.20) is the trace on $H^1(P-S, L_\chi) = H^1(X, \mathbb{C})_\chi$ of the Hodge decomposition of $H^1(X, \mathbb{C})$.

2 Reformulation of Picard's Theorem

2.1

We now let the punctures s move, while the monodromy remains the same. The starting data will be :

- P : a complex projective line ;
- N : an integer, ≥ 3 ;
- S : a set with N elements, for instance $[1, N] \subset \mathbb{N}$;
- $\alpha = (\alpha_s)_{s \in S}$: a family of complex numbers indexed by S , satisfying $\prod \alpha_s = 1$, and such that none of the α_s is 1.

We will be mainly interested in the case $|\alpha_s| = 1$.

For any space X mapping to $\mathbb{P}^S$, we will write P_X for the pull back on X of the universal punctured line

$$P_0 = \{(p, m) \in \mathbb{P} \times \mathbb{P}^S \mid p \notin m(S)\},$$

and π for the projection $P_X \rightarrow X$. Suppose a group H acts freely on X , with quotient Y , and that a lifting of the action to P_X is given. We will then write P_Y for P_X/H and π for the projection $P_Y \rightarrow Y$. The H -equivariant fiber space P_X over X is then the pull back of P_Y over Y .

Let $M \subset \mathbb{P}^S$ be the space of injective maps $m : S \rightarrow \mathbb{P}$. Then,

$$P_M = \{(p, m) \in \mathbb{P} \times M \mid p \notin m(S)\}$$

is a topologically locally trivial fiber space above M . On each fiber $P_m = P - m(S)$, there is a rank one local system with monodromy α , i.e. with monodromy α_s around $m(s)$, for $s \in S$. However, this local system not being unique up to unique isomorphism, one cannot conclude that on P_M there is a unique rank one local system L such that

$$\text{For } m \text{ in } M, \text{ the local system } L_m \text{ induced by } L \text{ on } P_m \text{ has monodromy } \alpha. \quad (2.1)$$

If L satisfies (3.1.1), all local systems satisfying (3.1.1) are of the form $L \otimes \pi^*T$, for T a rank one local system M . We will see in (3.12), (3.13) that there is an L satisfying (3.1.1). If N were 2, the analogous existence assertion would be false for $\alpha \neq \pm 1$.

2.2

Suppose L satisfies (3.1.1). The projection π being topologically locally trivial, the $H^1(P_m, L_m)$ ($= H_c^1(P_m, L_m)$ by (2.6.1)) organize themselves into a local system $R^1\pi_!L$ ($= R^1\pi_*L$) on M .

We will be interested mainly in the corresponding flat projective space bundle $B(\alpha)_M := \mathbb{P}R^1\pi_!L$, the fiber space with fiber at $m \in M$ the projective space $\mathbb{P}H^1(P_m, L_m) := (H^1(P_m, L_m) - \{0\})/\mathbb{C}^*$, and with flat structure, that deduced from $R^1\pi_!L$. For any vector space V and non-zero element $v \in V$, we will denote by $\mathbb{P}V$ the projective space of one-dimensional subspaces of V and by $\mathbb{P}_v$ the image of v in $\mathbb{P}V$.

If $L' = L \otimes \pi^*T$ is another local system satisfying (3.1.1),

$$R^1\pi_!L' = (R^1\pi_!L) \otimes T,$$

and $\mathbb{P}R^1\pi_!L'$ is canonically isomorphic to $\mathbb{P}R^1\pi_!L$, i.e. the flat projective bundle $B(\alpha)_M$ depends only on α . Another explanation of the same fact : the automorphisms of a local system L_m on P_m with monodromy α act trivially on $\mathbb{P}H^1(P_m, L_m)$. Hence, although L_m is unique up to only a non unique isomorphism, $\mathbb{P}H^1(P_m, L_m)$ is defined up to unique isomorphism.

2.3

Locally on M , the existence of L satisfying (3.1.1) poses no problem : if U is a contractible neighborhood of $m \in M$, a local system L_m on P_m with monodromy α extends uniquely to a local system L_U on $\pi^{-1}(U)$ and this extension has, fiber by fiber, monodromy α . The flat projective space bundle $\mathbb{P}R^1\pi_!L_U$ on U is independent (up to unique isomorphism) of the choice of L_U ; hence for variable U , they glue into a flat projective space bundle on the whole of M . This enables us to define $B(\alpha)_M$ without having to assume the existence of a global L .

For $o \in M$, the flat structure of $B(\alpha)_M$ defines an action of $\pi_1(M, o)$ by projective transformations on the fibre $B(\alpha)_o$. A choice of L enables one to lift this projective representation of $\pi_1(M, o)$ on $\mathbb{P}H^1(P_o, L_o)$ to a linear representation on $H^1(P_o, L_o)$.

2.4

Fix a system of complex numbers $(\mu_s)_{s \in S}$ such that $\alpha_s = \exp(2\pi i \mu_s)$, and that $\sum \mu_s = 2$. For each $m \in M$, there is then up to a factor a unique non-zero section ω of $\Omega^1(L_m)$ on P_m , meromorphic on P , and such that $v_s(\omega) > -\mu_s$ ($s \in S$) (2.12). Its cohomology class

is not trivial (2.13). This construction hence defines a section $w_\alpha(m)$ of the projection $B(\alpha)_M \rightarrow M$.

The flat structure of the projective space bundle $B(\alpha)$ induces a structure of holomorphic projective space bundle. With respect to it, we have :

Lemme 2.1. *The section w_α is holomorphic.*

Démonstration. The question is local on M . Fix $m \in M$, and choose on the fibre P_m a system of cycles C_i with coefficient in $L^\vee$ giving rise to a basis of $H_1(P_m, L^\vee)$. They have support in some compact K and, for U a contractible neighborhood of m , small enough so that $K \times U$ is isomorphic to the pull back of $L|_K$ by a unique isomorphism which is the identity above m . This enables us, for $m' \in U$, to consider C_i as a cycle with coefficients in $L^\vee$ on $P_{m'}$; and each linear form $\langle -, C_i \rangle$ on the $H_1(P_{m'}, L_{m'}^\vee)$ ($m' \in U$) is a horizontal linear form on $R^1\pi_!L$. This provides a horizontal system of projective coordinates on $\mathbb{P}R^1\pi_!L$.

Fix a coordinate $z : \mathbb{P} \rightarrow \mathbb{P}^1$, such that, for m' in a possibly smaller neighborhood U of m , $z^{-1}(\infty) \subset P_{m'}$, and let us trivialize L along $z^{-1}(\infty)$ by a section e . We can take (cf. (2.12))

$$\omega = \prod_{s \in S} (z - z(m'(s)))^{-\mu_s} \cdot dz \cdot e,$$

and the projective coordinates of w are the

$$\int_{C_i} \prod_{s \in S} (z - z(m'(s)))^{-\mu_s} \cdot dz \cdot e.$$

This is clearly holomorphic in m' . □

Remarque 2.2. A more general method to get horizontal linear forms on $R^1\pi_!L$ is to start with a C^∞ diffeomorphism $\varphi_{m,m'} : (P_m, L) \xrightarrow{\sim} (P_{m'}, L_{m'})$, and with $L^\vee$ -valued cycle or current C on P_m , and to take $\int_{\varphi_{m,m'}(C)}$. For instance, if C is a path from $m(s)$ to $m(t)$, together with a section e of $L^\vee$ on it, one deforms it with m' so that $C(m')$ remains a path from $m'(s)$ to $m'(t)$.

2.5

The group G of automorphisms of $\mathbb{P}$ is isomorphic to $\mathrm{PGL}(2)$. Its action on $\mathbb{P}$ induces an action on $M \subset \mathbb{P}^S$, on the space P_M , and on the flat projective bundle $B(\alpha)_M$ on M . The section w_α is preserved. The action on M is free. Let $Q = M/G$. The fiber bundle P_M , the flat bundle $B(\alpha)_M$ and w_α being equivariant, they descend to a fiber bundle P_Q , a flat projective space bundle $B(\alpha)_Q$ and a section w_α of $B(\alpha)_Q \rightarrow Q$.

For $\mathbb{P} = \mathbb{P}^1$, and for a, b, c distinct in S , let $M_0 \subset M$ be the space of m such that $m(a) = 0$, $m(b) = 1$, $m(c) = \infty$. One has $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$. The quotient map hence induces an isomorphism $M_0 \xrightarrow{\sim} Q$, and via this isomorphism, $B(\alpha)_Q$ and w_α can be identified with the restriction of $B(\alpha)_M$ and w_α to M_0 .

2.6

Fix a base point $o \in Q$. The section w_α can then be viewed as a multivalued map from Q to $B(\alpha)_o$. More precisely, let $\tilde{Q}$ be the covering of Q corresponding to the kernel

of the monodromy action ρ of $\pi_1(Q, o)$ on $B(\alpha)_o$, i.e. let $\tilde{Q}$ be the quotient by (ρ) of the universal covering of (Q, o) . The pull back of $B(\alpha)_Q$ on $\tilde{Q}$ is $\tilde{Q} \times B(\alpha)_o$, and w_α becomes a $\pi_1(Q, o)$ -equivariant map $\tilde{w}$ from $\tilde{Q}$ to $B(\alpha)_o$.

Proposition 2.3. $\tilde{w} : \tilde{Q} \rightarrow B(\alpha)_o$ is
'etale.

Démonstration. We may take $\mathbb{P} = \mathbb{P}^1$ and replace Q by M_0 , as in (3.7). The problem is local. Take $m \in M_0$, let e be a trivialization of L —near m —on $\mathbb{R}^+$ near ∞ . We may take

$$\omega = \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz \cdot e$$

(cf. (2.8)). Choose cycles C_i as in (3.5, 3.6). Differentiating in m , one has, with $d_{M_0}\omega = \sum \frac{\partial \omega}{\partial m(s)} dm(s)$,

$$d_{M_0}\omega = \sum_s \mu_s \frac{dm(s)}{z - m(s)} \cdot \omega.$$

The spaces $\tilde{Q}$ and $B(\alpha)_o$ have the same dimension $N - 3$. The map $\tilde{w}$ is hence 'etale at m if and only if $d(\tilde{w})$ is injective, i.e. if and only if for no tangent vector $v \neq 0$ at m is the family $\langle \partial_v \omega, C_i \rangle$ proportional to the family of integrals $\langle \omega, C_i \rangle$, i.e. if and only if for no v the cohomology class of $\partial_v \omega$ is proportional to that of ω . This means that the cohomology class of any non-trivial linear combination

$$\sum b_s \frac{\omega}{z - m(s)} \quad (b_s \in \mathbb{C}, \sum b_s = 0)$$

should be non-trivial. If $b_s \neq 0$, $\frac{\omega}{z - m(s)}$ is of valuation exactly $v_s(\omega) - 1$ at s , and hence $\neq 0$. If $b_s = 0$, the valuation is $\geq v_s$. This shows that $\sum b_s \frac{\omega}{z - m(s)} \neq 0$ and that

$$\sum_s v_s \left(\sum_t b_t \frac{\omega}{z - m(t)} \right) = \sum_{s: b_s \neq 0} (v_s(\omega) - 1) + \sum_{s: b_s = 0} v_s(\omega) \geq \sum v_s(\omega) - (N - 2) > 1 - N,$$

so that the non vanishing of the class of $\tilde{\omega}$ results from (2.14). $\square$

2.7

We will now assume that $|\alpha_s| = 1$ and that the numbers μ_s defined by $\alpha_s = \exp(2\pi i \mu_s)$, $0 < \mu_s < 1$, satisfy $\sum \mu_s = 2$.

Locally on M , L admits a horizontal positive definite Hermitian form $(,)$, unique up to a positive real factor (2.18, 3.3). It induces on $R^1\pi_1 L$ a horizontal Hermitian form $(,) = \frac{-1}{2\pi i} \psi$ (2.18), of signature $(1, N - 3)$ (2.21). The vectors on which $(,)$ is positive define a horizontal family of complex balls $B^+(\alpha)_M \subset B(\alpha)_M$, and the section w_α is in $B^+(\alpha)_M$ (2.20) (for the meaning of the term “complex ball”, see §5).

The family $B^+(\alpha)_M$ is equivariant, hence descends to $B^+(\alpha)_Q \subset B(\alpha)_Q$. The 'etale map of (3.9) is a map

$$\tilde{w} : \tilde{Q} \rightarrow B^+(\alpha)_o \tag{2.2}$$

equivariant with respect to the action of $\pi_1(Q, o)$ on $B^+(\alpha)_o$. This action is a morphism

$$\rho : \pi_1(Q, o) \rightarrow \text{PU} \left(H^1(P_o, L_o), \frac{-1}{2\pi i} \psi \right). \tag{2.3}$$

Notation being as above, our main result is the

Théorème 2.4. *Assume that*

(INT) *The numbers μ_s defined by $\alpha_s = \exp(2\pi i \mu_s)$, $0 < \mu_s < 1$, satisfy $\sum \mu_s = 2$ and, for all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer.*

Then, the image Γ of (3.10.2) is a lattice in the projective unitary group

$$\mathrm{PU} \left(H^1(P_o, L_o), \frac{-1}{2\pi i} \psi \right) = \mathrm{PU}(1, N - 3).$$

The theorem will be proved in section 11, where it is restated as theorem (11.4).

The strategy of the proof is presented in the next paragraphs.

In §4, we define a partial compactification Q_{st} of Q . We also define a compactification Q_{sst} , with $Q \subset Q_{\mathrm{st}} \subset Q_{\mathrm{sst}}$ and $Q_{\mathrm{sst}} - Q_{\mathrm{st}}$ finite. In §8, we recall a construction of R. H. Fox to define the completion $\tilde{Q}_{\mathrm{st}}$ (resp. $\tilde{Q}_{\mathrm{sst}}$) of $\tilde{Q}$ over Q_{st} (resp. Q_{sst}). When the condition (INT) is satisfied, one can show that each point y of Q_{st} admits open neighborhoods U such that the inverse image of U in $\tilde{Q}$ is a disjoint sum of finite coverings of $U \cap Q$. The completion $\tilde{Q}_{\mathrm{st}}$ is then a normal analytic space; it is the normal ramified covering of Q_{st} extending $\tilde{Q}$.

The results of §6 allow us, in §8, to extend $\tilde{w}$ to a map from $\tilde{Q}_{\mathrm{st}}$ to $B^+(\alpha)_o$. We will again write $\tilde{w}$ for the extended map. In §9, we show that the condition INT is tantamount to requiring this extended map to be

'etale in codimension one. Since the projection $\tilde{Q}_{\mathrm{st}} \rightarrow Q_{\mathrm{st}}$ is locally (on Q_{st}) finite to one, and Q_{st} is a normal analytic space, it follows from §6 that the extended map $\tilde{w} : \tilde{Q}_{\mathrm{st}} \rightarrow B^+(\alpha)_o$ is locally finite to one. By the purity of the branch locus theorem, if $\tilde{w}$ is

'etale in codimension one, it is then

'etale everywhere. Actually, we give two proofs of this fact in §10, the second proof not requiring the theorem on purity of branch locus. Additional work, relying on the compactness of Q_{sst} , shows that $\tilde{w} : \tilde{Q}_{\mathrm{st}} \rightarrow B^+(\alpha)_o$ is a topological covering map. The ball being simply connected, it is an isomorphism. In other words, the inverse of the multivalued map w_α is a single valued map $w_\alpha^{-1} : B^+(\alpha)_o \rightarrow \tilde{Q}_{\mathrm{st}}$.

The homeomorphism $\tilde{w}$ transforms the fibers of the projection $\tilde{Q}_{\mathrm{st}} \rightarrow Q_{\mathrm{st}}$ into the orbits of Γ . Those orbits are hence discrete, and Γ is a discrete subgroup of the Lie group $\mathrm{PU}(1, N - 3)$ of isometries of $B^+(\alpha)_o$. When $Q_{\mathrm{st}} = Q_{\mathrm{sst}}$, the quotient $B^+(\alpha)_o / \Gamma \cong Q_{\mathrm{sst}}$ is compact, and Γ is cocompact. In the general case, a local analysis near the cusps $p \in Q_{\mathrm{sst}} - Q_{\mathrm{st}}$ shows that it has finite covolume.

For $N = 3$, the group $\mathrm{PU}(1, 0)$ is trivial and the statement (3.11) uninteresting. For $N \geq 4$, it is not at once clear that $(1 - \mu_i - \mu_j)^{-1}$ must be an integer for all $i \neq j$ under condition INT (cf. (14.2.1)). It is easy to see however that condition INT implies that all μ_i are rational. We show this, even though it is not used in our proof.

Lemme 2.5. *Assume that $\mathrm{card} S > 4$. If positive real numbers $(\nu_s)_{s \in S}$ sum up to 2 and if for each of the pairs of elements $s \neq t$ of S for which $\nu_s + \nu_t \leq 1$, $\nu_s + \nu_t$ is rational, then all ν_s are rational.*

We may and shall assume that S is the set of integer from 1 to N , $N \geq 4$, and that $\nu_1 \leq \dots \leq \nu_N$.

Case 1. — $\nu_1 + \nu_2 > 1$. For any distinct $i, j < N$ one can find $k < N$ distinct from i and j . One has $\nu_k + \nu_N \geq \nu_1 + \nu_2 > 1$, hence

$$\nu_i + \nu_j = 2 - (\nu_k + \nu_N) < 1$$

is rational. If three numbers a, b, c are such that $a + b$, $b + c$ and $c + a$ are rational, they are rational. Applying this to the ν_i ($i < N$), we find that they are rational. So is ν_N , because $\sum \nu_i = 2$.

Case 2. — $\nu_1 + \nu_2 \leq 1$. In this case, for any $i \neq 1$, $\nu_1 + \nu_i \leq \nu_1 + \nu_2 \leq 1$ is rational. Summing, one concludes that $(N - 2)\nu_1 + \sum \nu_i = (N - 2)\nu_1 + 2$ is rational : ν_1 is rational, and so are the ν_i ($i \neq 1$).

2.8

Next, we explain how local systems L with the prescribed monodromy in the fiber direction can be constructed globally above M on P_M . Fix three distinct elements $a, b, c \in S$. For each m in M , there is then a unique isomorphism $\mathbb{P} \xrightarrow{\sim} \mathbb{P}^1$ mapping $m(a), m(b), m(c)$ respectively to $0, 1, \infty$. Let $z : P_M \rightarrow \mathbb{P}^1$ be the resulting map. When no ambiguity results, we will write $z(x)$ for $z(x, m)$. Let $A : M \rightarrow \mathbb{R}_+$ be a large enough continuous function. What is needed is that $[A(m), \infty[$ be disjoint from $z(m(S))$. For any $U \subset M$, let I_U denote the locus of $(x, m) \in \pi^{-1}(U)$ such that $z(x, m) \in [A(m), \infty[$. If $U \subset M$ is contractible, then $I_U \subset P_M$ is too. For $U \subset M$ contractible, if L is a local system on $\pi^{-1}(U)$ with the prescribed monodromy α (cf. (3.3)), then L_U admits an horizontal section $e_U \neq 0$ on I_U . The pair (L_U, e_U) is unique up to unique isomorphism, hence there is no problem in glueing. One gets a global local system L on P_M , provided with an horizontal section e on I_M .

Let $M_0 = \{m \in M \mid m(a) = 0, m(b) = 1, m(c) = \infty\}$ as in (3.7). The product decomposition $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$ induces an isomorphism $\mathrm{PGL}(2) \times P_{M_0} \xrightarrow{\sim} P_M$; and if one takes $A(m)$ constant on the $\mathrm{PGL}(2)$ orbits, so that I_M is the pullback of I_{M_0} , the system (L, e) is the pull back of its restriction to P_{M_0} .

2.9

We give two other descriptions of the same L . Fix numbers μ_s such that $\alpha_s = \exp(2\pi i \mu_s)$. Let L_μ be the subsheaf of $\mathcal{O}$ (cf. §2) consisting of those functions which are constant multiples of $\prod_{s \in S} (z - z(m(s)))^{\mu_s}$. This function is multivalued but the ratio of two determinations is a constant, so that the definition of L_μ makes sense. If μ' is another choice of μ_s with $\alpha_s = \exp(2\pi i \mu'_s)$, the multiplication by the univalued function $\prod_{s \in S} (z - z(m(s)))^{\mu'_s - \mu_s}$ is an isomorphism of L_μ with $L_{\mu'}$.

The principal determination of the multivalued function $\prod_{s \in S} (z - z(m(s)))^{\mu_s}$ on I_M is defined as $\exp(\sum \mu_s \log(z - z(m(s))))$ with $|\arg \log(z - z(m(s)))| < \pi$ for z real and large enough. This provides a section of L_μ on I_M . One easily checks that L_μ has the monodromy α on each fibre.

An identical but more algebraic description of L_μ is : the local system of horizontal sections of $\mathcal{O}$, provided with the connection ∇_μ for which

$$f^{-1} \nabla_\mu f = \frac{df}{f} - \sum_s \mu_s \frac{d(z - z(m(s)))}{z - z(m(s))}.$$

2.10

We close section 3 with some remarks about liftings of the projective representation

$$\rho : \pi_1(Q, o) \rightarrow \mathrm{PGL}(H^1(P_o, L_o))$$

to a linear representation.

As in (3.7) and (3.13), fix three elements a, b, c in S and denote by M_0 the subset of M with $m(a) = 0$, $m(b) = 1$ and $m(c) = \infty$. Let $M(c)$ denote the subset of $m \in M$ with $m(c) = \infty$, and let B denote the stabilizer of ∞ in $\mathrm{PGL}(2)$. One has $M_0 \xrightarrow{\sim} Q$ and $B \times M_0 \xrightarrow{\sim} M(c)$, $\mathrm{PGL}(2) \times M_0 \xrightarrow{\sim} M$.

As is well known,

$$\pi_1(\mathrm{PGL}(2)) = \pi_1((3)) = \pi_1((2)/\{\pm 1\}) = \mathbb{Z}/2$$

and $\pi_1(B) = \pi_1(\mathbb{C}^*) = \mathbb{Z}$, so that

$$\pi_1(M) \cong \pi_1(Q, o) \oplus \mathbb{Z}/2, \quad (2.4)$$

$$\pi_1(M(c)) \cong \pi_1(Q, o) \oplus \mathbb{Z}. \quad (2.5)$$

Both those isomorphisms depend on the choices of a, b, c .

For X any of Q , $M(c)$ or M , each choice of a local system L on P_X with fiberwise monodromy α leads to a lifting

$$\rho' : \pi_1(X, o) \rightarrow \mathrm{GL}(H^1(P_o, L_o))$$

of ρ . If L carries a unitary structure, ρ' has values in $(H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$. For instance, the L of (3.13) carries a unique unitary structure for which e is of length one, and its restriction to $M_0 \cong Q$ provides a lifting $\rho' : \pi_1(Q, o) \rightarrow (H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$.

The local system L itself provides a lifting $\rho' : \pi_1(M, o) \rightarrow (H^1(P_o, L_o), \psi)$ which is trivial on the second factor of the decomposition (3.15.1).

Let $L(c)$ denote the local system on $P_{M(c)}$ which is given by the subsheaf of $\mathcal{O}$ consisting of those functions which are constant multiples of

$$\prod_{s \in S - \{c\}} (z - m(s))^{-\mu_s}.$$

Here, z is a fixed coordinate on $\mathbb{P}$ composed with $\mathbb{P} \times M(c) \rightarrow \mathbb{P}$ and not as in (3.13) and (3.13). The local system $L(c)$ provides a lifting $\rho' : \pi_1(M(c)) \rightarrow (H^1(P_o, L_o), \frac{-1}{2\pi i}\psi)$ of ρ .

Inasmuch as $M(c)$ is the space of distinct $(N - 1)$ -uples of points in the plane ($N = \mathrm{card} S$), $\pi_1(M(c))$ is by definition the colored braid group on $(N - 1)$ strands. For any $s, t \in S - \{c\}$, let $\gamma_{s,t}$ in $\pi_1(M(c))$ be a path in which t comes near s , makes a positive turn around s and comes back to its original position. The colored braid group is generated by such elements. The lifting ρ' provided by $L(c)$ has the virtue that each $\rho'(\gamma_{s,t})$ is a pseudo-reflection (i.e. $\rho'(\gamma_{s,t}) - I$ is of rank one)—cf. (9.1), (12.3).

3 The compactified quotient space Q_{st}

3.1

As in section 3, we fix a complex projective line $\mathbb{P}$, an integer $N \geq 3$ and a finite set S with N elements. We further fix a family $\mu = (\mu_s)_{s \in S}$ of real numbers with $0 < \mu_s < 1$ and $\sum \mu_s = 2$. The complex numbers $\alpha_s := \exp(2\pi i \mu_s)$ satisfy $|\alpha_s| = 1$, $\alpha_s \neq 1$ and $\prod \alpha_s = 1$.

As in section 3, $M \subset \mathbb{P}^S$ is the space of injective maps $\gamma : S \rightarrow \mathbb{P}$. We denote by $\mathrm{PGL}(2)$ the group of automorphisms of $\mathbb{P}$.

These conventions will hold throughout this section as well as in sections 6 to 12. In this section, from (4.2) on, $\mathbb{P}$ will be the standard projective line $\mathbb{P} := \mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ and M the space of S -uples of distinct points of the Riemann sphere. This is no loss of generality.

3.2

The group $\mathrm{PGL}(2)$ operates diagonally on $\mathbb{P}^S$. We shall define a compactification of the quotient space $Q := \mathrm{PGL}(2) \backslash M$. The definition is taken from D. Mumford's theory of quotient varieties of reductive groups (cf. (4.6)). The account here is self-contained.

A point $y \in \mathbb{P}^S$ is called μ -stable (resp. μ -semi-stable) if and only if for all $z \in \mathbb{P}$,

$$\sum_{y(s)=z} \mu_s < 1 \quad (\text{resp. } \leq 1). \quad (3.1)$$

The set of all μ -stable points (resp. μ -semi-stable points) is denoted M_{st} (resp. M_{sst}) and we set $M_{0\mathrm{usp}} = M_{\mathrm{sst}} - M_{\mathrm{st}}$.

For each partition $\{S_1, S_2\}$ of S with $\sum_{s \in S_i} \mu_s = 1$ ($i = 1, 2$), the points y in $\mathbb{P}^S$ for which $y(S_1) \cap y(S_2) = \emptyset$ and y constant on S_1 or S_2 are in $M_{0\mathrm{usp}}$. All points in $M_{0\mathrm{usp}}$ are obtained in this way, each from a unique partition.

On M_{sst} we define a relation $\sim$ via

$y \sim y'$ if and only if either

- a) $y, y' \in M_{\mathrm{st}}$ and $y' \in \mathrm{PGL}(2)y$; or
- b) $y, y' \in M_{0\mathrm{usp}}$ and the partitions of S corresponding to y and y' coincide.

It is clear that $\sim$ is an equivalence relation. Set

$$Q_{\mathrm{st}} = M_{\mathrm{st}} / \mathrm{PGL}(2), \quad Q_{\mathrm{sst}} = M_{\mathrm{sst}} / \sim, \quad Q_{0\mathrm{usp}} = M_{0\mathrm{usp}} / \sim$$

each with its quotient topology. The elements of $Q_{0\mathrm{usp}}$ are uniquely determined by their partitions. Thus $Q_{0\mathrm{usp}}$ is a finite set.

Exemple 3.1. The cross-ratio $c(z_1, z_2, z_3, z_4) = \frac{z_1 - z_3}{z_3 - z_2} \cdot \frac{z_1 - z_4}{z_4 - z_2}$ of four points in $\mathbb{P} := \mathbb{P}^1$ is defined and is a continuous map into $\mathbb{P}^1$ as long as no three of z_1, z_2, z_3, z_4 are equal. When two of the z_i are equal, the cross-ratio takes one of the values $0, 1, \infty$, the value depending only on the partition of $\{1, 2, 3, 4\}$ consisting of the subset $\{i, j\}$ with $z_i = z_j$ and its complement.

Suppose $S = \{1, 2, 3, 4\}$ and $\mu_s = \frac{1}{2}$ for all $s \in S$. One can show that the cross-ratio map $M_{\mathrm{sst}} \rightarrow \mathbb{P}^1$ descends to Q_{sst} and yields a homeomorphism onto the projective line.

3.3

Let M' be the space of $y \in \mathbb{P}^S$ such that $y(S)$ has at least three distinct points, and let $Q' = \mathrm{PGL}(2) \backslash M'$ be the corresponding quotient orbit space. Fix a, b, c in S , let U be the set of all $y \in M'$ taking distinct values on a, b, c , and let U_0 be the subset of all $y \in M$ for which $y(a) = 0, y(b) = 1, y(c) = \infty$. Then U is in M , stable under the action of $\mathrm{PGL}(2)$, and $\mathrm{PGL}(2) \times U_0 \xrightarrow{\sim} U$. Hence

$$U_0 \xrightarrow{\sim} \mathrm{PGL}(2) \backslash U \hookrightarrow Q'.$$

As one varies a, b, c , the images of U_0 in Q' cover Q' . Consequently Q' is a manifold. However, this manifold is not Hausdorff if $\mathrm{card} S > 3$. For $\mathrm{card} S = 4$ (cf. (4.2)) one gets the projective line with $0, 1$ and ∞ doubled. One infers from (4.1.1) that $M_{\mathrm{st}} \subset M'$. The quotient Q_{st} being open in Q' is a manifold. The following Lemma implies that it is Hausdorff.

Lemme 3.2. Q_{sst} is Hausdorff.

Démonstration. Let F denote the set of injective maps of $\{1, 2, 3, 4\}$ into S . For any $f \in F$, let M_f denote the set of all $y \in M_{\text{sst}}$ such that for all $z \in \mathbb{P}$, $\text{card}(yf)^{-1}(z) < 2$ and additionally, if $y \in M_{0\text{usp}}$, $\text{card } f^{-1}(S_i) = 2$ for $i = 1, 2$ where $\{S_1, S_2\}$ is the partition of S corresponding to y . The additional condition guarantees that $M_f \cap M_{f'}$ as well as $M_f \cap M_{f'}$ is $\sim$ -saturated.

For any $f \in F$, define $c_f : M_f \rightarrow \mathbb{P}$ via the cross-ratio :

$$c_f(y) = c(yf(1), yf(2), yf(3), yf(4));$$

the function c_f is constant on $\sim$ -equivalence classes of the $\sim$ -saturated set M_f . Inasmuch as c_f is continuous on M_f , it descends to a continuous function on $M_f / \sim$. To prove the lemma, it suffices to prove :

For any $y \not\sim y' \bmod \sim$, there is an $f \in F$ such that y and y' are in M_f , and $c_f(y) \neq c_f(y')$. (3.2)

One is free to replace y and y' by y_1 and y'_1 , with $y \sim y_1$, $y' \sim y'_1$.

To each y in M_{sst} we attach a partition of S as follows : for $y \in M_{0\text{usp}}$, attach the partition defined in (4.1); for $y \in M_{\text{st}}$, attach the partition $T = \{C_1, \dots, C_k\}$ such that $y(s) = y(t)$ if and only if s and t lie in the same coset of T . If $y \sim y_1$, the same partition is attached to y and y_1 . In the proof of (4.3.1), we may and shall assume that if y or y' is in $M_{0\text{usp}}$, then it is constant on the cosets of its partition.

We first treat the case that y and y' are in $M_{0\text{usp}}$. Let $\{S_1, S_2\}$ and $\{S'_1, S'_2\}$ be their corresponding partitions of S . Since $y \not\sim y' \bmod \sim$, $S_i \cap S'_j$ is not empty for $i, j \in \{1, 2\}$; otherwise, if say $S_1 \cap S'_2 = \emptyset$, we would have $S_1 \subset S'_1$, $S_1 \neq S'_1$ and

$$\sum_{s \in S'_1 - S_1} \mu_s = 0,$$

contradicting $0 < \mu_s$ for each s . Choose f so that $\text{card } f^{-1}(S_i \cap S'_j) = 1$ for each choice of $i, j \in \{1, 2\}$. Then for suitable choice of f we have

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, a, b, b, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', b', a', b', \end{aligned}$$

with $a \neq b$, $a' \neq b'$. Hence $c_f(y) = \infty$, $c_f(y') = 0$.

Suppose next that $y \in M_{\text{st}}$ and $y' \in M_{0\text{usp}}$. Let $T = \{C_i\}$ be the partition of S corresponding to y' by (4.2). Each coset in the partition of S corresponding to y must meet at least two distinct cosets of T . We can thus choose an $f \in F$ so that

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, a, b, b, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', b', c, d, \end{aligned}$$

with $a \neq b$, $a' \neq b'$, and $c \neq d$. It follows at once that y and y' are in M_f , $c_f(y) = 1$, $c_f(y') \neq 1$.

It remains only to consider the case where y and y' are in M_{st} ; let T and T' be the corresponding partitions of S . If $T = T'$, then for any $f \in F$, $y \in M_f$ if and only if $y' \in M_f$. Replacing y' by an element in its $\text{PGL}(2)$ -orbit, we can assume that there are

three distinct cosets C_1, C_2, C_3 of T with $y(s) = y'(s)$ for $s \in C_1 \cup C_2 \cup C_3$. If $c_f(y) = c_f(y')$ for all f with $y \in M_f$ then, clearly $y = y'$. One is thus reduced to the case where $T \neq T'$. Then there is a $C_1 \in T$ and $C'_1 \in T'$ such that $C_1 \cap C'_1$ is not empty and $C_1 \neq C'_1$. Either $C_1 - C'_1$ or $C'_1 - C_1$ is not empty—say $C'_1 - C_1 \neq \emptyset$. Choose $C'_2 \in T'$ so that $C'_2 \cap C_1 \neq \emptyset$. Choose $C'_3 \in T'$ with $C'_3 \neq C'_2$. Next choose $C_2 \in T$ so that $C'_2 \cap C_2 \neq \emptyset$. Then $C'_3 \cup C_2 \neq S$; otherwise, since y' and y are in M_{st} ,

$$2 < \sum_{s \in C'_1} \mu_s + \sum_{s \in C_2} \mu_s.$$

Choose $f : \{1, 2, 3, 4\} \rightarrow S$ with

$$f(1) \in C_1 \cap C'_1, \quad f(2) \in C'_2 \cap C_1, \quad f(3) \in C'_2 \cap C_2, \quad f(4) \in C'_3 \cup C_2.$$

Any such f is injective, and

$$\begin{aligned} yf(1), yf(2), yf(3), yf(4) &= a, b, c, d \quad \text{with } a \neq b, c \neq d, \\ y'f(1), y'f(2), y'f(3), y'f(4) &= a', a', b', c' \quad \text{with } a' \neq b', a' \neq c'. \end{aligned}$$

Hence $c_f(y') = 1$, $c_f(y) \neq 1$. This completes the proof. $\square$

3.4

Lemme 3.3. *Fix a Riemannian metric on $\mathbb{P}$. For $y \in \mathbb{P}^S$, define*

$$d(y) = \inf_{\sum_{s \in T} \mu_s > 1} \text{diam } y(T)$$

and

$$a(y) = \sup_{g \in \text{PGL}(2)} d(gy).$$

Then there is $a > 0$ such that $(y) \geq a$ for all $y \in M_{\text{st}}$.

Démonstration. The validity of the lemma is independent of the chosen metric. We will use the Fubini metric (for which $\mathbb{P}$ is isometric to the sphere of radius 1) and prove the lemma with $a = \pi/4$.

If $y \in M_{0\text{usp}}$ corresponds to the partition $\{S_1, S_2\}$ of S , with y constant on S_1 , let us take g to fix $y(S_1)$ and the antipodal point u of $\mathbb{P}$ and to carry the complement of the ε -neighborhood of $y(S_1)$ into the ε -neighborhood of u . Taking $\varepsilon \rightarrow 0$, one gets $d(y) = \pi$.

For any ε there is a compact $K \subset \text{PGL}(2)$ such that for $g \notin K$, g maps the complement of the ε -neighborhood of some point u into the ε -neighborhood of some point v . If $y \in M_{\text{st}}$ and if ε is small enough, then, for all u , $\{s \in S \mid (y(s), u) > \varepsilon\}$ is in $\mathcal{C}$. This implies $d(gy) \rightarrow 0$ for $g \rightarrow \infty$ in $\text{PGL}(2)$. The sup defining (y) is hence attained. Fix $y \in M_{\text{st}}$ with $(y) = d(y)$. We have to prove $d(y) > a = \pi/4$.

Suppose to the contrary that $d(y) < a$. Fix $T_1 \in \mathcal{C}$ with $\text{diam } y(T_1) < a$ and $t \in T_1$. For any T_1, T_2 in $\mathcal{C}$, $T_1 \cap T_2 \neq \emptyset$. If S' is the union of all $T \in \mathcal{C}$ with $\text{diam } y(T) < a$, $y(S')$ is hence contained in a half-sphere of radius $r < 2a = \pi/2$ with center $y(t)$. In particular, $y(S')$ is contained in an open halfsphere. Identify $\mathbb{P}$ with the Riemann sphere $\mathbb{C} \cup \{\infty\}$, with the unit disc the halfsphere, in such a way that $y(S')$ lies in $D = \{z \mid |z| < b\}$, $b < 1$. Take for g the multiplication by $1 + \varepsilon$ ($\varepsilon > 0$). For ε small enough (relative to b), distances in D are increased by g . For any $T \in \mathcal{C}$ with $\text{diam } y(T) < a$, one hence has $\text{diam } gy(T) > \text{diam } y(T)$. For $T \in \mathcal{C}$ with $\text{diam } y(T) > a$, $\text{diam } y(T)$ can decrease, but not much for ε small, so that $d(gy) > d(y)$. This contradicts the choice of y . $\square$

Lemme 3.4. Q_{sst} is compact.

Démonstration. Fix $a > 0$ as in (4.4.1), and define $M' \subset \mathbb{P}^S$ to be the set of y such that $\text{diam } y(T) > a$ for each $T \subset S$ with $\sum_{s \in T} \mu_s > 1$. The subset M' of $\mathbb{P}^S$ is closed, hence compact. It is contained in M_{sst} and, by (4.4.1), it maps onto Q_{sst} . Compactness follows. $\square$

3.5

By contrast with Q_{st} which is a manifold, the points of $Q_{0\text{usp}}$ may be singular.

Let π denote the natural projection of M_{sst} onto Q_{sst} . Let $\{S_1, S'_1\}$ be the partition corresponding to a point $y_0 \in M_{0\text{usp}}$. We shall describe a neighborhood of $\pi(y_0)$ in Q_{sst} . We can assume that $y_0(s) = 0$ for all $s \in S_1$ and $y_0(s) = \infty$ for all $s \in S'_1$. Fix elements $a \in S_1$ and $b \in S'_1$, and define V as the set of all $y \in M_{\text{sst}}$ such that $y(a) = 0$, $y(b) = \infty$ and

$$\sup_{s \in S_1} |y(s)| < \inf_{s \in S'_1} |y(s)|.$$

Then, $\pi(V)$ is an open neighborhood of $\pi(y_0)$ in Q_{sst} and is the quotient of V by the equivalence relation on V induced by $\sim$. The punctured neighborhood $\pi(V) - \pi(y_0)$ is the quotient of $V - \pi^{-1}\pi(y_0)$ by the stabilizer in $\text{PGL}(2)$ of the points 0 and ∞ :

$$\pi(V) - \pi(y_0) = (V - \pi^{-1}\pi(y_0))/\mathbb{C}^*.$$

Set $S_2 := S_1 - \{a\}$, $S'_2 := S'_1 - \{b\}$, $m = \text{card}(S_2)$, $m' = \text{card}(S'_2)$ and consider the composite map

$$\begin{aligned} \Phi : V &\rightarrow \mathbb{C}^{S_2} \times \mathbb{C}^{S'_2} = \mathbb{C}^m \times \mathbb{C}^{m'} \rightarrow \mathbb{C}^{mm'} \\ y &\mapsto ((y(s))_{s \in S_2}, (y(s)^{-1})_{s \in S'_2}) \mapsto (y(s) \cdot y(t)^{-1})_{s \in S_2, t \in S'_2}. \end{aligned}$$

The map Φ descends to an embedding of $\pi(V)$ into $\mathbb{C}^{mm'}$, whose image is the intersection of the cone over the Veronese embedding of $\mathbb{P}^{m-1} \times \mathbb{P}^{m'-1}$ in $\mathbb{P}^{mm'-1}$ with the polydisc (of radius < 1 , $|x_{s,t}| < 1$).

If $m = 1$ or $m' = 1$, the Veronese embedding is an isomorphism with $\mathbb{P}^{mm'-1}$. As a consequence

If $\text{card}(S_1) = 2$ or $\text{card}(S'_1) = 2$, the cusp point with partition $\{S_1, S'_1\}$ is non singular. (3.3)

One can define an algebraic structure on Q_{sst} by using as a chart at a point $\text{PGL}(2)y$ of Q_{st} the orbit space

$$\text{PGL}(2) \backslash \{y' \in M_{\text{st}} \mid y'(i) \neq y'(j) \text{ if } y(i) \neq y(j)\}$$

and at a cusp point with partition $\{S_1, S'_1\}$ the Zariski-open subset of the Veronese cone of $\mathbb{P}^{m-1} \times \mathbb{P}^{m'-1}$ in $\mathbb{C}^{mm'}$ given by $x_{s,t} = y(s)/y(t) \neq 1$, for all $s \in S_1$, $t \in S'_1$.

Exemple 3.5. 1.— $S = \{1, 2, 3, 4, 5\}$, $\mu_s = \frac{2}{5}$ for all $s \in S$. Here there are no cusp points : $Q_{\text{sst}} = Q_{\text{st}} = \text{PGL}(2) \backslash M_{\text{st}}$. Each pair s, t in S defines the diagonal line $L_{s,t} : y(s) = y(t)$ in M_{st} and under the projection to Q_{st} , one gets 10 lines. The map of M_{st} to $(\mathbb{P}^1)^2$ given by

$$(y_1, y_2, y_3, y_4, y_5) \mapsto (\infty, 0, 1, c(y_1, y_2, y_3, y_4)^{-1}, c(y_1, y_2, y_3, y_5)^{-1})$$

(where c denotes cross-ratio) descends to a map p of Q_{sst} which is biregular except at $L_{0,\infty} = p^{-1}(0, 0)$, $L_{0,\infty} = p^{-1}(1, 1)$, $L_{\infty,\infty} = p^{-1}(\infty, \infty)$. Here, Q_{sst} is a blowup of $\mathbb{P}^1 \times \mathbb{P}^1$.

Each of the curves $p^{-1}(0, 0)$, $p^{-1}(1, 1)$, $p^{-1}(\infty, \infty)$ are exceptional, that is, each has self-intersection -1 . Inasmuch as the set of lines $\{L_{ij}, i \neq j, i, j \in S\}$ are permuted transitively by permutation of coordinates in M_{st} , their images in Q_{st} are also permuted transitively and thus each of the 10 lines is exceptional. The 10 lines of Q_{st} consist of $p^{-1}(0, 0)$, $p^{-1}(1, 1)$, $p^{-1}(\infty, \infty)$, and the 7 lifts of the seven lines on $\mathbb{P}^1 \times \mathbb{P}^1$:

$$x = 0, x = 1, x = \infty, \quad y = 0, y = 1, y = \infty, \quad x = y$$

where $x = c(y_1, y_2, y_3, y_4)^{-1}$, $y = c(y_1, y_2, y_3, y_5)^{-1}$.

2.— $S = \{1, 2, 3, 4, 5\}$, $\mu = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3})$. Here there are four points in Q_{0usp} corresponding to the partitions $\{S_i, S_i^c\}$ where $S_i = \{i, 5\}$, $1 \leq i \leq 4$. Q_{sst} is a non-singular manifold by (4.5.1). This Q_{sst} can be obtained from the Q_{sst} of Example 1 by blowing down the four lines $L_{i,5}$ ($1 \leq i \leq 4$). It can be obtained from $\mathbb{P}^1 \times \mathbb{P}^1$ by blowing up the point (∞, ∞) and then blowing down the lines $x = \infty$ and $y = \infty$. The resulting space is $\mathbb{P}^2$.

3.6

We conclude this section by relating our Q_{sst} to Mumford's when all the μ_s are rational.

Let $\mathcal{O}(n)$ be the line bundle on $\mathbb{P}$ of degree n , and let Ω be the canonical line bundle on $\mathbb{P}$; i.e. $\Omega \cong \mathcal{O}(-2)$. For any line bundle we write $^{\otimes n}$ for the dual line bundle and $^{\otimes n} = (^{\otimes(-1)})^{\otimes(-n)}$ for $n < 0$.

Let D be the common denominator for $\{\mu_s, s \in S\}$. On $\mathbb{P}^S$ define the line bundle

$$\bigotimes_{s \in S}^* (\Omega^{\otimes \mu_s \cdot D} \otimes \mathcal{O}(2)^{\otimes D}).$$

It admits a $\text{PGL}(2)$ -action coming from the action on the tangent bundle $\Omega^{\otimes(-1)} \cong \mathcal{O}(2)$. The stable and semi-stable points of $\mathbb{P}^S$ defined in (4.1) are the same as those defined in Mumford's "Geometric Invariant Theory" for the action of $\text{PGL}(2)$ on the linear space of holomorphic sections of this line bundle; Q_{sst} is the underlying topological space of Mumford's "quotient variety" for $\text{PGL}(2) \backslash \mathbb{P}^S$ [?].

4 The complex ball

4.1

Let V be a complex vector space, and $(,)$ be a Hermitian form on V , of signature $(1, \dim V - 1)$. The complex ball B^+ in $\mathbb{P}(V) := \{1\text{-dimensional subspaces of } V\}$ corresponding to $(,)$ is the set of lines in V which are spanned by a vector v with $(v, v) > 0$. The form $(,)$ is determined, up to a positive real factor, by B^+ . The closure $\overline{B^+}$ of B^+ in $\mathbb{P}(V)$ is the set of lines spanned by a non zero vector v such that $(v, v) \geq 0$.

4.2

The hyperbolic angle $\theta \in \mathbb{R}_+$ between lines $\ell_1, \ell_2 \in B^+$, spanned by vectors v_1, v_2 , is defined by

$$|(v_1, v_2)| = \cosh(\theta)(v_1, v_1)^{1/2}(v_2, v_2)^{1/2}. \quad (4.1)$$

It remains unchanged when $(,)$ is replaced by a positive multiple. The angle $\theta(\ell_1, \ell_2)$ is a Riemannian distance on B^+ . It is invariant by the action of the unitary group (V) on B^+ and renders B^+ a Hermitian symmetric space. By continuity we extend the function $\theta(\ell_1, -)$ to a function $\theta(\ell_1, -)$ from $\overline{B^+}$ to $\mathbb{R}_+ \cup \{\infty\} : \theta(\ell_1, \ell) = \infty$ for $\ell \in \partial B^+$.

4.3

Fix $\ell \in \partial B^+$. For $v \neq 0$ in ℓ , we define a “distance from ℓ ” function on B^+ by

$$d_\ell(\ell_1) = |(v, v_1)|/(v_1, v_1)^{1/2} \quad \text{for } \ell_1 \in B^+, \text{ any } v_1 \neq 0 \text{ in } \ell_1; \quad (4.2)$$

$$d_\ell(\ell) = 0, \quad d_\ell(\ell') = \infty \text{ for } \ell' \in \partial B^+, \ell' \neq \ell. \quad (4.3)$$

The function d_ℓ is continuous and ≥ 0 on $\overline{B^+} - \{\ell\}$. One has $d_{\lambda\ell}(\ell_1) = |\lambda| d_\ell(\ell_1)$, and the family of functions d_ℓ ($v \in \ell - \{0\}$) is stable by the stabilizer of ℓ in (V) .

4.4

On $\overline{B^+}$, we will use the coarsest topology for which the functions d_ℓ and $\theta(\ell_1, -)$ are continuous. A fundamental system of neighborhoods of $\ell \in B^+$ (resp. of $\ell \in \partial B^+$ —on which we choose $v \neq 0$) consists of the V_ε ($\varepsilon > 0$) defined by $V_\varepsilon = \{\ell_1 \in B^+ \mid \theta(\ell, \ell_1) < \varepsilon\}$ (resp. $V_\varepsilon = \{\ell_1 \mid \ell_1 = \ell \text{ or } \ell_1 \in B^+ \text{ and } d_\ell(\ell_1) < \varepsilon\}$).

This topology is finer than the one induced by the topology of $\mathbb{P}(V)$. It induces the usual topology on B^+ , and the discrete one on ∂B^+ . It is respected by (V) .

Proposition 4.1. *Fix $\ell \in \partial B^+$ and let A be a discrete subgroup of the projective unitary group $\text{PU}(V)$ stabilizing ℓ and respecting a “distance to ℓ ” function d_ℓ . If ℓ has a neighborhood V (in the topology 5.4) such that V/A is compact, the volume of $(V \cap B^+)/A$ is finite.*

Démonstration. The subgroup of $\text{PU}(V)$ respecting ℓ and d_ℓ is the isomorphic image of the stabilizer H of v in (V) . It is an extension of the unitary group $(v^\perp/v)$ by the unipotent subgroup N of H consisting of the $n \in (V)$ respecting the filtration $V \supset v^\perp \supset \mathbb{C}v \supset 0$ and acting trivially on the successive quotients. The extension is split : H and N act transitively on each horocycle $d_\ell(\cdot) = C$ and the stabilizer of ℓ_1 is a lifting of $(v_1^\perp/v_1)$.

Fix an isotropic vector v' such that $(v, v') = 1$, and let $v_1 = v \cdot t^{-2} - v'$. The line ℓ_1 spanned by v_1 is in B^+ on the horocycle $d_{\ell_1} = t$. Let V' be spanned by v and v' , and let V'' be the orthogonal complement ($\supset v^\perp/v$). Any $h \in H$ fixing ℓ_1 fixes each point of V' ; hence the stabilizer of ℓ_1 in H is $(V'') \subset (V)$. Let $A \subset (V') \subset (V)$ be the group of unitary transformations $a(\lambda) : v \mapsto \lambda^{-1}v, v' \mapsto \lambda v'$ ($\lambda \in \mathbb{R}_+$) and fixing each point of V'' . It normalizes H : one has $d_\ell \circ a(\lambda) = d_\ell \cdot |\lambda|^{-1} = \lambda d_\ell$. Let $A_R = \{a(\lambda) \mid \lambda < R\}$. The map $g \mapsto g\ell_1$ maps isomorphically $A_R H/(V'') = A_R N$ onto $\{\ell_1 \in B^+ \mid d_\ell(\ell_1) < R\}$. The horocycles are closed in $\overline{B^+}$ (for the topology 5.4). The compactness assumption implies that for R small enough, $\{\ell_1 \mid d_\ell(\ell_1) = R\}/A$ is compact. One has

$$A \backslash a(R)H/(V'') \xrightarrow{\sim} \{\ell_1 \mid d_\ell(\ell_1) = R\}/A$$

and A is hence cocompact in H . It remains to evaluate the volume of

$$A \backslash A_R H/(V'') \xrightarrow{\sim} \{\ell_1 \in B^+ \mid d_\ell(\ell_1) < R\}/A$$

for a volume element on $A_R H / (V'')$ which is A - H -invariant, hence a multiple of the image of $da \cdot dh$ on $A_R H$.

$$\int_{A \setminus A_R H} da' dh = \int_{\lambda \leq R} \int_{A \setminus a(\lambda) H} d(a(\lambda)^{-1} h a(\lambda)) d\lambda$$

and

$$\int_{A \setminus a(\lambda) H} dh = \int_{A \setminus H} d(a(\lambda)^{-1} h a(\lambda)) = |\det(a(\lambda)^{-1}, H)| \int_{A \setminus H} dh.$$

This leads us to evaluate $|\det(a(\lambda)^{-1}, H)|$. One checks that the character $\det(a(\lambda)^{-1}, H)$ of $\lambda \in \mathbb{R}_+^*$ tends to zero for $\lambda \rightarrow 0$, and the integral is hence finite.

The computation above is of course exactly the one used to prove that Siegel domains have finite volume ([?], Lemma 1.9). $\square$

5 Stable points

5.1

Notation and assumptions (4.0) are in force in this section as well as the assumption $N \geq 4$. Our aim is to investigate the asymptotic behaviour of the section w_α of $B(\alpha)_M$, for $y \in M$ converging to a stable point $m \in M_{\text{st}}$.

Fix $m \in M_{\text{st}}$. To make a local study of w_α near m , it will be convenient to identify $\mathbb{P}$ with $\mathbb{P}^1$, in such a way that $\infty \notin m(S)$. Let U be an open neighborhood of m in M_{st} , such that $\infty \notin y(S)$ for $y \in U$. On the inverse image P_U of U in P_M (3.1), there is a rank one local system, trivialized at ∞ by a section e over $\{\infty\} \times U$, and with monodromy α on P_U for $y \in U \cap M$. A model of it is : the local system of constant multiples of $\prod_{s \in S} (z - y(s))^{\mu_s}$; the μ_s summing up to an integer, there is no ramification at ∞ . A trivialisation for large positive real z is provided by the principal determination of $\prod_{s \in S} (z - y(s))^{\mu_s}$; it extends by continuity to a trivialisation e at ∞ . We normalize the Hermitian form $(,)$ on L so that $(e, e) = 1$.

For a general $y \in U$, the monodromy of L_U on P_y around $p \in y(S)$ is $\prod_{y(s)=p} \alpha_s = \exp(2\pi i \sum_{y(s)=p} \mu_s)$. The stability of y ensures that $0 < \sum_{y(s)=p} \mu_s \leq 1$, so that this local monodromy is never trivial.

Fiber by fiber, we take as section ω of $\Omega^1(\sum \mu_s \cdot y(s))(L)$

$$\omega_y = \prod_{s \in S} (z - y(s))^{-\mu_s} \cdot e dz.$$

On $U \cap M$, it depends holomorphically on y . The assumption of stability of y amounts to saying that this form is of the first kind.

5.2

The sheaf $R^1 \pi_! L_U$, for π the projection $P_U \rightarrow U$, is no longer a local system if $m \notin M$. Its fiber at m is $H_c^1(P_m, L_m)$ ($\xrightarrow{\sim} H^1(P_m, L_m)$ by (2.6.1)). For each $h \in H_c^1(P_m, L_m)$, there is hence a unique germ $h^\sim$ of a section of $R^1 \pi_! L$ inducing h on P_m . The vector space $H_c^1(P_m, L_m)$ being finite dimensional, this provides us on U suitably small, with a map

$$(\text{constant sheaf } H_c^1(P_m, L_m)) \hookrightarrow R^1 \pi_! L_U. \quad (5.1)$$

This map is compatible with the non degenerate Hermitian form $(\cdot, \cdot)$, and hence injective. Compatibility can be checked on the description given below. In particular, each $h \in H_c^1(P_m, L_m)$ defines a horizontal section $h^\sim$ of $R^1\pi_!L$ on $U \cap M$.

We describe $h^\sim$, for h the class of a L_m -valued compactly supported C^∞ form η , whose support is contained in a connected compact K containing ∞ , and for U connected and small enough for K to be disjoint from $y(S)$ for $y \in U$. On $K \times U \subset P_U$, L is uniquely isomorphic to ${}^*_1(L_m)$, by an isomorphism which is the identity above m . It hence make sense, in each fibre P_y , to take the same form η ; it defines $h^\sim$.

5.3

If we apply this to ω_m , we get an horizontal section $(\omega_m)^\sim$ of $R^1\pi_!L_U$ on $U \cap M$. As $(\omega_m, \omega_m) > 0$ (2.20), the same positivity holds for $(\omega_m)^\sim$, and, projecting from $R^1\pi_!L_U$ to $B(\alpha)$, we get a section $(\omega_m)^\sim$ of $B^+(\alpha)_M$. We will compare it with the holomorphic section w_α , using fiber by fiber the metric (5.2).

Proposition 5.1. *$\theta(w_\alpha, (\omega_m)^\sim)$ tends to 0 for $y \in M$ tending to m .*

Démonstration. This will result at once from the two statements

$$(\omega_y, \omega_y) \rightarrow (\omega_m, \omega_m) \quad \text{for } y \rightarrow m; \quad (5.2)$$

$$(\omega_y, (\omega_m)^\sim) \rightarrow (\omega_m, \omega_m) \quad \text{for } y \rightarrow m. \quad (5.3)$$

If it made sense, we would like to say that $\omega_y \rightarrow \omega_m$ in the L^2 -norm, for $y \rightarrow m$, but ω_y and ω_m do not live in the same space, having different ramification. Let K_R be the complement in $\mathbb{P}^1$ of discs of radius R around each point of $m(S)$. One assumes R small enough for these discs to be disjoint; on a connected neighborhood U of y , small enough for $y(S)$ to stay disjoint from K_R , one can compare $\omega_y|_{K_R}$ and $\omega_m|_{K_R}$: on $K_R \times U$, $L \xrightarrow{\sim} {}^*_1 L_m$. There, one has a C^∞ isotopy of ω_y to ω_m . To prove (6.4.1), it hence suffices to obtain a bound r uniform in y , close to m , for the L^2 -norm of the restriction of ω_y to a disk D_R of radius R around $p \in m(s)$, for any $s \in S$, this bound being such that $r \rightarrow 0$ for $R \rightarrow 0$.

We may assume R and U small enough so that $y(s) \subset D_{2R}$ if $m(s) \neq p$. This ensures

$$\prod_{m(s) \neq p} |z - y(s)|^{-\mu_s} < \text{constant} \cdot \prod_{m(s) \neq p} |z - y(s)|^{-\mu_s}$$

and the required estimate follows from the following lemma.

Lemme 5.2. *Let D be a disc of radius R , $(a_i)_{i \in I}$ be a family of points of D and $(\mu_i)_{i \in I}$ a family of real numbers, such that $0 < \mu_i < 1$ and $\sum \mu_i < 1$. Then, writing $z = x + iy$,*

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy < \int_{|z| < R} |z|^{-2\sum \mu_i} dx dy = \frac{2\pi R^{2(1-\sum \mu_i)}}{2(1-\sum \mu_i)}.$$

Démonstration. We may and shall assume D centered at 0. If the real numbers p_i are such that $1 > \frac{1}{p_i} > \mu_i$ and $\sum \frac{1}{p_i} = 1$, Hölder's inequality gives

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy \leq \prod_i \left[\int_D |z - a_i|^{-2\mu_i p_i} dx dy \right]^{1/p_i}.$$

One has that the integral of $|z - a_i|^{-2\mu_i p_i} dx dy$ is bigger when taken on the disc $D_R(a_i)$ of radius R and center a_i than when taken on $D_R(0)$: results from the fact that $|z - a_i|^{-2\mu_i p_i}$ is bigger on $D_R(a_i) - D_R(0)$ than on $D_R(0) - D_R(a_i)$: one has $|z - a_i| < R$ on $D_R(a_i) - D_R(0)$, and $|z - a_i| \geq R$ on $D_R(0) - D_R(a_i)$.

Using that $\sum 1/p_i = 1$, we have

$$\int_D \prod_i |z - a_i|^{-2\mu_i} dx dy \leq \prod_i [2\pi \dots]^{1/p_i} = 2\pi \cdot R^{2(1-\sum \mu_i)} \cdot C^{-1}$$

with $C = \prod_i (2 - 2\mu_i p_i)^{1/p_i}$. Taking the p_i such that the $\mu_i p_i$ are all equal, i.e. $p_i = \mu_i^{-1} \sum \mu_i$, one gets the result. $\square$

Proof of (6.4.2).— For each $p \in m(S)$, let D_p be a disc of radius R around p , u_p the solution on $D_p^* = D_p - \{p\}$ of $\omega_m = du$ and φ_p the characteristic function of D_p . The current with compact support $\omega_m - \sum d(\varphi_p u_p)$ can be used to compute $(\omega_m)^\sim$. One gets

$$(\omega_y, (\omega_m)^\sim) = \int_{P \cup D_p} \omega_y \wedge \overline{\omega_m} - \sum_p \int \omega_y \wedge \overline{d(\varphi_p u_p)}.$$

For $y \rightarrow m$, and $R \rightarrow 0$ in such a way that the disc of radius $2R$ around p contains all $y(s)$ for $m(s) = p$, the first integral converges to (ω_m, ω_m) , while the others each are $O(R^{2\beta})$, $\beta = 1 - \sum_{m(s)=p} \mu_s$. This concludes the proof. $\square$

5.4

In the neighborhood of a stable point $y \in Q_{\text{st}}$, the projection of $\tilde{Q}_{\text{st}}$ to Q_{st} has a section. Taking a pull-back by such a section, we get the following :

Corollaire 5.3. *For $y \in Q_{\text{st}}$, there exists a neighborhood V of y in Q_{st} and a horizontal section b_V of $B(\alpha)$ on $Q \cap V$ such that $\theta(b_V(y'), w_\alpha(y')) \rightarrow 0$ for $y' \rightarrow y$ ($y' \in Q$).*

For $y \in Q$, the corollary just tells once more that the section w is continuous at y .

5.5

A partition T of S is *stable* if for each $C \in T$, $\mu(T)_C := \sum_{s \in C} \mu_s < 1$.

Define $\alpha(T)_C = \exp(2\pi i \mu(T)_C) = \prod_{s \in C} \alpha_s$. For T stable, let $M_T \subset M_{\text{st}}$ be the space of maps y from S to $\mathbb{P}$, such that $y(s) = y(t)$ if and only if s and t are in the same coset of T . It is the inverse image in M_{st} of a subspace Q_T of Q_{st} . The M_T (resp. Q_T) for T stable form a partition of M_{st} (resp. Q_{st}) and each M_T (resp. Q_T) can be identified with the analogue $M(T)$ of M (resp. $Q(T)$ of Q) for T , $\mu(T)$, and its closure with the analogue $M_{\text{st}}(T)$ (resp. $Q_{\text{st}}(T)$) of M_{st} (resp. Q_{st}).

Fix m, U, L as in (6.1), with $m \in M_T$. On $M_T \cap U$, the $H_c^1(P_y, L_y)$ form a local system. By (6.2.1), this local system, which is trivial as U is small enough, extends as a trivial sublocal system of $R^1 \pi_! L_U$. The ω_y ($y \in M_T \cap U$) define a non-zero holomorphic section of this local system on $M_T \cap U$. The corresponding ray can be identified with the section $w_{\alpha(T)}$ of $B(\alpha(T))$ on $M(T)$.

On Q_{st} , this gives us in a neighborhood V of $y \in Q_T$:

- a) A horizontal sub-ball-bundle of $B(\alpha)$, which extends as a constant horizontal bundle on V . Its restriction to Q_T can be identified with $B(\alpha(T))$ on $Q(T)$.

b) Via this identification, the horizontal section b_V of (6.8) is the horizontal section through $w_{\alpha(T)}(y)$.

Proposition 5.4. *If μ satisfies condition INT of (3.11) : $0 < \mu_s < 1$, $\sum \mu_s = 2$ and for all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer, then any subset $C \subset S$ for which $\sum_{s \in C} \mu_s < 1$ has at most three elements, and for any such subset with at least two elements, $(1 - \sum_{s \in C} \mu_s)^{-1}$ is an integer.*

The condition INT implies that $\mu_s + \mu_t$ is always $> 1/2$ for $s \neq t$. Averaging over the pairs in $C \subset S$, we get that for $\text{card}(C) \geq 2$, the mean of the μ_s ($s \in C$) is $> 1/4$. From $\sum_{s \in C} \mu_s < 1$ there results $\text{card}(C) < 4$.

The case $\text{card}(C) = 2$ is trivial by INT. Suppose $C = \{a, b, c\}$. The relations

$$1 - \mu_a - \mu_b = \frac{1}{n_{ab}}, \quad 1 - \mu_b - \mu_c = \frac{1}{n_{bc}}, \quad 1 - \mu_c - \mu_a = \frac{1}{n_{ca}}$$

sum up to $3 - 2(\mu_a + \mu_b + \mu_c) = \frac{1}{n_{ab}} + \frac{1}{n_{bc}} + \frac{1}{n_{ca}}$, hence $\frac{1}{n_{ab}} + \frac{1}{n_{bc}} + \frac{1}{n_{ca}} > 1$.

We now use the elementary fact that a triple of integers whose reciprocals have their sum strictly greater than 1 must be one of

$$(2, 2, n), \quad (2, 3, 3), \quad (2, 3, 4), \quad (2, 3, 5).$$

The excesses of the sums over 1 are respectively $1/n$, $1/6$, $1/12$, $1/30$.

It follows at once that $(1 - \mu_a - \mu_b - \mu_c)^{-1}$ is an integer.

Corollaire 5.5. *If μ satisfies INT and T is a stable partition of S (cf. (6.9)), the family $\mu(T)$ defined by*

$$\mu(T)_C = \sum_{s \in C} \mu_s \quad (C \in T)$$

again satisfies INT.

6 Semi-stable points

6.1

The notation and assumptions (4.0) are in force in this section ; we assume also $N \geq 4$. Let $(S(1), S(2))$ be a partition of S , such that $\sum_{s \in S(i)} \mu_s := 1$. We want to investigate the asymptotic behavior on Q of w_α , for y converging to a point in $Q_{0\text{usp}}$ of type $(S(1), S(2))$.

Fix $a \in S(1)$ and $b \in S(2)$, identify $\mathbb{P}$ with $\mathbb{P}^1$, and let W be the space of $y \in \mathbb{P}^S$ such that $y(a) = 0$, $y(b) = \infty$, and

$$\sup\{|y(s)| \mid s \in S(1)\} < \inf\{|y(s)| \mid s \in S(2)\}$$

(as in section 4 where we wrote V instead of W). We put $A(y) := \sup\{|y(s)| \mid s \in S(1)\}$, $B(y) = \inf\{|y(s)| \mid s \in S(2)\}$, $C(y) = A(y)/B(y)$. The quotient map $\pi : W \rightarrow Q_{\text{sst}}$ sends a point y with $C(y) = 0$ to the semistable point J of type $(S(1), S(2))$ and identifies $(W - \pi^{-1}(J))/\mathbb{C}^*$ with a punctured neighborhood of J . One has convergence of $\pi(y)$ to J if and only if $C(y)$ converges to 0.

On P_W , there is, up to a unique isomorphism, a unique rank one local system L , trivialized on the annulus $A(y) < |z| < B(y)$ by a section e , and which for $y \in M$, has

fiberwise monodromy α (hence trivial monodromy around the annulus, as $\prod_{s \in S(i)} \alpha_s = 1$). A model of it is the local system of constant multiples of

$$\prod_{s \in S(1)} (z - y(s))^{\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{\mu_s}.$$

We normalize the Hermitian form $(\cdot, \cdot)$ on L by $(e, e) = 1$.

For any y , fix R such that $A(y) < R < B(y)$, and let γ be the L -valued homology class represented by the cycle $|z| = R$ in P_y , positively oriented and provided with e , the trivializing section of L . It can also be viewed as a class in $H_1^{\text{lf}}(P_y, L_y)$. For $y = j$, defined by $j(S(1)) = 0$, $j(S(2)) = \infty$, it is a generator of this H_1^{lf} . The γ provide a horizontal section of $R^1\pi_!L$. Since $(\omega_j, \omega_j) = 0$, the line (ω_j) spanned by ω_j is in the boundary of the ball $B^+(\alpha)$ and the functions d_{ω_j} of (5.3) provide a horizontal family of “distance to $\mathbb{P}\omega_j$ ” functions. We will use them to compare $\mathbb{P}\omega_j$ with the holomorphic section w_α of $B^+(\alpha)$.

Proposition 6.1. *For y variable in W , if $C(y)$ tends to 0, the $d_{\omega_j}(w_\alpha(y))$ tend to 0.*

Démonstration. Fiber by fiber w_α is the line spanned by the cohomology class of

$$\omega_y = \prod_{s \in S(1)} (z - y(s))^{-\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{-\mu_s} \cdot dz \cdot e.$$

The form ω_y , as a function of y , is invariant by the action of $\mathbb{C}^*$ on (W, L) . It hence suffices to treat the case where $A(y) \rightarrow 0$, $B(y) \rightarrow \infty$. In that case, the numerator of $d_{\omega_j}(w_\alpha)$ clearly tends to 1 : it is the absolute value of

$$\frac{1}{2\pi i} \int_{|z|=R} \prod_{s \in S(1)} (z - y(s))^{-\mu_s} \cdot \prod_{s \in S(2)} \left(1 - \frac{z}{y(s)}\right)^{-\mu_s} \cdot \frac{dz}{z}$$

which tends to 1.

We have to prove that $(\omega_y, \omega_y) \rightarrow \infty$. Indeed,

$$(\omega_y, \omega_y) = \int_{\mathbb{C}} \prod_{s \in S(1)} |z - y(s)|^{-2\mu_s} \cdot \prod_{s \in S(2)} \left|1 - \frac{z}{y(s)}\right|^{-2\mu_s} |dz \wedge d\bar{z}|$$

and this integral is bounded below by

$$\int_{A(y) < |z| < B(y)} |z|^{-2} |dz \wedge d\bar{z}| = 2\pi \cdot 2 |\log C(y)|.$$

□

On Q , the proposition (7.2) has the following corollary :

Corollaire 6.2. *For each semi-stable point y of Q_{usp} , there is on a neighborhood V of y a horizontal family of functions d on the balls $B(\alpha)_{y'}$ ($y' \in V \cap Q$)—each a “distance to a point of $\partial B(\alpha)_y$ ” function—such that $d(w_\alpha(y')) \rightarrow 0$ for $y' \rightarrow y$.*

7 Extending $\tilde{w}$ by continuity

Let X, Y be complex algebraic varieties which are separated as algebraic varieties, i.e. which are Hausdorff in their complex topology.

7.1

One knows that any quasi-finite map $f : X \rightarrow Y$ —i.e. a map whose fibers are finite—admits a factorization $X \xrightarrow{j} \mathcal{X} \xrightarrow{\bar{f}} Y$, with j an open embedding, and $\bar{f}$ a finite map (= quasi-finite and proper). If X and Y are connected and normal, of the same dimension, there is an unique such factorization for which $\mathcal{X}$ is normal. One calls this $\mathcal{X}$ the normalization of Y in X .

For more general spaces, and for some maps with totally disconnected but possibly infinite fibers, a simple topological generalization of this construction has been given by R. H. Fox [?].

The spaces X and Y are taken to be Hausdorff and locally connected. A continuous map $f : X \rightarrow Y$ is called a *spread* if, for any x in X , one obtains a fundamental system of neighborhoods of x by taking, for each neighborhood V of $f(x)$, the connected component of $f^{-1}(V)$ containing x . A spread f is called *complete* if, for any $y \in Y$,

$$f^{-1}(y) \cong \varprojlim \pi_0(f^{-1}(V)),$$

the projective limit being taken on the neighborhoods V of y .

Fox proves that each spread $f : X \rightarrow Y$ can be embedded in a complete spread $\bar{f} : \bar{X} \rightarrow Y$ with the following universal property : any commutative diagram of solid arrows

$$\begin{array}{ccc} X & \longrightarrow & X' \\ f \downarrow & \circlearrowleft & \downarrow f' \\ Y & \xlongequal{\quad} & Y \end{array}$$

with f' a complete spread, can be completed as shown. For any $y \in Y$, one has

$$\bar{f}^{-1}(y) \cong \varprojlim \pi_0(f^{-1}(V)),$$

the projective limit being taken on the neighborhoods V of y . It is clear from the universal property that $\bar{X}$ is unique up to unique isomorphism, and functorial. We will call it the *completion of X over Y* .

It contains X as a dense subspace, and the topology of X is induced by that of $\bar{X}$. The space $\bar{X}$ is locally connected.

7.2

We define a space X to be connected if it is non-empty and has no proper open and closed subset.

A map $f : X \rightarrow Y$ is called a *covering map* if each $y \in Y$ has a neighborhood V for which there exists a discrete set D (possibly empty), and a commutative diagram

$$\begin{array}{ccc} D \times V & \longrightarrow & f^{-1}(V) \\ & \searrow & \downarrow \\ & & V \end{array}$$

If Y is connected and X non-empty, such a map is onto.

A map $f : X \rightarrow Y$ is called a *local homeomorphism* at $x \in X$ if x has a neighborhood U which maps homeomorphically onto a neighborhood of $f(x)$. The map f is called a *local homeomorphism* if this holds at all points $x \in X$.

Assume that X is a connected covering space of a connected locally simply connected open subset U of Y . The composed map $X \rightarrow U \hookrightarrow Y$ is then a spread. Assume further that each y in Y has a fundamental system $\mathcal{y}$ of open neighborhoods such that

$$\text{for } V \text{ in } \mathcal{y}, V \cap U \text{ is connected;} \quad (7.1)$$

$$\text{for } V' \subset V'' \text{ in } \mathcal{y}, \pi_1(V' \cap U) \rightarrow \pi_1(V'' \cap U). \quad (7.2)$$

As f is a local homeomorphism, X is open in $\bar{X}$. As X is a covering of U , one has further $\bar{X} = f^{-1}(U)$.

Fix a base point $o \in U$. One knows that the functor $(g : \tilde{U} \rightarrow U) \mapsto g^{-1}(o)$ is an equivalence of the category of coverings of U with that of discrete sets on which $\pi_1(U, o)$ acts. In particular, the connected components of $\tilde{U}$ correspond to the orbits of $\pi_1(U, o)$ on $g^{-1}(o)$. For W open in U , connected and containing o , the same applies to W and the restriction functor $(g : \tilde{U} \rightarrow U) \mapsto (g^{-1}(W) \rightarrow W)$ from coverings of U to coverings of W corresponds to the functor : restriction of the action of $\pi_1(U, o)$ to $\pi_1(W, o)$.

In particular, if $\tilde{U}$ corresponds to $\pi_1(U, o)/K$, $\pi_0(g^{-1}W) = \pi_1(W, o) \backslash \pi_1(U, o)/K$. This applies to X . If a lifting $\tilde{o}$ of o is chosen in X , the map $\pi_1(U, o) \rightarrow f^{-1}(o) : \sigma \mapsto \sigma \cdot \tilde{o}$ identifies $f^{-1}(o)$ with a suitable homogeneous space $\pi_1(U, o)/K$ of $\pi_1(U, o)$.

Fix $y \in Y$, and a neighborhood $V \in \mathcal{y}$. Let us assume at first that $o \in V$. The decomposition group D_V is then defined as the image of $\pi_1(V \cap U, 0)$ in $\pi_1(U, 0)$ and one has

$$\pi_0(f^{-1}(V \cap U)) = D_V \backslash \pi_1(U, o)/K.$$

For $V' \subset V$ in $\mathcal{y}$, (8.2.1) and (8.2.2) imply that $\pi_0(f^{-1}(V' \cap U)) \rightarrow \pi_0(f^{-1}(V \cap U))$ hence

$$f^{-1}(y) = D_V \backslash \pi_1(U, o)/K. \quad (7.3)$$

For a $\sigma \in \pi_1(U, o)/K$, $\sigma \cdot \tilde{o}$ is a lifting of o in the connected component $f^{-1}(V)_\sigma$ of $f^{-1}(V)$ corresponding to the double coset $D_V \sigma K$. If we use it as a base point, the fibre at 0 of the covering $f^{-1}(V)_\sigma$ of $V \cap U$ is identified with $D_V \sigma K / K \cong D_V / D_V \cap \sigma K \sigma^{-1}$. To the component $f^{-1}(V)_\sigma$ there corresponds a point $\tilde{x}$ in $f^{-1}(y)$. The $f^{-1}(V)_\sigma \cap f^{-1}(V')$, for V' a neighborhood of y , are the trace on X of a fundamental system of neighborhood of $\tilde{x}$ in $\bar{X}$.

For o not necessarily in V , one chooses o' in $V \cap U$ and a path p from o to o' in U . This path lifts as a path from $\tilde{o}$ to some lifting $\tilde{o}'$ of o' . It defines an isomorphism of $\pi_1(U, 0)$ with $\pi_1(U, o')$ via which the above constructions, which make sense for o' , can be pushed back into $\pi_1(U, o)$. For instance, the decomposition group $D_V \subset \pi_1(U, o)$ is defined as the image of the decomposition group $D_V \subset \pi_1(U, o')$. It depends on the path p . One still has (8.2.3).

(8.2.4) A covering map $f : X \rightarrow U$ is normal if the $\pi_1(U, o)$ homogeneous space $f^{-1}(o)$ is isomorphic to a homogeneous space $\pi_1(U, o)/K$, with K a normal subgroup. If $f : X \rightarrow U$ is a normal covering map, the group $\pi_1(U, 0)$ acts on X by deck-transformations, and this action extends to $\bar{X}$ by functoriality. We leave to the reader to check that, under the conditions (8.2.1), (8.2.2), one has

1. $\pi_1(U, o) \backslash \bar{X} \xrightarrow{\sim} Y$;
2. for each $y \in Y$, the stabilizer in $\pi_1(U, o)$ of a point $\tilde{x} \in \bar{f}^{-1}(y)$ is the conjugate of the decomposition group D_V determined by a path in U corresponding to $\tilde{x}$.

Remarque 7.1. Let K denote the kernel of the action of $\pi_1(U, o)$ on X . Then $\pi_1(U, o)/K$ acts properly discontinuously on X but not necessarily on $\bar{X}$.

7.3

Let w be a continuous function from X to a topological space B . We assume B regular, i.e. that any neighborhood of any point contains a closed neighborhood. The function w then extends as a continuous function on $\bar{X}$ if and only if, for any $\tilde{y} \in \bar{Y}$, $w(x)$ has a limit for $x \rightarrow \tilde{y}$ in X (convergence of w on the filter of traces on X of neighborhoods of $\tilde{y}$ in $\bar{X}$).

In the situation (8.2) (with $o \in V$ for simplicity), one can treat w as a multivalued function on U . If $\tilde{x}$ in $f^{-1}(y)$ corresponds to the connected component $f^{-1}(V)_\sigma$ of $f^{-1}(V)$, the restriction of w to $f^{-1}(V)_\sigma$ is a multivalued function w_σ on $V \cap U$. That $w(x)$ has a limit for $x \rightarrow \tilde{x}$ means that there is b in B , and for each neighborhood W of b a neighborhood V' of y , such that all determinations of $w_\sigma(u)$, for $u \in V' \cap U$, are in W .

Exemple 7.2. If we take for Y the unit disk $D \subset \mathbb{C}$, for U the punctured unit disk D^* and for X the universal covering of D^* , the completion of X over D is deduced from X by adding one point above 0 : $\bar{X} = X \cup \{0\}$. For α , real > 0 , the function z^α has a limit for $z \rightarrow 0$ in X , but it has no limit for non real α , with $(\alpha) > 0$. Similarly, $z^\alpha \log z$ has no limit.

7.4

With the assumptions (8.2.1) and (8.2.2) on (Y, U) , let π be the flat fiber space on U defined by an action θ of $\pi_1(U, o)$ on a regular topological space B , and let (X, o) be the smallest covering of (U, o) on which π becomes trivial : one has $f^{-1}(o) = \pi_1(U, o)/(\theta)$. A continuous section w of π defines by pull-back to X an equivariant continuous map from X to B (and conversely). We write w for this map. When does it extend to $\bar{X}$? Unravelling (8.3), we find that w has a limit for $x \rightarrow \tilde{x}$ above y if and only if there is a horizontal section b of π over the trace on U of a neighborhood $V \in \mathcal{V}_y$ of y , toward which w converges for $y' \rightarrow y$ ($y' \in U$), in the following sense. For $y_1 \in V \cap U$ and W_1 a neighborhood of $b(y_1)$ in the fibre $\pi^{-1}(y_1)$ of π at y_1 , there is a neighborhood V' of y such that for any $y_2 \in V' \cap U$, and for W_2 a neighborhood of $b(y_2)$ in the fiber of π at y_2 , obtained by horizontal transport of W_1 along any path in $V \cap U$ from y_1 to y_2 , one has $w(y_2) \in W_2$. This holds if b has a system of horizontal neighborhoods W which are fiber arbitrarily small, such that $w(y_1)$ is in W for y_1 close enough to y . Of course, b defines the limit of w .

In the rest of section 8, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are in force.

7.5

As in (3.8), fix a base point $o \in Q$, and let $p : (\tilde{Q}, o) \rightarrow (Q, o)$ be the smallest covering of (Q, o) on which the flat projective space bundle $B(\alpha)_Q$ becomes trivial. For $\tilde{Q} \subset Y \subset Q_{\text{sst}}$ and Y open in Q_{sst} , the composed map $\tilde{Q} \rightarrow Y$ is a spread. We will write $\tilde{Y}$ for the completion of $\tilde{Q}$ over Y . The assumptions (8.2.1) and (8.2.2) are satisfied by (Y, Q) .

It follows from (8.2.4) that

$$\pi_1(Q, o) \text{ acts on } \tilde{Q}_{\text{sst}}; \quad (7.4)$$

$$\pi_1(Q, o) \backslash \tilde{Q}_{\text{sst}} = Q_{\text{sst}}. \quad (7.5)$$

Proposition 7.3. *The map $\tilde{w}$, of (3.8) extends as a continuous map from $\tilde{Q}_{\text{sst}}$ to the closed ball $\overline{B^+(\alpha)_o}$, provided with the topology (5.4). The inverse image of $\partial B^+(\alpha)_o$ consists of the stable points.*

Démonstration. We are in the situation envisioned in (8.5), if we take for the flat fiber space with fiber $\overline{B^+(\alpha)}$, completing $B^+(\alpha)$. That the extendability criterion of (8.6) applies is (6.8) for y stable and (7.3) for y semi-stable. Further, the limit occurring in criterion (8.5) lies in $\partial B^+(\alpha)_o$ if and only if y is in Q_{usp} . $\square$

The extended map $\tilde{Q}_{\text{sst}} \rightarrow \overline{B^+(\alpha)_o}$ is also denoted $\tilde{w}$. It is $\pi_1(Q, o)$ -equivariant.

7.6

For T a stable partition of S (cf. (6.9)), the system of the $Q_{T'}$, for T' finer than T , is topologically constant along Q_T , and the inverse image $\tilde{Q}_T$ of Q_T in $\tilde{Q}_{\text{sst}}$ is hence a covering of Q_T ($\cong Q_*(T)$). By (6.9), on each connected component $\tilde{Q}_T^c$ of $\tilde{Q}_T$, the flat ball bundle $B(\alpha(T))^+$ trivializes, and can be identified with a flat sub-ball bundle of the constant bundle $B(\alpha)^+$, in such a way that

$$\tilde{w}|_{\tilde{Q}_T^c} = w_{\alpha(T)}. \quad (7.6)$$

From (3.9), we hence get :

Proposition 7.4. *For any component $\tilde{Q}_T^c$ of $\tilde{Q}_T$, $\tilde{w}|_{\tilde{Q}_T^c}$ is an étale map from $\tilde{Q}_T^c$ to a sub-ball of $B(\alpha)^+$.*

8 Codimension 1 'etateness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

Let T be a stable partition of S (cf. (6.9)), with $\text{card } T = \text{card } S - 1$; one coset of T has two elements and all others only one. Our aim in this section is to describe the flat bundle $B(\alpha)_Q$ over Q (cf. (3.7)) and the holomorphic section w_α in the neighborhood of a point in $Q_T \subset Q_{\text{st}}$ (cf. (6.9)).

8.1

The subspace Q_T of Q_{sst} , defined in (6.9), is a locally closed purely one codimensional complex submanifold of the complex manifold Q_{sst} and $Q \cap Q_T$ is open in Q_{st} . The monodromy around Q_T is the following conjugacy class in $\pi_1(Q)$.

Let $\varphi : D \rightarrow Q \cup Q_T$ be an embedding of the unit disc with $\varphi(D)$ transversal to Q_T and $\varphi^{-1}(Q_T) = \{0\}$; φ induces an embedding of $D^* := D - \{0\}$ in Q . The fundamental group $\pi_1(D^*)$ is canonically isomorphic to $\mathbb{Z}$, generated by the loop $[0, 1] \rightarrow D^* : t \mapsto z_0 \cdot \exp(2\pi it)$. The decomposition group at $\varphi(0)$ is the image of $\pi_1(D^*)$ in $\pi_1(Q)$ and the monodromy around Q_T is defined as the image of the positive generator of $\pi_1(D^*)$. Both are well defined up to conjugacy and independent of φ . Let us fix a base point o in Q . The fundamental group $\pi_1(Q, o)$ acts on $B(\alpha)_o$. We will compute the action of the monodromy around Q_T .

Let $\{s_1, s_2\}$ be the two element coset of T and choose additional elements $s_3, s_4 \in S$. We assume that $\mathbb{P}$ is the standard projective line $\mathbb{P}^1$ and we choose in it two distinct points $b, c \neq 0$. Let M_0 be the space of injective maps $y : S \rightarrow \mathbb{P}$ with $y(s_1) = 0$, $y(s_2) = b$, $y(s_4) = c$. Let M_{0T} be the space of maps $S \rightarrow \mathbb{P}$ with $y(s) = y(t)$ if and only if s and t are in the same coset of T , and $y(s_1) = y(s_2) = 0$, $y(s_3) = b$, $y(s_4) = c$. The quotient map $M_{\text{st}} \rightarrow Q_{\text{st}}$ induces isomorphisms $M_0 \xrightarrow{\sim} Q$, $M_{0T} \xrightarrow{\sim} Q_T$ and $M_0 \cup M_{0T} \xrightarrow{\sim} Q \cup Q_T$. We will work on $M_0 \cup M_{0T}$, rather than on $Q \cup Q_T$.

The monodromy around Q_T is represented by the following loop in M_0 : it starts from a point y_0 with, for some r , $y_0(S_1)$ in the disc D_r of radius r around 0 and the $y_0(s_i)$ ($i \neq 1, 2$) outside of this disc ; it is given by the map from $[0, 1]$ to M_0 , $t \mapsto y_t$, with $y_t(s_i)$ constant for $i \neq 2$, and $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$.

Above this loop, one can construct on P_{y_t} a local system L with fiberwise monodromy α , as follows : we fix a base point d on $\mathbb{P}$, outside of $y_0(S)$ and of D_r , and take for L the local system trivialized on the constant section d and with the required monodromy. The action of monodromy on $B(\alpha)_{y_0} = \mathbb{P}H^1(P_{y_0}, L)$ is deduced from the action of monodromy on $H^1(P_{y_0}, L)$.

Proposition 8.1. *In a suitable basis of $\mathbb{P}H^1(P_{y_0}, L)$, the above action of the monodromy around Q_T is given by a diagonal matrix $\text{diag}(\alpha_1\alpha_2, 1, \dots, 1)$.*

Démonstration. Set $S_1 = \{s_1, s_2\}$ and $S_2 = S - S_1$. Let T_1, T_2 be trees as in (2.5) with the vertices of T_i in S_i ($i = 1, 2$) and let $\beta : T_1 \amalg T_2 \hookrightarrow \mathbb{P}$ be an embedding with $\beta|_S = y_0$. We may and shall assume that $\beta|_{T_1}$ is the straight line segment from $y_0(s_1) = 0$ to $y_0(s_2)$ and that $\beta(T_2)$ is outside of the disc D_r . For each (open) edge a of T_1 or T_2 , we choose one of its orientations, and a non zero section $t(a)$ of β^*L above it. As in (2.4), (2.5), each edge then defines a closed L -valued current $t(a) \cdot \beta|_a$, and the cohomology classes of those currents form a basis of $H^1(P_{y_0}, L)$. It is the basis we will use.

To transport our basis elements around the loop y_t ($0 \leq t \leq 1$) horizontally, it suffices to deform β with $\beta_t|_S = y_t$, dragging along the $t(a)$. We will keep $\beta_t|_{T_2}$ fixed and take for $\beta_t|_{T_1}$ the family of straight line segments from $y_t(s_1) = 0$ to $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$. Outside of $D_r \subset \mathbb{P}$, the fibers P_{y_t} and L remain constant, and the $t(a) \cdot \beta_t|_a$ for a an edge of T_2 are, as currents, independent of t . It follows that the monodromy is trivial on the $t(a) \cdot \beta_t|_a$ for a an edge of T_2 .

Let a be the unique edge of T_1 . When $\beta_t(a)$ ($0 \leq t \leq 1$) has made a complete turn around 0, $t(a)$ is multiplied by $\alpha_1\alpha_2$. Indeed, if $p_t(x) : [0, 1] \rightarrow \mathbb{P}$ is a path from the base point d on $\mathbb{P}$ to a point d_t on $\beta_t(a)$, and if this path deforms with t so as to always avoid $\beta_t(S)$, one can take $t(a)$ to be obtained at d_t by transporting along p_t the trivialising section e of L at d . As the following picture shows, p_1 can be chosen to differ from p_0 by a loop around $y_0(s_1)$ and $y_0(s_2)$. Around this loop, L has monodromy $\alpha_1\alpha_2$, and the claim follows.

$$\begin{array}{ccccc}
& y_0(s_1) & y_0(s_2) & d & \\
t = 0 : & \bullet & \bullet & \bullet & \\
&) & | & / & \\
& & y_t(s_2) & & \\
& & \bullet & & \\
t \text{ small} : & y_0(s_1) & | & d & \\
& \bullet & | & \bullet & \\
& y_0(s_1) & y_0(s_2) & d & \\
t = 1 : & \bullet & \bullet & \bullet & \\
&) & | & / &
\end{array}$$

□

Remarque 8.2. In the computation (9.2), we have made use of the assumption $\mu_1 + \mu_2 < 1$ only in the weaker form : each $\mu_i < 1$. In the case $\mu_1 + \mu_2 > 1$, the monodromy when s_2 turns around s_1 would be given by the same formula, but the basis e in which it holds would have $(e_1, e_1) > 0$ instead of $(e_1, e_1) < 0$: it would be a pseudo-reflection with respect to a hyperplane not meeting the ball $B^+(\alpha)$.

More precisely, one checks that if the section $t(a)$ of L , for a the only edge of T_1 , is chosen of length one : $(t(a), t(a)) = 1$, then

$$(e_1, e_1) = 2\pi i \cdot \frac{2}{1 - \alpha_1 \alpha_2} (1 + \mu_1 + \mu_2 - 1)$$

We will not need this formula.

Remarque 8.3. For later computations, it will be more convenient to work with the basis of the dual space $H_1^{\text{lf}}(P - y_0(S), L^\vee)$ to $H^1(P - y_0(S), L)$ given by a choice of non zero sections $t^\vee(a)$ of $\beta^* L^\vee$ above the edges of T_1 or T_2 . The same arguments as in (9.2), show that on such a basis of the dual space, monodromy around Q_T acts by $\text{diag}((\alpha_1 \alpha_2)^{-1}, 1, \dots, 1)$.

8.2

Fix $y \in Q_T$. We will describe the section w_α of $B(\alpha)_Q$ near y . A neighborhood of y in $Q \cup Q_T$ and local coordinates are chosen as follows. First, one replaces $Q \cup Q_T$ by $M_0 \cup M_{0T}$, as in (9.1). It is convenient to choose b and c , in the definition of M_0 , so that $\infty \notin y(S)$ and that the discs of radius 1 around the points in $y(S)$ are disjoint. We do so. The chosen neighborhood U is then identified with the space of $m \in M_0 \cup M_{0T}$ such that $|m(s) - y(s)| < 1$. The chosen coordinates are the $m(s) - y(s)$ ($s \neq s_1, s_2, s_4$). If $S' := S - \{s_1, s_2, s_3, s_4\}$, this system of coordinates identifies U with $D \times D^{S'}$, and $U \cap M_0$ with $D^* \times D^{S'}$.

We choose L as in (9.1), using a base point d on $\mathbb{P}$ at distance > 1 from $y(S)$ —for instance d a large positive real number. In $U \cap M_0$, we choose a base point m_0 —for instance with $m_0(s_2) = -1$ and $m_0(s) = y(s)$ for $s \neq s_2$. We view w_α as a multivalued map from $U \cap M_0$ to $B(\alpha)_{m_0} = \mathbb{P}H^1(\mathbb{P} - m_0(S), L)$, and on $H^1(\mathbb{P} - m_0(S), L)$ we use the coordinates provided by a basis (9.4) of the dual space. The multivalued map w_α then lifts to the multivalued map with values in $\mathbb{C}^{N-3}$ having as coordinates (up to multiplicative

constants)

$$\int_{C_{s_1}} t^\vee(z) \cdot \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

as first coordinate and as the other coordinates the integrals

$$\int \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

taken along a path remaining outside the disc $|z| < 1$ from one $m(s)$ to another ($s \neq s_1, s_2$). When $m(s_2)$ turns once around $m(s_1)$, the first coordinate gets multiplied by $(\alpha_1 \alpha_2)^{-1} = \exp(2\pi i(1 - \mu_1 - \mu_2))$ while the others regain their value (9.4). This allows us to write the first coordinate as $(m(s_2) - m(s_1))^{1-\mu_1-\mu_2} I(m)$, with $I(m)$ an ordinary function on $U \cap M_0$. When m tends to $m_1 \in M_{0T}$, $I(m)$ tends to the limit

$$\int_{C_{s_1}} \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot z^{-\mu_1-\mu_2} dz,$$

which is non zero (2.17.1), hence $I(m)$ extends as a holomorphic function on U , invertible on $U \cap M_{0T}$. The other coordinates extend as holomorphic functions on U , having as restriction to $U \cap M_{0T}$ integrals

$$\int z^{-\mu_1-\mu_2} \cdot \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} dz.$$

By (3.9) applied to $\mu(T)$, these functions on $U \cap M_{0T}$ are the projective coordinates of an 'etale map from $U \cap M_{0T}$ to the projective $[(N-2)-3]$ -space (cf. (8.9)).

8.3

In suitable local coordinates, the multivalued map w_α from $U \cap M_0$ to $B(\alpha)_{m_0}$ has the form

$$(z_0, z) \mapsto (z_0^{1-\mu_1-\mu_2} I(z_0, z), J(z_0, z));$$

here, (z_0, z) are local coordinates on U coming from the product decomposition above, $U \cong D \times D^n$, $U \cap M_0 \cong D^* \times D^n$; the local coordinates on the image of U in $B(\alpha)_{m_0}$ are selected on an open set V with $V \cong D \times D^n$ and $\{0\} \times D^n$ lying in a hyperplane; and finally, $I : D \times D^n \rightarrow \mathbb{C}$ and $J : D \times D^n \rightarrow D^n$ are holomorphic maps, with $I(0, z) \neq 0$ and $J(0, z) : D^n \rightarrow D^n$

'etale.

At a point of $\{0\} \times D^n \subset D \times D^n$, the holomorphic functions $z_0 \cdot I(z_0, z)^{1/(1-\mu_1-\mu_2)}$ and $J(z_0, z)_i$ ($1 \leq i \leq n$) form a system of coordinates. In those coordinates, the multivalued map w_α is

$$(z_0, z_1, \dots, z_n) \mapsto (z_0^{1-\mu_1-\mu_2}, z_1, \dots, z_n).$$

Suppose $1 - \mu_1 - \mu_2$ is rational, set $1 - \mu_1 - \mu_2 = k/\ell$, in reduced terms. If $\tilde{D}$ is another copy of the disc, with coordinate u , mapping to D by $u \mapsto u^\ell$, the pull back to $\tilde{D} \times D^n$ of the multivalued map above is the holomorphic map

$$(u, z_1, \dots, z_n) \mapsto (u^k, z_1, \dots, z_n).$$

This map is

'etale in a neighborhood of $\{0\} \times D^n \subset \tilde{D} \times D^n$ if and only if $k = 1$, i.e. if and only if $(1 - \mu_1 - \mu_2)^{-1}$ is an integer.

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7.1

As in (3.8), fix a base point $o \in Q_*$ and let $p : (\tilde{Q}_*, o) \rightarrow (Q_*, o)$ be the smallest covering of (Q_*, o) on which the flat projective space bundle $B(\alpha)_q$ becomes trivial. For $Q_* \subset Y \subset Q_{*st}$ and Y open in Q_{*st} , the composed map $\tilde{Q}_* \rightarrow Y$ is a spread. We will write $\tilde{Y}$ for the completion of $\tilde{Q}_*$ over Y . The assumptions (8.2.1) and (8.2.2) are satisfied by (Y, Q_*) .

It follows from (8.2.4) that

$$\pi_1(Q_*, o) \text{ acts on } \tilde{Q}_{*st}, \quad (1)$$

$$\pi_1(Q_*, o) \backslash \tilde{Q}_{*st} = Q_{*st}. \quad (2)$$

Proposition 7.1. *The map $\tilde{w}$ of (3.8) extends as a continuous map from $\tilde{Q}_{*st}$ to the closed ball $B(\alpha)_0^+$, provided with the topology (5.4). The inverse image of $B(\alpha)_0^+$ consists of the stable points.*

Démonstration. We are in the situation envisioned in (8.5), if we take for $\mathcal{E}$ the flat fiber space with fiber $B(\alpha)^+$, completing $B(\alpha)^+$. That the extendability criterion of (8.6) applies is (6.8) for y stable and (7.3) for y semi-stable. Further, the limit occurring in criterion (8.5) lies in $\partial B(\alpha)^+$ if and only if y is in Q_{cusp} . $\square$

The extended map $\tilde{Q}_{*st} \rightarrow B(\alpha)_0^+$ is also denoted $\tilde{w}$. It is $\pi_1(Q_*, o)$ -equivariant.

7.2

For T a stable partition of S (cf. (6.9)), the system of the $Q_{T'}$, for T' finer than T , is topologically constant along Q_T , and the inverse image $\tilde{Q}_T$ of Q_T in $\tilde{Q}_{*st}$ is hence a covering of Q_T ($\cong Q_*(T)$). By (6.9), on each connected component $\tilde{Q}_T^0$ of $\tilde{Q}_T$, the flat ball bundle $B(\alpha(T))^+$ trivializes, and can be identified with a flat sub-ball bundle of the constant bundle $B(\alpha)^+$, in such a way that

$$\tilde{w} | \tilde{Q}_T^0 = I. \quad (3)$$

From (3.9), we hence get :

Proposition 7.2. *For any component $\tilde{Q}_T^0$ of $\tilde{Q}_T$, $\tilde{w} | \tilde{Q}_T^0$ is an étale map from $\tilde{Q}_T^0$ to a sub-ball of $B(\alpha)^+$.*

8 Codimension 1 étaleness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

Let T be a stable partition of S (cf. (6.9)), with $\text{card } T = \text{card } S - 1$; one coset of T has two elements and all others only one. Our aim in this section is to describe the flat bundle $B(\alpha)_q$ over $\tilde{Q}_*$ (cf. (3.7)) and the holomorphic section $\tilde{w}_q$ in the neighborhood of a point in $\tilde{Q}_T \subset \tilde{Q}_{*st}$ (cf. (6.9)).

8.1

The subspace Q_T of Q_{*st} , defined in (6.9), is a locally closed purely one codimensional complex submanifold of the complex manifold Q_{*st} and $Q_* \cup Q_T$ is open in Q_{*st} . The monodromy around Q_T is the following conjugacy class in $\pi_1(Q_*)$.

Let $\varphi : D \rightarrow Q_* \cup Q_T$ be an embedding of the unit disc with $\varphi(D)$ transversal to Q_T and $\varphi^{-1}(Q_T) = \{o\}$; φ induces an embedding of $D^* := D - \{o\}$ in Q_* . The fundamental group $\pi_1(D^*)$ is canonically isomorphic to $\mathbb{Z}$, generated by the loop $[0, 1] \rightarrow D^* : t \mapsto z_0 \cdot \exp(2\pi it)$. The decomposition group at $\varphi(o)$ is the image of $\pi_1(D^*)$ in $\pi_1(Q_*)$ and the monodromy around Q_T is defined as the image of the positive generator of $\pi_1(D^*)$. Both are well defined up to conjugacy and independent of φ .

Let us fix a base point o in Q_* . The fundamental group $\pi_1(Q_*, o)$ acts on $B(\alpha)_o$. We will compute the action of the monodromy around Q_T .

Let $\{s_1, s_2\}$ be the two element coset of T and choose additional elements $s_3, s_4 \in S$. We assume that P is the standard projective line $\mathbb{P}^1$ and we choose in it two distinct points $b, c \neq o$. Let M_0 be the space of injective maps $y : S \rightarrow P$ with $y(s_1) = o$, $y(s_3) = b$, $y(s_4) = c$. Let M_{0T} be the space of maps $S \rightarrow P$ with $y(s) = y(t)$ if and only if s and t are in the same coset of T , and $y(s_1) = y(s_2) = o$, $y(s_3) = b$, $y(s_4) = c$.

The quotient map $M_0^{st} \rightarrow Q_{*st}$ induces isomorphisms $M_0 \xrightarrow{\sim} Q_*$, $M_{0T} \xrightarrow{\sim} Q_T$ and $M_0 \cup M_{0T} \xrightarrow{\sim} Q_* \cup Q_T$. We will work on $M_0 \cup M_{0T}$, rather than on $Q_* \cup Q_T$.

The monodromy around Q_T is represented by the following loop in M_0 : it starts from a point y_0 with, for some r , $y_0(S_1)$ in the disc D_r of radius r around o and the $y_0(s_i)$ ($i \neq 1, 2$) outside of this disc; it is given by the map from $[0, 1]$ to M_0 , $t \mapsto y_t$, with $y_t(s_i)$ constant for $i \neq 2$, and $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$.

Above this loop, one can construct on $\mathbb{P}^1$ a local system L with fiberwise monodromy, as follows: we fix a base point d on P , outside of $y_0(S)$ and of D_r , and take for L the local system trivialized on the constant section d and with the required monodromy. The action of monodromy on $B(\alpha)_{y_0} = \mathbb{P}H^1(\mathbb{P}_{y_0}, L)$ is deduced from the action of monodromy on $H^1(\mathbb{P}_{y_0}, L)$.

Proposition 8.1. *In a suitable basis of $H^1(\mathbb{P}_{y_0}, L)$, the above action of the monodromy around Q_T is given by a diagonal matrix $\text{diag}(\alpha_1\alpha_2, 1, \dots, 1)$.*

Démonstration. Set $S_1 = \{s_1, s_2\}$ and $S_2 = S - S_1$. Let T_1, T_2 be trees as in (2.5) with the vertices of T_i in S_i ($i = 1, 2$) and let $\beta : T_1 \cup T_2 \hookrightarrow P$ be an embedding with $\beta|_S = y_0$. We may and shall assume that $\beta|_{T_1}$ is the straight line segment from $y_0(s_1) = o$ to $y_0(s_2)$ and that $\beta(T_2)$ is outside of the disc D_r . For each (open) edge a of T_1 or T_2 , we choose one of its orientations, and a non zero section $\ell(a)$ of β^*L above it. As in (2.4), (2.5), each edge then defines a closed L -valued current $\ell(a) \cdot \beta_*[a]$, and the cohomology classes of those currents form a basis of $H^1(\mathbb{P}_{y_0}, L)$. It is the basis we will use.

To transport our basis elements around the loop y_t ($0 < t < 1$) horizontally, it suffices to deform β with $\beta_t|_S = y_t$, dragging along the $\ell(a)$. We will keep $\beta_t|_{T_2}$ fixed and take for $\beta_t|_{T_1}$ the family of straight line segments from $y_t(s_1) = o$ to $y_t(s_2) = y_0(s_2) \cdot \exp(2\pi it)$. Outside of $D_r \subset P$, the fibers $\mathbb{P}_{y_t}$ and L remain constant, and the $\ell(a) \cdot \beta_{t*}[a]$ for a an edge of T_2 are, as currents, independent of t . It follows that the monodromy is trivial on the $\ell(a) \cdot \beta_*[a]$ for a an edge of T_2 .

Let a be the unique edge of T_1 . When $\beta_t(a)$ ($0 \leq t \leq 1$) has made a complete turn around o , $\ell(a)$ is multiplied by $\alpha_1\alpha_2$. Indeed, if $p_t(x) : [0, 1] \rightarrow P$ is a path from the base point d on P to a point d_t on $\beta_t(a)$, and if this path deforms with t so as to always avoid $\beta_t(S)$, one can take $\ell(a)$ to be obtained at d_t by transporting along p_t the trivialising

section e of L at d . As the following picture shows, p_1 can be chosen to differ from p_0 by a loop around $y_0(S_1)$ and $y_0(S_2)$. Around this loop, L has monodromy $\alpha_1\alpha_2$, and the claim follows.

$$\begin{array}{rcccl}
& y_0(s_1) & y_0(s_2) & d & \\
t=0 : &) & I / \dots\dots & & \\
& y_0(s_2) & & & \\
& y_0(s_1) & & & \\
t \text{ small} : & d & & & \\
& -\circ-\circ-\circ-\circ-\circ- & & & \\
& d & & & \\
t=1 : & \circ-\circ-\circ-\circ-\circ- & & &
\end{array}$$

□

Remarque 8.2. In the computation (9.2), we have made use of the assumption $\alpha_1 + \alpha_2 < 1$ only in the weaker form $\alpha_i \neq 1$. In the case $\alpha_1 + \alpha_2 > 1$, the monodromy when s_2 turns around s_1 would be given by the same formula, but the basis e in which it holds would have $(e_1, e_1) > 0$ instead of $(e_1, e_1) < 0$: it would be a pseudo-reflection with respect to a hyperplane not meeting the ball $B(\alpha)^+$.

More precisely, one checks that if the section $\ell(a)$ of L , for a the only edge of T_1 , is chosen of length one : $(\ell(a), \ell(a)) = 1$, then

$$(e_1, e_1) = 2i \frac{\sin \pi(\alpha_1 + \alpha_2) \cdot \sin \pi(1 - \alpha_1 + 1 - \alpha_2)}{\sin \pi(2 - \alpha_1 - \alpha_2)}.$$

We will not need this formula.

Remarque 8.3. For later computations, it will be more convenient to work with the basis of the dual space $H_1(\mathbb{P} - y_0(S), L^\vee)$ to $H^1(\mathbb{P} - y_0(S), L)$ given by a choice of non zero sections $\ell'(a)$ of $\beta^* L^\vee$ above the edges of T_1 or T_2 . The same arguments as in (9.2), show that on such a basis of the dual space, monodromy around Q_T acts by $\text{diag}((\alpha_1\alpha_2)^{-1}, 1, \dots, 1)$.

8.2

Fix $y \in \tilde{Q}_T$. We will describe the section $\tilde{w}_q$ of $B(\alpha)_q$ near y . A neighborhood of y in $\tilde{Q}_* \cup \tilde{Q}_T$ and local coordinates are chosen as follows. First, one replaces $\tilde{Q}_* \cup \tilde{Q}_T$ by $\tilde{M}_0 \cup \tilde{M}_{0T}$, as in (9.1). It is convenient to choose b and c , in the definition of M_0 , so that $\infty \notin y(S)$ and that the discs of radius 1 around the points in $y(S)$ are disjoint. We do so. The chosen neighborhood U is then identified with the space of $m \in M_0 \cup M_{0T}$ such that $|m(s) - y(s)| < 1$. The chosen coordinates are the $m(s) - y(s)$ ($s = s_1, s_2, s_3, s_4$). If $S' := S - \{s_1, s_2, s_3, s_4\}$, this system of coordinates identifies U with $D \times D^{S'}$, and $U \cap M_0$ with $D' \times D^{S'}$.

We choose L as in (9.1), using a base point d on P at distance > 1 from $y(S)$ — for instance d a large positive real number. In $U \cap M_0$, we choose a base point m_0 — for instance with $m_0(s_2) = -1$ and $m_0(s) = y(s)$ for $s \neq s_2$. We view $\tilde{w}_q$ as a multivalued map from $U \cap M_0$ to $B(\alpha)_{m_0} = \mathbb{P}H^1(\mathbb{P} - m_0(S), L)$, and on $H^1(\mathbb{P} - m_0(S), L)$ we use the coordinates provided by a basis (9.4) of the dual space. The multivalued map $\tilde{w}_q$ then lifts to the multivalued map with values in $\mathbb{C}^{S-3}$ having as coordinates (up to multiplicative constants)

$$\int_{C(s_1, s_2)} \ell'(z) \cdot \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

as first coordinate and as the other coordinates the integrals

$$\int_{[s]} \prod_{s \in S} (z - m(s))^{-\mu_s} \cdot dz$$

taken along a path remaining outside of the disc $|z| < 1$ from one $m(s)$ to another ($s \neq s_1, s_2$). When $m(s_2)$ turns once around $m(s_1)$, the first coordinate gets multiplied by $(\alpha_1 \alpha_2)^{-1} = \exp(2\pi i(1 - \mu_1 - \mu_2))$ while the others regain their value (9.4). This allows us to write the first coordinate as $(m(s_2) - m(s_1))^{1-\mu_1-\mu_2} \cdot I(m)$, with $I(m)$ an ordinary function on $U \cap M_0$. When m tends to $m_1 \in M_{0T}$, $I(m)$ tends to the limit

$$\int_{C(s_1, s_2)} \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot z^{-\mu_1-\mu_2} (z-1)^{-\mu_2} dz,$$

which is non zero (2.17.1), hence $I(m)$ extends as a holomorphic function on U , invertible on $U \cap M_{0T}$. The other coordinates extend as holomorphic functions on U , having as restriction to $U \cap M_{0T}$ integrals

$$\int_{[s]} z^{-\mu_1-\mu_2} \cdot \prod_{s \neq s_1, s_2} (z - m(s))^{-\mu_s} \cdot dz.$$

By (3.9) applied to μ_T , these functions on $U \cap M_{0T}$ are the projective coordinates of an étale map from $U \cap M_{0T}$ to the projective $[(N-2)-3]$ -space (cf. (8.9)).

8.3

In suitable local coordinates, the multivalued map $\tilde{w}_q$ from $U \cap \widetilde{M}_0$ to $B(\alpha)_q$ has the form

$$(z_0, z) \mapsto (z_0^{1-\mu_1-\mu_2} \cdot I(z_0, z), J(z_0, z));$$

here, (z_0, z) are local coordinates on U coming from the product decomposition above, $U \cong D \times D^n$, $U \cap \widetilde{M}_0 \cong D^* \times D^n$; the local coordinates on the image of U in $B(\alpha)_q$ are selected on an open set V with $V \cong D \times D^n$ and $\{0\} \times D^n$ lying in a hyperplane; and finally, $I : D \times D^n \rightarrow \mathbb{C}$ and $J : D \times D^n \rightarrow D^n$ are holomorphic maps, with $I(0, z) \neq 0$ and $J(0, z) : D^n \rightarrow D^n$ étale.

At a point of $\{0\} \times D^n \subset D \times D^n$, the holomorphic functions $z_0 \cdot I(z_0, z)^{1/(1-\mu_1-\mu_2)}$ and $J(z_0, z)_i$ ($1 \leq i \leq n$) form a system of coordinates. In those coordinates, the multivalued map $\tilde{w}_q$ is

$$(z_0, z_1, \dots, z_n) \mapsto (z_0^{1-\mu_1-\mu_2}, z_1, \dots, z_n).$$

Suppose $1 - \mu_1 - \mu_2$ is rational, set $1 - \mu_1 - \mu_2 = t/k$, in reduced terms. If $\tilde{D}$ is another copy of the disc, with coordinate u , mapping to D by $u \mapsto u^k$, the pull back to $\tilde{D} \times D^n$ of the multivalued map above is the holomorphic map

$$(u, z_1, \dots, z_n) \mapsto (u^t, z_1, \dots, z_n).$$

This map is étale in a neighborhood of $\{0\} \times D^n \subset \tilde{D} \times D^n$ if and only if $t = 1$, i.e. if and only if $(1 - \mu_1 - \mu_2)^{-1}$ is an integer.

8.4

Let Y be a complex analytic manifold, $\mathcal{F}$ a family of disjoint closed complex analytic submanifolds purely of complex codimension 1, $Z = \bigcup\{H; H \in \mathcal{F}\}$, and $\kappa : \mathcal{F} \rightarrow \mathbb{N}$ a positive integer valued function.

A continuous map $\phi : R \rightarrow Y$ is called a *branched cover with branch locus Z of order κ* if and only if :

1. ϕ induces a covering map from $\phi^{-1}(Y - Z)$ to $Y - Z$.
2. For any $H \in \mathcal{F}$, if we put $k := \kappa(H)$, any $y \in H$ admits a neighborhood V in Y such that the restriction of ϕ to any connected component of $\phi^{-1}(V)$ is topologically equivalent to the map

$$D^n \rightarrow D^n : (z_1, \dots, z_n) \mapsto (z_1^k, z_2, \dots, z_n).$$

A branched cover $\phi : R \rightarrow Y$ has a unique complex analytic manifold structure such that ϕ is holomorphic.

Proposition 8.4. *Let $\phi : R \rightarrow Y$ be a branched cover with branch locus Z . It follows from the definitions that ϕ is the completion of $\phi^{-1}(Y - Z) \rightarrow Y - Z$ over Y . Moreover, given any commutative diagram*

$$\begin{array}{ccc} \tilde{Q}_* & \xrightarrow{\quad} & R \\ & \searrow & \swarrow \phi \\ & Y - Z & \end{array}$$

with p and ϕ spreads, then there is a diagram

$$\begin{array}{ccc} \tilde{Q}_{*st} & \xrightarrow{\quad} & \tilde{R} \\ & \searrow & \swarrow \\ & Y & \end{array}$$

with $\tilde{p}$ and $\tilde{\phi}$ the completions of the spreads above, which is also commutative by the universal property of completions (8.1.1).

8.5

Let Q be an open set in a locally connected space Y . Let $p : \tilde{Q} \rightarrow Q$ be a spread and denote by $\bar{p}$ also the completion $\tilde{Y} \rightarrow Y$ of p . Then for any open set U in Y , the restriction of $\bar{p}$ to a connected component $\bar{p}^{-1}(U)_C$ of $\bar{p}^{-1}(U)$ is the completion of the spread $\bar{p}^{-1}(U)_C \cap \bar{p}^{-1}(U \cap Q) \rightarrow U \cap Q$.

8.6

We now assume that the μ_s are rational (cf. (2.12)). Let $\mathcal{T}_1$ denote the family of stable partitions T of S such that $\text{card } T = \text{card } S - 1$. For T in $\mathcal{T}_1$, and $\{s, t\}$ the coset of T with two elements, let k_T be the denominator of $1 - \mu_s - \mu_t$. Set

$$Q_1 = Q_* \cup \bigcup_{T \in \mathcal{T}_1} Q_T,$$

$\tilde{Q}_1$ = completion of $\tilde{Q}_*$ over Q_1 ,

$$\kappa(Q_T) = k_T \text{ for each } T \in \mathcal{T}_1.$$

Proposition 8.5. *The map $\tilde{Q}_1 \rightarrow Q_1$ is a branched cover with branch locus $\bigcup Q_T$ of order κ .*

Démonstration. By (9.8), the problem is local around each $y \in Q_T$. The claim then results from (9.2) and the fact that the order in the projective unitary group of $\text{diag}(\alpha_s \alpha_t, 1, \dots, 1)$ is the denominator of $1 - \mu_s - \mu_t$. $\square$

Proposition 8.6. *The map $\tilde{w} : \tilde{Q}_1 \rightarrow B^+(\alpha)_0$ is holomorphic.*

Démonstration. For $\tilde{w}$ is holomorphic on $\tilde{Q}_*$ and is continuous on $\tilde{Q}_1$ by Proposition (8.7). Since $\tilde{Q}_1 - \tilde{Q}_*$ has $\mathbb{C}$ -codimension 1, the assertion follows from the theorem on removable singularities. $\square$

Proposition 8.7. *Assume that*

$$\text{INT : } 0 < \mu_s < 1, \quad \sum \mu_s = 2 \text{ and for all } s, t \in S \text{ distinct and with } \mu_s + \mu_t < 1, \\ (1 - \mu_s - \mu_t)^{-1} \text{ is an integer.}$$

Then, $\tilde{w} : \tilde{Q}_1 \rightarrow B^+(\alpha)_0$ is étale.

Démonstration. This follows from the local description (9.6). $\square$

9 Proof of discreteness

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued.

9.1

We will give two proofs that when condition INT is satisfied, the image Γ of $\pi_1(Q_*, o)$ in $\text{PU}(1, N - 2)$ is discrete.

9.2

First proof. We follow the strategy explained in (2.13). The key point in proving that the projection $\tilde{Q}_{*st} \rightarrow Q_{*st}$ is locally (on $\tilde{Q}_{*st}$) finite-to-one is the following lemma.

Lemme 9.1. *Let D_1, D_2 and D_3 be three distinct lines in $\mathbb{C}^2$ passing through o , and n_1, n_2, n_3 integers > 1 such that $\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} \geq 1$. Fix a base point $b \in \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$, and consider the coverings of $(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3), b)$, with ramification index along D_i dividing n_i ($i = 1, 2, 3$) : the monodromy permutation of the sheets, when turning around D_i , is of order dividing n_i . Then the universal such covering is a finite covering.*

Démonstration. There exists a spherical triangle Δ with angles $\frac{\pi}{n_1}, \frac{\pi}{n_2}, \frac{\pi}{n_3}$. Let W be the Coxeter group generated by the reflections along its edges. The $w\Delta$ ($w \in W$) form a tessellation of the sphere, and hence

$$|W| = \text{area}(S^2)/\text{area}(\Delta) = 2 \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}.$$

Let W^+ be the subgroup of index two in W consisting of the orientation preserving elements. If we identify the sphere with the Riemann sphere metrized by its Fubini metric, W^+ becomes a finite subgroup of $\text{PU}(2)$, of order

$$|W^+| = \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}.$$

The quotient $\mathbb{P}^1/W^+$ is of genus 0 and hence is a projective line. Fix an isomorphism $\mathbb{P}^1/W^+ \cong \mathbb{P}^1$ and let $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1/W^+ \cong \mathbb{P}^1$ be the quotient map. The pull-back by f of the line bundle $\mathcal{O}(-1)$ on $\mathbb{P}^1$ is of degree the negative of $d = \deg(f) = |W^+|$. Fix an isomorphism $\mathcal{O}(-1) \cong f^*\mathcal{O}(-1)$. The group W^+ acts naturally on $(\mathbb{P}^1, f^*\mathcal{O}(-1))$. Let H be the group of all automorphisms (h, r) of $(\mathbb{P}^1, f^*\mathcal{O}(-1))$ with $(h, r) \mapsto h \in W^+$. It is a central extension of W^+ by μ_d , the group of roots of unity of order dividing d . As a space, the line bundle $\mathcal{O}(-1)$, minus the 0 section, is $\mathbb{C}^2 - \{0\}$, the bundle map being the natural projection $\mathbb{C}^2 - \{0\} \rightarrow \mathbb{P}^1$. The action of H on $\mathbb{C}^2 - \{0\}$ is induced by a linear action on $\mathbb{C}^2$: the contragredient of its action on $H^0(\mathbb{P}^1, \mathcal{O}(1))$. The quotient map of the space $(\mathbb{P}^1, f^*\mathcal{O}(-1))$ to its μ_d orbits may be identified with a d -th power map of $(\mathbb{P}^1, \mathcal{O}(-1))$ to $(\mathbb{P}^1, \mathcal{O}(-1))$, and the quotient map $g : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ to its H orbits is the composite of the d -th power map to the total space of $f^*\mathcal{O}(-1)$ minus its zero section, followed by the quotient map to W^+ orbits. Thus the quotient maps by H and W^+ give rise to a commutative diagram

$$\begin{array}{ccccc} \mathbb{C}^2 & \longrightarrow & \mathbb{C}^2 - \{0\} & \longrightarrow & \mathbb{P}^1 \\ g \downarrow & & \downarrow & & \downarrow \\ \mathbb{C}^2 & \longrightarrow & \mathbb{C}^2 - \{0\} & \longrightarrow & \mathbb{P}^1 \end{array}$$

The stability groups in W^+ of the vertices P_1, P_2, P_3 of the original spherical triangle are cyclic of order n_1, n_2, n_3 , and the points in the orbits of P_1, P_2, P_3 are the only points with a non trivial stabilizer (this results from the fact that the stabilizer in W of any point of Δ is generated by the reflections along the edges passing through the point ([4] V (3.3) prop. 1), or simply from the fact that Δ is a fundamental domain for W ([20])).

Let $d_i = f(P_i)$ and $D_i \subset \mathbb{C}^2$ be the corresponding line. One easily checks that g ramifies only along D_1, D_2, D_3 and that at each point above a point of $D_i - \{o\}$, the ramification index is exactly n_i . If X is a covering of $\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$ as in (10.3), its pull-back by g can hence be extended to unbranched covering of $\mathbb{C}^2 - \{o\}$. As $\mathbb{C}^2 - \{o\}$ is simply connected, each connected component of this covering of $\mathbb{C}^2 - \{o\}$ is trivial. This implies that the universal covering of type (10.3) is $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)) \rightarrow \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$. It is finite. $\square$

Remarque 9.2.

- (i) The proof showed that the universal covering (10.3) is of order $2 \left(\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} - 1 \right)^{-1}$.

It is a Galois covering, with group a central extension of W^+ by a cyclic group of order $|W^+|^{-1}$.

- (ii) If $\alpha_1 + \alpha_2 + \alpha_3 = 1 - \frac{1}{n_1}$, $\alpha_2 + \alpha_3 = 1 - \frac{1}{n_2}$ and $\alpha_3 + \alpha_1 = 1 - \frac{1}{n_3}$, one has $\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} > 1$ amounts to $\alpha_1 + \alpha_2 + \alpha_3 < 1$, and when it holds $(1 - \alpha_1 - \alpha_2 - \alpha_3)^{-1} \cdot |W^+|$ is an integer, as observed in (6.10).
- (iii) The fundamental group of $\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)$ is generated by three elements $\gamma_1, \gamma_2, \gamma_3$ (conjugate to a small positive loop around D_i) with relations those expressing that $\gamma_1 \gamma_2 \gamma_3$ (conjugate to a small loop around o , on a general line D through o) is central. To get H , one imposes the additional relations $\gamma_i^{n_i} = 1$.
- (iv) In the commutative diagram

$$\begin{array}{ccc} g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3)) & \hookrightarrow & \mathbb{C}^2 \\ \downarrow & & \downarrow g \\ \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3) & \hookrightarrow & \mathbb{C}^2 \end{array}$$

$\mathbb{C}^2$, at the upper right corner, is the completion of $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3))$ over $\mathbb{C}^2$ — at points of $\mathbb{C}^2 - \{o\}$ by (9.7.1) and at o by direct verification. Such a fact is generally true for a finite morphism g of normal analytic spaces. The completion of $g^{-1}(\mathbb{C}^2 - (D_1 \cup D_2 \cup D_3))$ over $\mathbb{C}^2$ thus acquires a structure of a normal (in fact non-singular) analytic space — indeed of a non-singular analytic manifold; this structure is in fact the unique one for which the projection to $\mathbb{C}^2$ is analytic.

Remarque 9.3. Let $J = J' \cap J'' \cap J'''$ be a finite index set. For $j \in J'$, let $n_1(j), n_2(j), n_3(j)$ satisfy $\frac{1}{n_1(j)} + \frac{1}{n_2(j)} + \frac{1}{n_3(j)} > 1$; for $j \in J''$, let $n(j)$ be any integer > 1 . We define

$$X(j) = \begin{cases} \mathbb{C}^2 - (D_1 \cup D_2 \cup D_3) \subset \mathbb{C}^2 & \text{for } j \in J', \\ \mathbb{C} - \{o\} \subset \mathbb{C} & \text{for } j \in J'', \\ \mathbb{C} & \text{for } j \in J'''. \end{cases}$$

The product X of the $X(j)$ is the complement of a divisor in $Y := (\mathbb{C}^2)^{J'} \times \mathbb{C}^{J'' \cup J'''}$. The covering $\tilde{X}$ of X , with ramification index along $\text{pr}_j^{-1}(D_i)$ dividing $n_i(j)$, ($j \in J'$), with ramification index along $\text{pr}_j^{-1}(\{o\})$ dividing $n(j)$ ($j \in J''$), and universal with respect to those properties, is the product of the corresponding coverings of each $X(j)$. Hence $\tilde{X}$ is a finite covering of X . The completion $\bar{Y}$ of $\tilde{X}$ over Y is similarly a product.

By (10.4) (iv), $\bar{Y}$ has a unique structure of a normal (in fact non-singular) analytic space for which the map to Y is analytic. If X_1 is a covering space of X and a quotient of $\tilde{X}$, then $X_1 = \tilde{X}/G$ with G a finite group, and the completion $\bar{Y}_1$ of X_1 over Y is $\bar{Y}/G$. It inherits the structure of a normal analytic space from $\bar{Y}$.

Similar remarks apply with X replaced by the trace on X of a ball around o in $(\mathbb{C}^2)^{J'} \times \mathbb{C}^{J'' \cup J'''}$.

Lemme 9.4. *If condition (INT) is satisfied, $\tilde{Q}_{*st}$ admits a structure of normal analytic space, such that each $y_0 \in Q_{*st}$ has an open neighborhood U whose inverse image in $\tilde{Q}_{*st}$ is a disjoint sum of finite ramified coverings of U .*

Démonstration. It suffices to show that each $y_0 \in Q_{*st}$ has an open neighborhood U whose inverse image in $\tilde{Q}_{*st}$ is a disjoint sum of finite (topological) coverings of $U \cap Q_*$ which, in suitable local coordinates at y_0 , are of the type considered in (10.5). This we proceed to show, using the control over ramification in codimension 1 provided by § 9.

Fix $y_0 \in Q_{*st}$, and let T be the corresponding stable partition of S . Fix a, b, c in distinct cosets. One can identify a neighborhood of y_0 in Q_{*st} with a subspace of P^S whose elements take prescribed values on a, b, c . We arrange these values so that $\infty \notin y_0(S)$. Near y_0 , we then have the following system of local coordinates, depending on the choice of a representative $d(C)$ in each coset C :

- a) the $y(d(C)) - y_0(d(C))$, for C not the coset of a, b or c ;
- b) for each coset C , the $y(e) - y(d(C))$ with $e \in C - \{d(C)\}$.

In terms of these coordinates, the condition for y near y_0 to be in Q_* is :

- a) for C a coset with three elements, $(y(e) - y(d(C)))_{e \in C - \{d(C)\}} \in \mathbb{C}^2$ is not on any of the lines $z_1 = 0, z_2 = 0, z_1 = z_2$;
- b) for C a coset with two elements, $y(e) - y(d(C)) \neq 0$ ($e \in C, e \neq d(C)$).

We recall that C has at most three elements (6.10.1), and that if C has three elements x, y, z , the sum of the reciprocal of the integers $(1 - \mu_x - \mu_y)^{-1}, (1 - \mu_y - \mu_z)^{-1}, (1 - \mu_z - \mu_x)^{-1}$ is > 1 (6.10).

For U a suitable open neighborhood of y in Q_{*st} , we can hence apply (10.5) to the covering of $U \cap Q_*$ induced by $\tilde{Q}_*$. We find that this covering breaks up into a disjoint sum of finite covering of the type required. $\square$

Lemme 9.5. *If condition INT is satisfied, the map $\tilde{w} : \tilde{Q}_{*st} \rightarrow B^+(\alpha)_0$ is étale.*

Démonstration. The map $\tilde{w}$ is holomorphic on $\tilde{Q}_*$ by (3.5) and continuous on $\tilde{Q}_{*st}$ by (8.7). It is hence holomorphic on $\tilde{Q}_{*st}$. By (6.9), it has an injective differential on the strata of a suitable analytic stratification. The fibers of $\tilde{w}$ hence have no component of dimension > 0 and $\tilde{w}$ is locally finite-to-one.

We know from § 9 that $\tilde{w}$ is étale outside of a closed analytic subset of Q_{*st} of complex codimension > 2 . By the purity of the branch locus theorem (the fact that the branch locus is always purely of codimension one), it follows that $\tilde{w}$ is étale everywhere. $\square$

9.3

Let us now assume, in addition to INT, that $Q_{*st} = Q_{*sst}$, i.e. that for no $S(1) \subset S$ is $\sum_{s \in S(1)} \mu_s = 1$. Choose a metric on Q_{*st} . We provide $\tilde{Q}_{*st}$ with the metric for which $d(x, y)$ is the infimum of the lengths of paths from x to y , the length being measured by its projection into Q_{*st} . What we need is a metric invariant by the action of $\pi_1(Q_*, o)$; any such metric will do. The space $\tilde{Q}_{*st}$ is locally compact, and the projection to Q_{*st} is open. Since Q_{*st} is assumed to be compact, there is a compact $K \subset \tilde{Q}_{*st}$ mapping onto Q_{*st} . Since $\tilde{w}$ is étale, there are numbers $r, R > 0$ such that for $k \in K$, the restriction of $\tilde{w}$ to the ball $B_r(k)$ of radius r around k in $\tilde{Q}_{*st}$ is an embedding, and

$$\tilde{w}(B_r(k)) \supset B_R(\tilde{w}(k)). \quad (4)$$

The $\pi_1(Q_*, o)$ translates of K cover $\tilde{Q}_{*st}$. The map $\tilde{w}$ being equivariant, and the action on the ball $B^+(\alpha)_0$ being via isometries, (10.8.1) holds for any $k \in \tilde{Q}_{*st}$. The map $\tilde{w}$ is hence a covering map. The ball being simply connected, $\tilde{w}$ is an isomorphism. This concludes the proof of (3.11) in the cocompact case ($Q_{*st} = Q_{*sst}$), following the strategy outlined there.

9.4

Our second proof will bypass lemma (10.6). The purity of the branch locus theorem can be deduced from the fact that, if Y is a complex submanifold of a complex manifold X of complex codimension ≥ 2 , and B a small ball around $y \in Y$, then $B - Y$ is simply connected. It is the latter fact which we will use directly to prove (10.7).

The reader who merely wants to get the idea of how we complete the proof when $Q_{*st} \neq Q_{*sst}$ can skip to Proposition (10.18.1).

9.5

Our second proof of (10.1.1) comes after proving that $\tilde{w}$ is étale on $\tilde{Q}_{*sst}$, that in turn is proved in stages.

Set $N = \text{card } S$. For i between 0 and $N - 3$, set

$$Q_i = \bigcup Q_T$$

where T runs through all stable partitions of cardinality $\geq \text{card } S - i$. Then $Q_0 = Q_*$, $Q_{N-3} = Q_{*st}$. Set $Q_{N-2} = Q_{*sst}$. Let $\tilde{Q}_i$ denote the completion of $\tilde{Q}_*$ over Q_i ($i = 1, \dots, N - 2$). We shall prove inductively that $\tilde{w}$ is étale on $\tilde{Q}_i$.

Proposition (10.17) below will be used repeatedly to show at each stage that as one adds on a submanifold, the extended map $\tilde{w}$ remains étale. We lead up to it via some general topological remarks, especially Proposition (10.15.1) and Corollary (10.15.4).

Proposition 9.6. *Let $\varphi : X \rightarrow Y$ be a continuous map with Y locally connected and X Hausdorff. Assume that each $y \in Y$ admits a neighborhood V such that each $x \in \varphi^{-1}(V)$ has an open neighborhood U with $\varphi(U) \supset V$ such that φ induces a homeomorphism from U to $\varphi(U)$. Then φ is a covering map.*

We first prove the

Lemme 9.7. *Let $\varphi : X \rightarrow Y$ be a continuous map with Y connected and X Hausdorff. Let X_1 and X_2 be open in X and such that φ induces homeomorphisms from X_i to Y , $i = 1, 2$. Then $X_1 = X_2$ or $X_1 \cap X_2 = \emptyset$.*

Démonstration. Let $s_i : Y \rightarrow X$ be the inverse of $\varphi \mid X_i$. The set of $y \in Y$ with $s_1(y) = s_2(y)$ is closed. It is also open, being the image of $X_1 \cap X_2$ by $\varphi : X_i \rightarrow Y$. It is hence the whole of Y or empty; i.e. $X_1 = X_2$ or $X_1 \cap X_2 = \emptyset$. $\square$

Proof of (10.11). For $V \subset Y$, let $\bar{\varphi}$ be the map induced by φ from $\varphi^{-1}(V)$ to V . We have to show that each $y \in Y$ has a neighborhood V such that $(\varphi^{-1}(V), \bar{\varphi}) \cong (V \times D, \text{pr}_1)$ with D discrete. Take V as in (10.11), open and connected. Replacing the U of (10.11) by $U \cap \varphi^{-1}(V)$, we find that each $x \in \varphi^{-1}(V)$ is contained in an open set U such that φ induces a homeomorphism from U to V . By (10.11.1) applied to $\varphi^{-1}(V) \rightarrow V$, they form a partition of $\varphi^{-1}(V)$ and the claim follows. $\square$

Proposition 9.8. *Let $\varphi : (X, o) \rightarrow (Y, p)$ be a continuous map of metric spaces, with $Y - \{p\}$ locally connected. We write d for the metric on both X and Y . Assume that there exists a neighborhood U of o such that for each neighborhood V of o , one has*

- a) *there exists $r > 0$ such that for all $x \in U - V$, the open ball $B_r(x)$ maps homeomorphically onto an open set of Y ;*

b) for all r there is $\varepsilon > 0$ such that for each $x \in U - V$, $\varphi B_r(x) \supset B_\varepsilon \varphi(x)$.

Then, there exist open neighborhoods X' of o and Y' of p such that $\varphi : X' - \varphi^{-1}(p) \rightarrow Y' - \{p\}$ is a covering map.

Roughly speaking, a), b) mean that on $U - \{o\}$, φ is a local homeomorphism with some uniformity. The uniformity is allowed to get worse for $x \approx o$. It is not assumed that o is isolated in $\varphi^{-1}(p)$. Under additional assumptions, this will be a consequence; see (10.13).

Proof of (10.12). Choose U as in (10.12). Select a neighborhood V of o and $\varepsilon(1) > 0$ such that $B_{\varepsilon(1)}(V) := \{x \in X \mid \exists v \in V d(x, v) < \varepsilon(1)\}$ is contained in U . If $B_r(o) \subset U$, one can take $r = r$ and $V = B_r(o)$. By a), b) there is ε and η such that

for all $x \in U - V$, φ induces a homeomorphism of $B_\varepsilon(x)$ with an open set of Y containing $B_\eta(\varphi(x))$.
(*)

Set $Y' = B_\eta(p)$ and take X' to be the set of all x with $\varphi(x) \in Y'$ such that either

(i) $x \in V$; or

(ii) there is $x_1 \in U - V$ with $d(x, x_1) < \varepsilon$, and $\varphi(x_1) \in Y'$.

In case (ii), $x \in B_\varepsilon(x_1) \cap \varphi^{-1}(Y') \subset X'$ and φ induces a homeomorphism of $B_\varepsilon(x_1) \cap \varphi^{-1}(Y')$ with Y' . This results from (*).

Fix $y \in Y' - p$ and let W be a neighborhood of y , in $Y' - \{p\}$, disjoint from a neighborhood of p . Then $\varphi^{-1}(W)$ is disjoint from a neighborhood of o and by a), b) for r small enough there is $\delta(2)$ such that

for all $x \in U$ with $\varphi(x) \in W$, φ induces a homeomorphism from $B_{\delta(2)}(x)$ to an open subset of Y containing $B_{\delta(2)}(\varphi(x))$.
(**)

We may and shall assume that $\delta(2) < \varepsilon$ and that $B_{\delta(2)}(y) \subset W$. Proposition (10.12) now follows from (10.11) applied to $X' - \varphi^{-1}(p) \rightarrow Y' - \{p\}$ and from the

Lemme 9.9. *Each x in $X' \cap \varphi^{-1}(B_{\delta(2)}(y))$ has an open neighborhood B' in X' which maps homeomorphically onto an open subset of Y containing $B_{\delta(2)}(y)$.*

As $x \in X'$, one of the conditions (i) (ii) holds. In the first case ($x \in V$), one has $B_{\delta(2)}(x) \subset U$, and we claim that $B_{\delta(2)}(x) \cap \varphi^{-1}(Y') \subset X'$. For if $x' \in B_{\delta(2)}(x)$ and $\varphi(x') \in Y'$, then either $x' \in X'$ by virtue of (i) or $x' \in U - V$ and $x' \in X'$ by virtue of (ii) with $x_1 = x'$. In the first case, $B' = B_{\delta(2)}(x) \cap \varphi^{-1}(Y')$ and one uses (*). In the second case, take $B' = B_\varepsilon(x_1) \cap \varphi^{-1}(Y')$ and use (*). $\square$

The assumptions of (10.12) remain valid if we replace X and Y by open neighborhoods of o and p . The X' and Y' of (10.12) can hence be chosen arbitrarily small.

From this, we will deduce the

Corollaire 9.10. *The point o has a fundamental system of open neighborhoods X' such that for Y' a suitable open neighborhood of p , $X' - \varphi^{-1}(p)$ is open and closed in $\varphi^{-1}(Y' - \{p\})$ and is a covering of $Y' - \{p\}$.*

First take X_0, Y_0 as in (10.12), and W a neighborhood of o such that $\bar{W} \subset X_0$. If X', Y' are as in (10.12), with $X' \subset W$, $Y' \subset Y_0$ and if $X'_1 := X_0 \cap \varphi^{-1}(Y')$, then both $X'_1 - \varphi^{-1}(p)$ and $X' - \varphi^{-1}(p) (\subset X'_1 - \varphi^{-1}(p))$ are coverings of $Y' - \{p\}$; hence $X' - \varphi^{-1}(p)$ is open and closed in $X'_1 - \varphi^{-1}(p)$. As $(X' - \varphi^{-1}(p)) \subset \bar{W} \subset X_0$, $X' - \varphi^{-1}(p)$ is also open and closed in $\varphi^{-1}(Y' - \{p\})$.

Corollaire 9.11. *If o is not isolated in X , and if p has a fundamental system of neighborhoods V such that $V - \{p\}$ is connected (in particular, non empty), then φ is open at o .*

If o is not isolated, X' cannot be contained in $\varphi^{-1}(p)$; for applying (10.12) a) to x in X' near o would contradict $\varphi(X') \subset \{p\}$. It follows that $\varphi(X' - \varphi^{-1}(p)) = Y' - \{p\}$ and hence that $\varphi = Y'$. Openness follows.

Proposition 9.12. *In addition to the hypothesis of (10.12), assume that $Y - \{p\}$ is locally simply connected and that*

- c) *p has a fundamental system of open neighborhoods $\mathcal{V}$, such that each $V - \{p\}$ (V in $\mathcal{V}$) is connected and that for $V \subset V'$, both in $\mathcal{V}$, $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(V' - \{p\})$.*
- d) *o has a fundamental system of open neighborhoods N with $N - \{o\}$ connected (non-empty).*

Then, one can find X' and Y' as in (10.12) such that $\varphi' : X' \rightarrow Y'$ has the following additional properties : Y' is in $\mathcal{V}$, $\varphi'^{-1}(\{p\}) = \{o\}$, $X' - \{o\}$ is connected, and X' is the completion of $X' - \{o\}$ above Y' .

Proof of (10.13). Shrinking X and Y , we may and do assume that Y is in $\mathcal{V}$, that $\varphi : X - \varphi^{-1}(p) \rightarrow Y - \{p\}$ is a covering map, and that φ is a local homeomorphism at each point $x \neq o$. Let N be an open neighborhood of o such that $N - \{o\}$ is connected.

Lemme 9.13. *$N - \varphi^{-1}(p)$ is connected.*

Let $A \subset N - \varphi^{-1}(p)$ be open and closed. Each $x \in \varphi^{-1}(p) - \{o\}$ has a neighborhood $W \subset N$ with $W - \{x\} \cong \varphi(W) - \{p\}$ connected. For such a W , $W - \{x\} = W - \varphi^{-1}(p)$ lies either in A or in its complement. It follows that the closure of A in $N - \{o\}$ is again open and closed and, by hypothesis, is empty or the whole of $N - \{o\}$. If it is empty (resp. the whole of $N - \{o\}$), A is empty (resp. the whole of $N - \varphi^{-1}(p)$).

Since $Y - \{p\}$ is locally connected, so is the covering space $X - \varphi^{-1}(p)$. Each connected component of $X - \varphi^{-1}(p)$ is open and closed in $X - \varphi^{-1}(p)$ and is a covering of $Y - \{p\}$. Let N_1 be the component of N containing $N - \varphi^{-1}(p)$.

Lemme 9.14. *$N_1 \cap \varphi^{-1}(p) = \{o\}$.*

Any $x \neq o$ in $\varphi^{-1}(p)$ has an open neighborhood W' such that φ induces a homeomorphism of W' with an open neighborhood of p in Y . Take V in $\mathcal{V}$ with $V \subset \varphi(W')$ and define $W := W' \cap \varphi^{-1}(V)$. One has $\varphi : W \xrightarrow{\sim} V$.

The assumption $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(Y - \{p\})$ implies that for any covering map $\psi : Z \rightarrow Y - \{p\}$, one has $\psi_*(\psi^{-1}(V - \{p\})) \xrightarrow{\sim} \pi_0(Z)$. In particular, as N_1 is connected, so is $N_1 \cap \varphi^{-1}(V)$. Both $N_1 \cap \varphi^{-1}(V)$ and $W - \{x\}$ are connected coverings of $V - \{p\}$, contained in the covering $\varphi^{-1}(V - \{p\})$. They are not equal : x is the only point of $\varphi^{-1}(p)$ in W , while $o \in N_1$. They are hence disjoint and $x \notin N_1$.

Lemma (10.13.2) is proved.

We take $Y' = Y$ and $X' := N_1$. It follows from (10.13.2) that X' is open and closed in X and that $\varphi^{-1}(p) = \{o\}$. By construction, $X' - \varphi^{-1}(p) = X' - \{o\}$ is connected. For each V in $\mathcal{V}$, $\pi_1(V - \{p\}) \xrightarrow{\sim} \pi_1(Y' - \{p\})$ implies as above that $\varphi^{-1}(V - \{p\})$ is connected. It then follows from (10.12.2) that the $\varphi^{-1}(V) \cap X'$ ($V \in \mathcal{V}$) form a fundamental system of neighborhoods of o in X' . Consequently X' is the completion of $X' - \{o\}$ over Y' . $\square$

Corollaire 9.15. *If, in addition to the assumptions of (10.12) and (10.13), the $V - \{p\}$ for V in $\mathcal{V}$ are simply connected, then φ is a local homeomorphism at o .*

Indeed, with the notation of (10.13), $Y' - \{p\}$ is simply connected; hence $\varphi : X' - \{o\} \rightarrow Y' - \{p\}$ is a homeomorphism as well as $\varphi' : X' \rightarrow Y'$.

9.6

We shall require a simpler variant of (10.12) in our extension of the map $\tilde{w}$ to cusp points. In that situation, we shall be dealing with a continuous map $\varphi : (X, o) \rightarrow (Y, p)$. Metrics are given only on $X - \{o\}$ and $Y - \{p\}$, and we assume

- a) The $\varphi^{-1}(V)$, for V a neighborhood of p in Y , form a fundamental system of neighborhoods of o . In particular, $\varphi^{-1}(p) = \{o\}$.
- b) There is a neighborhood U of o such that for any neighborhood $V \subset U$, the conditions a), b) of (10.12) are satisfied.

Fix a neighborhood V_1 of p such that $\varphi^{-1}(V_1) \subset U$. For any neighborhood V_2 of p and $x \in V_1 - V_2$, the map $\varphi_1 : \varphi^{-1}(V_1 - V_2) \rightarrow V_1 - V_2$ induces a homeomorphism of $B_r(x)$ with an open set containing $B_\varepsilon(\varphi(x))$ for suitable ε and δ . The map φ_1 is hence a covering map. As this holds for any V_2 , $\varphi : \varphi^{-1}(V_1 - \{p\}) \rightarrow V_1 - \{p\}$ is also a covering map.

If in addition

- c) p has a fundamental system of neighborhoods V such that $V - \{p\}$ is connected and simply connected,
 - d) o has a fundamental system of neighborhoods N such that $N - \{o\}$ is connected,
- then an easy argument shows that φ is a local homeomorphism at o .

9.7

Actually, we require (10.13.3) for the more general case where o and p are replaced by strata. The analogue of (10.12) is :

Proposition 9.16. *Let $\varphi : (X, X_1, o) \rightarrow (Y, Y_1, p)$ be a continuous map of metric spaces, with $Y - Y_1$ locally connected. Assume that there exists a neighborhood U of o such that for each neighborhood V of X_1 , the conditions a) b) of (10.12) hold. Assume further that X_1 is locally compact and that the map $\varphi_1 : X_1 \rightarrow Y_1$ induced by φ is a local homeomorphism at o .*

Then there are open neighborhoods X' of o and Y' of p such that $\varphi : X' - \varphi^{-1}(Y_1) \rightarrow Y' - Y_1$ is a covering map.

Démonstration. Fix U as above, open and small enough so that for some open set R disjoint from U , $K := X_1 - R$ is compact and so that φ is injective on K . Note that if we take $R = X - \bar{U}$, then $K = X_1 \cap \bar{U}$.

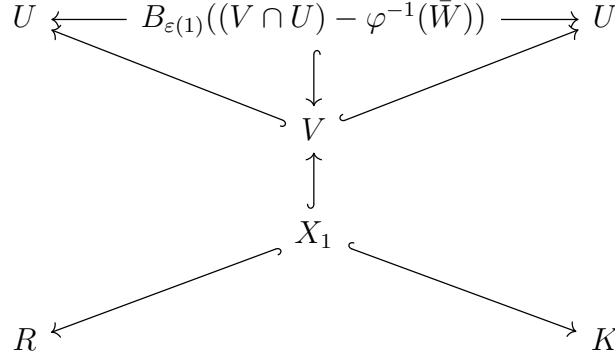
The subset $\varphi(K - U)$ of $Y_1 \subset Y$ is compact and does not contain p . Let W be an open neighborhood of $\varphi(K - U)$ whose closure does not contain p . Then, $\varphi^{-1}(W)$ is an open neighborhood of $K - U$, and $\varphi^{-1}(\bar{W})$ does not contain o .

Select a neighborhood V of X_1 , and $\varepsilon(1) > 0$ such that

$$B_{\varepsilon(1)}((V \cap U) - \varphi^{-1}(\bar{W})) \subset U;$$

this can be done as follows. Since $K - \varphi^{-1}(W)$ is compact and in U , we choose $\varepsilon(1) > 0$ so that $B_{\varepsilon(1)}(K - \varphi^{-1}(W)) \subset U$. One takes $V = B_{\varepsilon(1)}(K - \varphi^{-1}(W)) \cup \varphi^{-1}(W) \cup R$. Then

$(V \cap U) - \varphi^{-1}(W) \subset B_{\varepsilon(1)}(K - \varphi^{-1}(W))$ and $B_{\varepsilon(1)}((V \cap U) - \varphi^{-1}(\bar{W})) \subset U$ by choice of $\varepsilon(1)$.



By a), b), there are r and ε such that

for all $x \in U - V$, φ induces a homeomorphism of $B_\varepsilon(x)$ with an open set of Y containing $B_{2\eta}(\varphi(x))$. (*)

We may and do choose η small enough so that $B_\eta(p)$ is disjoint from $\bar{W}$. Set $Y' = B_\eta(p)$ and take X' to be the set of all x with $\varphi(x) \in Y'$ (hence $x \notin \varphi^{-1}(\bar{W})$) such that either $x \in U \cap V$, or there is $x_1 \in U - V$ with $d(x, x_1) < \varepsilon$ and $\varphi(x_1) \in Y'$.

Any $y \in Y - Y_1$ has a neighborhood W_y disjoint from a neighborhood of Y_1 ; hence, as in (10.12), for r small enough there is $\delta(2)$ such that

for all $x \in U$ with $\varphi(x) \in W_y$, φ induces a homeomorphism from $B_{\delta(2)}(x)$ to an open set of Y containing $B_{\delta(2)}(\varphi(x))$. (**)

One can assume that $\delta(2) < \varepsilon$ and that $B_{\delta(2)}(y) \subset W_y$ and one completes the proof just as in (10.12), using that $\varphi(x) \in Y'$ implies $x \notin \varphi^{-1}(\bar{W})$. □

9.8

As in (10.12.2), X' can be taken as small as we please. If o is not an interior point of X_1 , and if p has a fundamental system of neighborhoods V with $V - Y_1$ connected, φ is open at o . The proof is as in (10.12.3).

Proposition 9.17. *Assume that $Y - Y_1$ is locally simply connected and that*

- c) *p has an open neighborhood V_0 with $V_0 - Y_1$ connected, and each $p' \in Y_1$ close enough to p has a fundamental system of neighborhoods $V \subset V_0$ with $V - Y_1$ connected and $\pi_1(V - Y_1) \xrightarrow{\sim} \pi_1(V_0 - Y_1)$.*
- d) *Each $o' \in X_1$ close enough to o has a fundamental system of neighborhoods N with $N - X_1$ connected.*

Then one can find X' and Y' as in (10.15.1) such that $\varphi' : X' \rightarrow Y'$ has the following additional properties : c) holds with $V_0 = Y'$ for all $p' \in Y_1 \cap V_0$, $X' - X_1$ is connected, $X'_1 := X_1 \cap X'$ is the inverse image of $Y'_1 := Y_1 \cap Y'$, φ' induces a homeomorphism from X'_1 to Y'_1 , and X' is the completion of $X' - X'_1$ above Y' .

Démonstration. Shrinking X and Y , we may and do assume that c) holds for $V_0 = Y$ and for all $p' \in Y_1 \cap V_0$, that $\varphi_1 : X_1 \rightarrow Y_1$ is a homeomorphism, that d) holds for all $o' \in X_1$, that $\varphi : X - \varphi^{-1}(Y_1) \rightarrow Y - Y_1$ is a covering map, and that φ is a local homeomorphism at each point $x \notin X_1$. Let N be an open neighborhood of o with $N - X_1$ connected. As in

(10.13.1), one checks that $N - \varphi^{-1}(Y_1)$ is connected. Let N_1 be the connected component of $X - \varphi^{-1}(Y_1)$ containing $N - \varphi^{-1}(Y_1)$. It is a covering of $Y - Y_1$.

If we shrink Y again without changing the $\pi_1(Y - Y_1)$, and replace X , N , N_1 by their traces on the pull-back of the new Y , we get $N - X_1 \subset N_1$ with N_1 connected and $\bar{N} \supset X_1$.

As in (10.13.2), one sees that $N_1 \cap \varphi^{-1}(Y_1) = X_1$. One takes $Y' = Y$, $X' = \bar{N}_1$ and $\varphi' : X' \rightarrow Y'$ induced by φ . The set X' is open and closed in the (shrunk) X and $\varphi'^{-1}(Y'_1) = X'_1$. All the listed properties of φ' are clear from the construction except for X' being the completion of $X' - X'_1$ over Y' . For $p' \in Y'_1$, $p' = \varphi'(o')$, and V an open neighborhood of p' with $\pi_1(V - Y'_1) \xrightarrow{\sim} \pi_1(Y - Y'_1)$, one has $\pi_0(\varphi'^{-1}(V - Y'_1)) \cong \pi_0(X' - X'_1)$, i.e. $\varphi'^{-1}(V - Y'_1)$ is connected. The $\varphi'^{-1}(V)$ for such V form a fundamental system of neighborhoods of o' and the assertion about completion follows. $\square$

In particular,

Corollaire 9.18. *In addition to the hypothesis of (10.15.1) assume that $Y - Y_1$ is locally simply connected and that*

- c') Each $p' \in Y_1$ close to p has a fundamental system of neighborhood V with $V - Y_1$ connected and simply connected.*
- d) Each $o' \in X_1$ close to o has a fundamental system of neighborhood V with $V - X_1$ connected.*

Then φ is a local homeomorphism at o .

9.9

We will apply (10.15.4) iteratively to some stratified spaces. In the next proposition, “stratified space” means a Hausdorff topological space U , provided with a partition $\mathcal{S}$ into locally closed subsets, the strata, where

- a) each stratum is a manifold ;
- b) the closure of any stratum S is the union of strata ;

$(U, \mathcal{S})$ is conical in the following sense and locally constant along each stratum : each point x of any stratum S has a neighborhood V isomorphic (with the induced partition) to the product of $V \cap S$ with the cone over $(\Sigma, \mathcal{S}(\Sigma))$, where Σ is a topological space, provided with a partition $\mathcal{S}(\Sigma)$. For simplicity, we assume in addition the existence of a metric d for which $d(x, y)$ is the greatest lower bound of lengths of paths from x to y . The case we will need is $U = Q_{*st}$, $\mathcal{S}$ = the partition of Q_{*st} by the Q_T for T varying over the stable partitions.

Proposition 9.19. *Consider a diagram*

$$\begin{array}{ccccc} \tilde{U}_0 & \hookrightarrow & \tilde{U}_1 & \hookrightarrow & \tilde{U} \\ & \searrow p & \downarrow & \swarrow \varphi & \\ & & U & & \end{array}$$

where

- a) U is locally compact stratified space ; U_0 is a connected open and dense stratum, and $U_1 - U_0$ is a union of strata ; each point has a fundamental system of neighborhoods whose traces on U_0 are connected.

- b) $\tilde{U}_0$ is a covering space of U_0 , and $\tilde{U}_1$ (resp. $\tilde{U}$) is the completion of $\tilde{U}_0$ over U_1 (resp. U).
- c) The covering $\tilde{U}_0$ of U_0 is Galois, with group Γ .
- d) Y is a manifold, provided with a metric and an action of Γ by isometries. The map φ is equivariant.
- e) $\varphi|_{\tilde{U}_1}$ is a local homeomorphism.
- f) For any stratum S , $\tilde{S} := p^{-1}(S)$ is a covering space of S (because U_0 is a stratum). We assume that for $S \subset U - U_1$, each point of $\tilde{S}$ has a neighborhood W in $\tilde{U}$ such that $\varphi|_W$ is a tame embedding, with $\varphi(W)$ a subvariety of codimension > 3 in Y .

Then, φ is a local homeomorphism and the (local) decomposition groups of $\tilde{U}/U$ are finite, i.e. $\tilde{U}$ is locally finite to one.

Démonstration. Let d be a Γ -invariant metric on $\tilde{U}$; for instance take $d(x, y) =$ the greatest lower bound of the lengths, computed in U , of paths in $\tilde{U}$ from x to y .

Let S be an open stratum of $U - U_1$. We first prove that φ is a local homeomorphism at any point $o \in \tilde{p}^{-1}(S)$. For this, replacing U by a suitable open neighborhood V of $p(o)$, $\tilde{U}$ by the connected component of $p^{-1}(V)$ containing o and Γ by the stabilizer of o , we may and shall assume that $U - U_1 \subset S$, that p induces an isomorphism $\tilde{S} \xrightarrow{\sim} S$ and that φ embeds $\tilde{S}$ in Y . Replacing Y by a Γ -stable open neighborhood of $\varphi(o)$, and $\tilde{U}_1$ by its inverse image, one may further assume that $Y_1 := \varphi(\tilde{S}')$ is a tame closed submanifold of Y . Its codimension is ≥ 3 . We will apply (10.15.4) with $X = \tilde{U}$, $X_1 = \tilde{S}$ and o, Y, Y_1, φ as above. Local connectedness and simple connectedness of $Y - Y_1$ results from Y being a manifold. Condition c') of (10.15.4) results from Y_1 being a tame submanifold of codimension ≥ 3 . Condition d) of (10.15.4) results from assumption a). By e) $\varphi|_{(X - X_1)}$ is a local homeomorphism, and it remains to check the uniformity conditions of (10.12). As neighborhood of o , we choose the pull back $p^{-1}(K)$ of a compact neighborhood K of $p(o) \in U$. By definition of the topology of $\tilde{U}$, any neighborhood of $X_1 = \tilde{S}$ contains $p^{-1}(V)$, for V an open neighborhood of S . The required uniformities will follow from the compactness of $K - V$. The function from $X - X_1$ to $\mathbb{R}_+^*$: $x \mapsto$ the greatest lower bound of the ε such that φ is an embedding on $B_\varepsilon(x)$, is > 0 , lower semi-continuous, and Γ -invariant; hence it is of the form $\bar{r} \circ p$ with $\bar{r}$ continuous. On $K - V$, $\bar{r}$ stays away from 0, and this gives (10.12 a)). If r is such that for any $x \in p^{-1}(K - V)$, $\varphi|_{B_{2r}(x)}$ is an embedding, the function from $p^{-1}(K - V)$ to $\mathbb{R}_+^*$: $x \mapsto$ the greatest lower bound of the δ such that $\varphi(B_r(x)) \supset B_\delta(\varphi(x))$ is > 0 , lower semi-continuous and Γ -invariant, hence of the form $\bar{r} \circ p$ and staying away from 0. This gives (10.12 b)).

We now prove that $\varphi : \tilde{U} \rightarrow Y$ is a local homeomorphism. We have just shown that φ is a local homeomorphism on $\tilde{p}^{-1}(U'_1)$, for U'_1 the union of U_1 and of the open strata of $U - U_1$. The assumptions of the Proposition hence hold with U_1 (resp. $\tilde{U}_1$) replaced by U'_1 (resp. $p^{-1}(U'_1)$), and one concludes its proof by induction on $\dim(U - U_1)$.

It remains to show that for any $x \in \tilde{U}$, the stabilizer $\Delta \subset \Gamma$ of x is finite. Fix a neighborhood V of x such that $\varphi|_V$ is an open embedding, fix a Δ -stable compact neighborhood K of $\varphi(x)$ in $\varphi(V)$, and define $V_1 = V \cap \varphi^{-1}(K)$. It is a Δ -invariant compact neighborhood of x . For $y \in V_1 \cap \tilde{U}_0$, $\Gamma \cdot y$ is closed and discrete. Therefore Δ_y , which is in $\Gamma \cdot y \cap V_1$, is discrete and compact, and hence finite. Thus Δ is finite. $\square$

We will use (10.16.1) to give an alternate proof of lemma (10.7), by passing the purity of the branch locus theorem (which was derived in the special case $Q_{*st} = Q_{*sst}$).

Lemme 9.20. *If condition (INT) is satisfied, the map $\tilde{w} : \tilde{Q}_{*st} \rightarrow B^+(\alpha)_0$ is a local homeomorphism.*

Démonstration. We apply (10.16.1) with $U = Q_{*st}$ stratified by the Q_T for T varying over the stable partitions, $U_0 = Q_*$, $U_1 =$ the union of Q_* and of the (complex) codimension one strata, $\tilde{U}_0 = \tilde{Q}_*$; hence $\tilde{U} = \tilde{Q}_{*st}$ and $\varphi = \tilde{w}$. Conditions a) to d) are clear, e) is (9.12) and f) follows from (6.9) and (3.9). $\square$

9.10

As shown in (8.7), the map $\tilde{w}$ extends to a continuous map from $\tilde{Q}_{*sst}$ to $B^+ := B^+(\alpha)_0$, where B^+ has the topology (5.4). Fix $j \in Q_{*sst} - Q_{*st}$ and $o \in \tilde{Q}_{*sst}$ above j . Define $p := \tilde{w}_{*sst}(o)$.

Let V be an open connected neighborhood of j , and $\tilde{V}$ be the connected component containing o of its inverse image in $\tilde{Q}_{*sst}$. For V small enough, one has : o is the only point of $\tilde{V}$ above j ; the stabilizer $\Delta \subset \Gamma$ of o acts on $\tilde{V}$, and $\tilde{V}/\Delta \xrightarrow{\sim} V$; if δ is a distance to p function (5.3) on B^+ , δ descends to a continuous function f on $\tilde{V}$, vanishing only at j . As $\tilde{V}$ is locally compact, it follows, for η small enough, that the open sets $\{v \mid f(v) < r\}$ form a fundamental system of neighborhood of j . As a consequence, the $\tilde{w}^{-1}(A)$, for A a neighborhood of p , form a fundamental system of neighborhood of o .

We now apply (10.14) to prove

Lemme 9.21. *If condition (INT) is satisfied, the map $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ is a local homeomorphism.*

Démonstration. It remains to prove it at a point o as above. We apply (10.14) to $\tilde{w} : (\tilde{V}, o) \rightarrow (B^+, p)$. Condition a) has been proved. Condition c) and d) are clear. The map is a local homeomorphism outside of o . The required uniformity (10.14 b)) follows as before from the local compactness of $\tilde{V}$ and Δ -equivariance. $\square$

Théorème 9.22. *Assume condition (INT). Then the map $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ restricts to a homeomorphism of $\tilde{Q}_{*st}$ onto B^+ and maps $\tilde{Q}_{*sst}$ homeomorphically onto an open subset of B^+ , in the (5.4) topology.*

Démonstration. By Proposition (10.18.1), $\tilde{w}$ is an étale map of $\tilde{Q}_{*sst}$ into B^+ . Let φ denote the restriction of $\tilde{w}$ to $\tilde{Q}_{*sst}$. It suffices to prove that φ is a covering map of $\tilde{Q}_{*sst}$ onto B^+ .

For any $x \in \tilde{Q}_{*sst}$, set

$$f(x) = \sup\{r \mid \varphi \text{ maps a neighborhood of } x \text{ homeomorphically onto } B_r(\varphi(x))\}.$$

Then as in the proof of Lemma (10.16.1), f is continuous, $f(x) > 0$ for $x \in \tilde{Q}_{*st}$ and $f(\gamma \cdot x) = f(x)$ for all $\gamma \in \pi_1(Q_*, o)$; that is, f descends to a positive-valued continuous function on Q_{*st} .

Since $\tilde{w}$ is étale at each point of $\tilde{Q}_{*sst}$ and $Q_{*sst} - Q_{*st}$ consists only of finitely many points, one can deduce from the (5.4) topology of B^+ the existence of an open neighborhood W of Q_{*sst} and an $\varepsilon_1 > 0$ such that $f(x) \geq \varepsilon_1$ for all $x \in p^{-1}(W)$. By § 4, Q_{*sst} is compact. From the compactness of $Q_{*sst} - W$ one deduces that f has a non-zero lower bound on $Q_{*sst} - p^{-1}(W)$. Consequently, f has a non-zero lower bound on Q_{*st} . From this it follows that $B_\eta(\varphi(Q_{*st})) = B^+$ for suitable η and therefore, $\varphi(\tilde{Q}_{*st}) = B^+$.

By (10.11), φ is a covering map. Inasmuch as B^+ is simply connected, φ is a homeomorphism.

From the fact that each point $y \in Q_{*sst}$ has a base of neighborhoods $\{V\}$ with $V - \{y\}$ connected, and the fact that $\tilde{w}$ is a homeomorphism on $\tilde{Q}_{*st}$, it follows that $\tilde{w}$ is injective on $\tilde{Q}_{*sst}$ and hence a homeomorphism onto its image, in the topology of (5.4). $\square$

9.11

Let S be a finite set with at least 3 elements, $N = \text{card } S$, $\mu = (\mu_s)_{s \in S}$ with $0 < \mu_s < 1$ and $\sum \mu_s = 2$. Set $\alpha_s = \exp 2\pi\sqrt{-1}\mu_s$ for each $s \in S$, P the projective line over $\mathbb{C}$, $G = \text{Aut } P$, M the subset of injective elements in P^S , $Q_* = M/G$, $o \in Q_*$ and $\theta : \pi_1(Q_*, o) \rightarrow \text{Aut } B^+(\alpha)_o = \text{PU}(1, N-2)$ the homomorphism of $\pi_1(Q_*, o)$ into the isometry group of the ball $B^+(\alpha)_o$, defined in (3.10.2). Set $\Gamma = \theta(\pi_1(Q_*, o))$, $B^+ = B^+(\alpha)_o$.

Théorème 9.23. *If μ satisfies condition (INT), then Γ is a discrete subgroup of $\text{Aut } B^+$ and discontinuous on B^+ .*

Démonstration. By Theorem (10.17.1), the action of Γ on B^+ is equivalent to the action of Γ on $\tilde{Q}_{*st}$. The latter action is discontinuous on $\tilde{Q}_*$ since Γ is the covering group of the covering map $\tilde{Q}_* \rightarrow Q_*$. It follows at once that Γ is discontinuous on a dense open subset of B^+ . Since Γ acts on B^+ via isometries, it follows that Γ is discrete in $\text{Aut } B^+$, the isometry group of B^+ . Since $\text{Aut } B^+$ operates transitively on B^+ with a compact isotropy group, Γ operates discontinuously on B^+ . $\square$

Corollaire 9.24. *For any $y \in \tilde{Q}_{*sst}$, the stabilizer Γ_y of y in Γ is a finite group.*

Démonstration. Set $p = \tilde{w}_*(y)$. Then $\Gamma_y = \Gamma_p = \Gamma \cap (\text{Aut } B^+)_p$. Since Γ is discrete in $\text{Aut } B^+$, and $(\text{Aut } B^+)_p$ is compact, Γ_p is finite. $\square$

Corollaire 9.25. *$\tilde{Q}_*$ is the normal covering of Q_* universal with respect to the property $[R]$: for each stable partition T of S with $\text{card } T = \text{card } S - 1$, if k_T is defined as in (9.9), the ramification index along Q_T divides k_T .*

Démonstration. Let $\tilde{Q}^*$ denote the universal covering with the property above. Then the map $\tilde{Q}^* \rightarrow \tilde{Q}_*$ extends to a covering map of $\tilde{Q}_1$. By Theorem (10.18.2), $\tilde{Q}_1$ is homeomorphic to the complement in the ball B^+ of a closed subset of complex codimension 2 and is simply connected. Consequently, $\tilde{Q}^* \xrightarrow{\sim} \tilde{Q}_*$. $\square$

Corollaire 9.26. *Above a suitably small neighborhood U of a point $x \in \tilde{Q}_{*sst}$, each connected component of $\tilde{Q}_* \cap p^{-1}(U)$ is the universal covering of $Q_* \cap U$ with respect to the ramification property $[R]$.*

The proof is the same as the preceding one, with the ball replaced by the trace of a neighborhood of a point in B^+ (in the topology of (5.4)).

Corollaire 9.27. *The stabilizer in Γ of a point x in the ball is generated by $\mathbb{C}$ -reflections.*

Démonstration. Choose U as in Corollary (10.19.3). Then $U \cap \tilde{Q}_1$ is simply connected. Hence $\pi_1(Q_* \cap U)$ is generated by loops around the ramification locus. These generators yield $\mathbb{C}$ -reflections by (9.2), generating Γ_x if $x \in B^+$. $\square$

10 Finite measure

In this section, the notation and assumptions (4.0) as well as the assumption $N \geq 4$ are continued. We assume moreover that the μ_s are rational.

10.1

Let $j \in Q_{*sst} - Q_{*st}$ be a semi-stable point of type $(S(1), S(2))$ where $\sum_{s \in S(1)} \mu_s = 1$. Let $a \in S(1)$ and $b \in S(2)$. Let W be the neighborhood of j in Q_{*sst} lying below the set of all $y \in P^S$ such that $y(a) = o$, $y(b) = \infty$ and such that

$$\sup\{|y(s)|; s \in S(1)\} \leq \inf\{|y(s)|; s \in S(2)\}.$$

Let $p : \tilde{Q}_{*sst} \rightarrow Q_{*sst}$ and let $o \in p^{-1}(j)$. By (8.2.4), $\pi_1(Q_*, o)$ operates on $\tilde{Q}_{*sst}$, and

$$\pi_1(Q_*, o)_o = D_j = \text{image of } \pi_1(W \cap Q_*) \text{ in } \pi_1(Q_*, o);$$

here, $\pi_1(Q_*, o)_o$ denotes the stabilizer of o and D_j the decomposition group of j .

By (4.5), $W - \{j\}$ is a simply connected manifold if $\text{card } S \geq 5$; hence $\pi_1(W \cap Q_*)$ is generated by circuits around the codimension 1 subspaces Q_T for all stable partitions T finer than $\{S(1), S(2)\}$ with $\text{card } T = \text{card } S - 1$ if $N \geq 5$. By (9.2) the monodromy of such generators are complex reflections which have finite order if μ_s is rational for all $s \in S$. Thus

If $\text{card } S \geq 5$, the quotient $\pi_1(Q_*, o)_o / \text{Ker } \theta$ is generated by elements of finite order; here θ denotes the
(5)

10.2

Assume that condition (INT) is satisfied. Set $V = H^1(P_o, L_o)$, let $\text{PU}(V)$ denote the projective unitary group on V with respect to the hermitian form of (3.10.2). Let $B^+ = B(\alpha)_o^+$, i.e. the set of all $v \in V$ with $(v, v) > 0$ modulo $\mathbb{C}^*$. Let $\tilde{w} : \tilde{Q}_{*sst} \rightarrow B^+$ be the map of Proposition (8.7). Set

$$t = \tilde{w}(o), \quad \Gamma_t = \text{stabilizer of } t \text{ in } \Gamma.$$

By Theorem (10.18.1), $\tilde{w}$ is a homeomorphism onto an open subset of B^+ . Consequently $\Gamma_t = \theta(\pi_1(Q_*, o)_o)$, the stabilizer of t in Γ . By (11.1.1) and (10.19.3), we get

Γ_t is generated by $\mathbb{C}$ -reflections of finite order if $\dim B^+ > 1$ and by a unipotent element if $\dim B^+ =$
(6)

10.3

Let $v \in t$, let $\text{PU}(V)_t$ (resp. $\text{PU}(V)_\delta$) denote the stabilizer in $\text{PU}(V)$ of t (resp. δ , (5.3)), and let N denote the unipotent radical of $\text{PU}(V)_t$. As pointed out in the proof of Proposition (5.5),

$$\text{PU}(V)_\delta / N \cong \text{U}(V'')$$

which is compact, and $\mathrm{PU}(V)_t/N = \mathbb{R} \ltimes \mathrm{U}(V'')$. Hence $\mathrm{PU}(V)_t/N$ contains all compact subgroups of $\mathrm{PU}(V)_\delta/N$. Therefore, as is well-known, $\mathrm{PU}(V)_t$ contains all compact subgroups of $\mathrm{PU}(V)_\delta$ (because $\mathrm{PU}(V)_\delta$ contains a maximal reductive subgroup and all maximal reductive subgroups of $\mathrm{PU}(V)_\delta$ are conjugate by an element of N).

From (11.2.1), one infers

$$\Gamma_t \subset \mathrm{PU}(V)_\delta. \quad (7)$$

Remarque 10.1. A more direct proof of (11.3.1) results from (7.3) — more precisely from the fact that on W , we have a horizontal family of functions δ . It follows at once that Γ_p preserves a “distance to t ” function (5.3).

Théorème 10.2. *Let S be a finite set, $0 < \mu_s < 1$, $\alpha_s = \exp 2\pi\sqrt{-1}\mu_s$, $\sum \mu_s = 2$ (all $s \in S$). Let $\theta : \pi_1(Q_*, o) \rightarrow \mathrm{PU}(V)$ be the monodromy action defined in (3.10.2). Assume*

(INT) *For all $s \neq t$ in S such that $\mu_s + \mu_t < 1$, $(1 - \mu_s - \mu_t)^{-1}$ is an integer.*

Then Γ , the image of θ , is a lattice in $\mathrm{PU}(V)$.

Démonstration. By Theorem (10.19), Γ is discrete in $\mathrm{PU}(V)$, and by Theorem (10.18.2)

$$\Gamma \backslash \tilde{w}(\tilde{Q}_{*\mathrm{sst}}) = \Gamma \backslash B^+$$

where $\tilde{w}(\tilde{Q}_{*\mathrm{sst}})$ is an open subset of B^+ with respect to the topology of (5.4). Hence $\Gamma \backslash \tilde{w}(\tilde{Q}_{*\mathrm{sst}})$ is the union of a compact quotient of a subset of B^+ and a finite set of neighborhoods W_δ (in the topology of (5.4)) of points $t_\delta \in \partial B^+$, where $t_\delta = \tilde{w}(p^{-1}(j))$, $p : \tilde{Q}_{*\mathrm{sst}} \rightarrow Q_{*\mathrm{sst}}$, and j varies over $Q_{*\mathrm{sst}}$. By § 4, $Q_{*\mathrm{sst}}$ is compact. Any compact subset of B^+ has finite measure. By (11.3.1) and Proposition (5.5), the image in $\Gamma \backslash B^+$ of $W_\delta - t_\delta$ has finite measure. It follows that $\Gamma \backslash B^+$ has finite measure. Thus Γ is a lattice in $\mathrm{PU}(V)$. $\square$

Lemme 10.3. *The subgroup Γ_μ is Zariski-dense in $\mathrm{PU}(V)$.*

The proof will be by induction on $\mathrm{card} S$. By (3.10.2), we can lift $\Gamma_\mu \subset \mathrm{PU}(V)$ to a group $\Gamma'_\mu \subset \mathrm{U}(V)$ which is generated by pseudo-reflections (cf. (9.2), (12.3.2)); recall the definition : u is a pseudo-reflection if and only if $u - 1$ has rank 1. Hence Γ'_μ is irreducible on the linear span L of the one dimensional subspaces of V corresponding to its pseudo-reflections.

It is easy to verify that $L = V$. Thus Γ'_μ stabilizes no proper vector subspace of V .

We start the induction at $N = 4$. Here Γ'_μ operates on the complex 1-ball as a triangle group, rotating through twice the angle at each vertex of a geodesic triangle. Since $0 < \mu_s < 1$ and $\sum \mu_s = 2$, the sum of the angles of the geodesic triangle is less than π and the orbit of each point in B^+ under Γ'_μ is infinite. Let $\bar{\Gamma}'_\mu$, $\bar{\Gamma}_\mu$ denote the Zariski closure of Γ'_μ , Γ_μ in $\mathrm{U}(V)$ and $\mathrm{PU}(V)$ respectively. The group $\bar{\Gamma}'_\mu$ can have no connected normal solvable subgroup non-trivial modulo the center of $\mathrm{U}(V)$, otherwise, some Γ'_μ orbits in B^+ would be finite. Consulting the short well known list of closed complex analytic subgroups in $\mathrm{PU}(1, 1) \cong \mathrm{PGL}(2, \mathbb{R})$, one sees that $\bar{\Gamma}_\mu = \mathrm{PU}(V)$ if $\mathrm{card} S = 4$. Suppose now $\mathrm{card} S > 4$.

By (6.21) and two induction hypothesis, $\bar{\Gamma}_\mu \supset \mathrm{PU}(V_T)$ where T is a stable partition of S with $\mathrm{card} T = \mathrm{card} S - 1$ and V_T is the corresponding subspace of V . Since Γ'_μ is irreducible on V , we infer $\bar{\Gamma}_\mu = \mathrm{PU}(V)$. $\square$

11 Arithmeticity and Integral Monodromy

11.1

Assume that for each $s \in S$, μ_s is a rational number and let d denote the least common denominator of $\{\mu_s \mid s \in S\}$. Set $F = \mathbb{Q}(\sqrt[d]{1})$, $\mathcal{O}_F$ the ring of integers in F , and $V = H^1(P_o, L_o)$ where $P_o = P - o(S)$ as in (3.1). Then the vector space V can be defined over the ring $\mathcal{O}_F$. To see this, let $L(F)$ (resp. $L(\mathcal{O})$) denote a local subsystem of L on P_o (suitably chosen using a base point on P_o) with fiber the subfield F (resp. subring $\mathcal{O}$) of $\mathbb{C}$. The pairing of L with its complex conjugate induces an $\mathcal{O}_F$ -valued pairing on $L(\mathcal{O})$. Then one can define cohomology combinatorially as in (2.2) with coefficients in $L(\mathcal{O})$; one can also define the skew-hermitian cup product

$$\psi_0 : H^1(P_o, L(\mathcal{O})_o) \times H^{2-1}(P_o, \bar{L}(\mathcal{O})_o) \rightarrow H^2(P_o, \mathcal{O}) \cong \mathcal{O}_F$$

combinatorially; set

$$V(\mathcal{O}) = H^1(P_o, L(\mathcal{O})_o), \quad V(F) = H^1(P_o, L(F)_o).$$

Define

$$\psi_0 : V(F) \times V(F) \rightarrow F \tag{8}$$

where $j = b - \bar{b}$ for some $b \in \mathcal{O}_F - \mathcal{O}_F \cap \mathbb{R}$. This ψ_0 is a hermitian form on V defined over $\mathcal{O}$ and may be identified with the form defined in (2.18), up to a real scalar factor.

We lift the map $\theta : \pi_1(Q_*, o) \rightarrow \text{PU}(V)$ of (3.10.2) via a local system L on $M(c)$ to $\theta' : \pi_1(M(c), o) \rightarrow \text{U}(V)$ by (3.14). The image of θ' is in $\text{U}(V)(\mathcal{O})$. Set

$$\Gamma_\mu := \Im \theta, \quad \Gamma'_\mu := \Im \theta' \tag{9}$$

where $\mu : \{\exp 2\pi i \mu_s \mid s \in S\}$. Then

$$\Gamma'_\mu \text{ stabilizes } V(\mathcal{O}). \tag{10}$$

11.2

We collect here some remarks and definitions pertaining to arithmeticity of lattices.

Let Γ be a Zariski-dense subgroup of $G(k)$, for G an adjoint connected semi-simple algebraic group G over k , a field of characteristic 0. Set $E = \mathbb{Q}[\text{Tr Ad } \Gamma]$, the field spanned over $\mathbb{Q}$ by $\{\text{Tr Ad } g \mid g \in \Gamma\}$.

Proposition 11.1.

- (i) *The group G has a faithful matrix representation ρ such that $\rho(\Gamma) \subset \text{GL}_n(E)$ (and hence G has E as a field of definition).*
- (ii) *E remains unchanged when Γ is replaced by a commensurable subgroup of G .*

Démonstration. Let T denote the function $g \mapsto \text{Tr Ad } g$ on G , and let W denote the $\mathbb{C}$ -linear span of the left G -translates of T (we define $(x \cdot f)(y) = f(yx)$ for any function $f : G \rightarrow \mathbb{C}$, x and y in G). Inasmuch as T is a sum of matrix coefficients, W is finite dimensional. Since Γ is Zariski dense in G , there is a finite set of elements $x_1, \dots, x_n$ in Γ such that $\beta = \{x_1 \cdot T, \dots, x_n \cdot T\}$ is a base of W . Let $\rho_W : G \rightarrow \text{Aut } W$ be the representation given by $x \mapsto x \cdot f$ for $x \in G$, $f \in W$. It is well known that the representation ρ_W is

faithful ([10], Theorem (2.1), p. 123). Let $\rho(x)$ denote the matrix of $\rho_W(x)$ with respect to the base β . Then $\rho(x)$ has all its entries in the field E for all $x \in \Gamma$.

To prove (ii) it clearly suffices to consider the case that Γ_0 is a subgroup of Γ of finite index. Replacing Γ_0 by $\bigcap_x x\Gamma x^{-1}$, we can suppose moreover that Γ_0 is normal in Γ . Then Γ_0 is also Zariski-dense in G . Set $E_0 = \mathbb{Q}[\text{Tr Ad } \Gamma_0]$ and select the base $\beta = \{x_1 \cdot T, \dots, x_n \cdot T\}$ with $x_i \in \Gamma_0$ ($i = 1, \dots, n$). Let $\sigma : E \rightarrow \mathbb{C}$ be a monomorphism which is the identity on E_0 . To prove (ii), it suffices to prove that σ leaves each element of E fixed. For any $y \in \Gamma$ and $x \in \Gamma_0$, we have $xyx^{-1} \in \Gamma_0$. Hence, by (12.2.1),

$$\sigma\rho(yxy^{-1}) = \sigma\rho(y) \cdot \sigma\rho(x) \cdot \sigma\rho(y)^{-1} = \rho(y) \cdot \sigma\rho(x) \cdot \rho(y)^{-1}.$$

that is $\rho(y)\sigma\rho(x)\rho(y)^{-1} = \sigma\rho(y)\sigma\rho(x)\sigma\rho(y)^{-1}$. Hence for all $y \in \Gamma_0$, $\sigma\rho(y)^{-1}\rho(y)$ centralizes $\sigma\rho(\Gamma_0)$ and therefore $\sigma\rho(G)$ since Γ_0 is Zariski-dense in G . The center of $\sigma\rho(G)$ is 1 and consequently $\sigma\rho = \rho(y)$ for all $y \in G$. Therefore σ fixes each element of E . $\square$

11.3

Take ρ as in (12.2.1). The Zariski closure of $\rho(E)$ in $\text{GL}_n(E)$ is an algebraic group over E . Because taking Zariski closure commutes with field extensions it is a form of G over E . This provides the group G with an E -structure, called the *natural E -structure* of G . If ρ' is another faithful matrix representation of G with $\rho'(\Gamma) \subset \text{GL}_n(E)$, it leads to the same E -structure. To check this, one compares ρ to ρ' via $\rho \oplus \rho'$.

If F is a subfield of k and G_F an F -structure on G for which $\Gamma \subset G_F(F)$, the field of traces E is contained in F and, by the above construction applied to G_F over F , the given F -structure on G is deduced from the natural E -structure.

11.4

Let A be a semi-simple algebraic linear group defined over the field $\mathbb{Q}$ of rational numbers, let V be a finite dimensional vector space defined over $\mathbb{Q}$, and $\rho : A \rightarrow \text{GL}(V)$ a faithful rational representation defined over $\mathbb{Q}$. By a theorem of Borel-Harish-Chandra [3], the subgroup $\Gamma = \rho^{-1}(\text{GL}(V_{\mathbb{Z}}))$ is a lattice in $A(\mathbb{R})$ for any lattice $V_{\mathbb{Z}}$ in $V_{\mathbb{Q}}$; that is, Γ is a discrete subgroup of $A(\mathbb{R})$ and $A(\mathbb{R})/\Gamma$ has finite measure.

11.5

Let G be an adjoint connected semi-simple real Lie group. It is the topological connected component $G(\mathbb{R})^0$ of $G(\mathbb{R})$, for G an adjoint connected semi-simple algebraic group over $\mathbb{R}$. By definition, a subgroup Γ of G is *arithmetic in G* if and only if there exists an algebraic group A over $\mathbb{Q}$, a compact group K and an analytic isomorphism θ of $A(\mathbb{R})^0$ onto $G \times K$ such that $\theta(A(\mathbb{Z}) \cap A(\mathbb{R})^0)$ has its projection into G commensurable to Γ . The group A is necessarily reductive. It can be assumed connected adjoint semi-simple; namely, replace A by $A' := A/Z(A)^0$. The group A is then a product of $\mathbb{Q}$ -simple groups A_i (cf. [23], p. 46 (3.1.2)). If G is a simple non-compact Lie group, it follows from the definition that all A_i but one, A_1 , are such that $A_i(\mathbb{R})$ is compact. Replacing A by A_1 , one can then assume that A is $\mathbb{Q}$ -simple, i.e. of the form (cf. [23], p. 46 (3.1.2))

$$A = \text{Rest}_{F/\mathbb{Q}} B$$

for B an absolutely simple group over a finite extension F of $\mathbb{Q}$. One has $F \otimes \mathbb{R} = \prod F_v$ (product of the completions of F at the infinite places v) and $A(\mathbb{R}) = B(F \otimes \mathbb{R}) = \prod B(F_v)$. The existence of θ as in (12.2.4) amounts to saying that for all v but one, v_1 , $B(F_v)$ is compact (hence if $v \neq v_1$ the place v is a real place) and that $B(F_{v_1})^0 \cong G$, the isomorphism carrying $B(\mathcal{O}_F)$ to a subgroup of G commensurable to Γ . If G is an absolutely simple Lie group, the place v_1 is a real place, F is totally real, and B is an F -form of G , F being identified with a subfield of $\mathbb{R}$ via v_1 .